Bozeman Climate Plan

2020


Our Climate
More Public Transport
That reduces dependency
on cars
Reduce transportation
emissions
DENVER


THESE HANDS HELP


OUR PLANET BY....


Our Climate Vision

ACKNOWLEDGEMENTS

The Bozeman Climate Plan was led by the City of Bozeman, but it is a plan for our entire community. It was made possible by an accomplished team of committed community members representing students, non-profits, small businesses, utilities, civic associations, Montana State University, farmers, health care professionals, neighbors, and City staff. Thank you for joining us in this effort to help define our pathway to a low-carbon future.

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Commissioner I-Ho Pomeroy
Commissioner Jennifer Madgic
Commissioner Michael Wallner

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[1] Britt Ide serves on the NorthWestern Energy Board of Directors, the Coalition for Green Capital Board of Directors, and co-authored the City of Boise's Energy Future, a plan to achieve 100% renewable energy.

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- Jay Sinnott, Bozeman Climate Partners
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- Claire Vlases, Bozeman High School-Student Body
- Edie Wiley, Bozeman Health

Bozeman Climate Plan

CLIMATE PLAN EXECUTIVE SUMMARY

Implementing this ambitious Climate Plan will lower Bozeman's contribution to climate change while making the community more resilient to the local impacts from an already changing climate. Moreover, it will advance Bozeman's strategic priorities to be a safe, inclusive community that fosters civic engagement, a thriving diversified economy, a strong environmental ethic, and a high quality of life as the community grows and changes.

In setting bold new goals and actionable strategies, the plan reaffirms and expands past commitments made in the 2011 Community Climate Action Plan (CAP) and in a 2017 resolution to uphold the goals of the Paris Climate Agreement through local action.

RELEVANCE DURING UNPRECEDENTED TIMES

During the course of developing this plan, COVID-19 has swept the globe, and more recently there have been widespread protests to address systemic racial inequalities in our country. These events reinforce the timeliness and relevance of this plan, in particular the resiliency goals to protect human health and increase social cohesion and the cross-cutting focus on equity. Further, COVID-19 may be an insightful test of the community resolve and cooperative frameworks needed to manage through this pandemic. The experience demonstrates the need to remain resilient and flexible to changing economic conditions while proactively addressing climate change. Aligning these pursuits will help ease suffering among Bozeman's most vulnerable populations and create synergies with related priorities like affordable housing.

As national policy frameworks are being considered for building back better and more equitably after COVID-19, climate action takes on new meaning and opportunity. For example, considering Bozeman community members currently spend over $100 million per year on electricity, natural gas, and transportation fuels, the transition to a carbon neutral economy over the next 30 years can and should be navigated with short-term economic conditions in mind. This includes prioritizing cost-effective solutions and aligning with the City's efforts to foster a strong economic base.

VISION FOR THE FUTURE AND STAYING THE COURSE

Given today's risks and a highly uncertain future, the plan's vison and guiding principles will help ensure relevance throughout Bozeman's 30-year journey to carbon neutrality.

Climate Vision

Through leadership and collaboration, the City of Bozeman will advance innovative solutions to cultivate a more equitable and resilient low-carbon community for current and future generations.

Guiding Principles

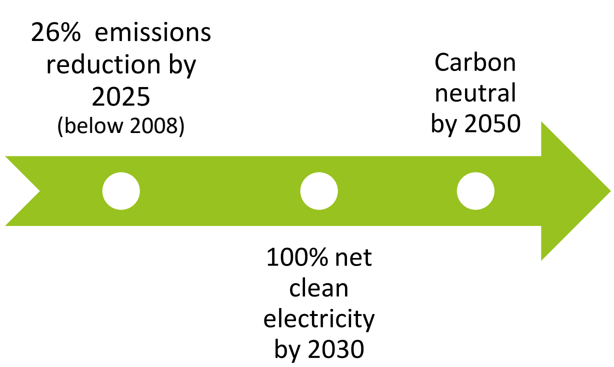
Bozeman will be a leader in addressing climate change by:

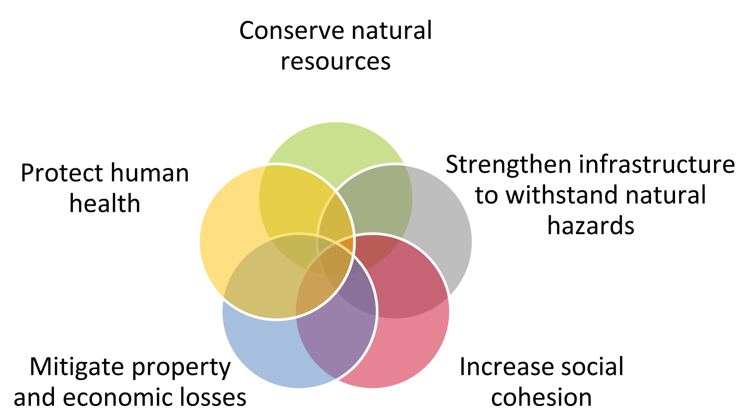
- Adopting bold targets for emissions reduction and renewable energy.
- Seeking innovative, actionable solutions to mitigate climate change.
- Weaving sustainability and resilience into decision-making processes.
- Pursuing partnerships with other municipalities and our utility provider.
- Inviting all Bozeman residents to join us, including current and future leaders.

BOLD TARGETS FOR EMISSIONS REDUCTION AND RENEWABLE ENERGY

Delivering on the first guiding principle, the bold emissions mitigation targets summarized below set a pathway to carbon neutrality by 2050, starting with significantly reducing emissions in the next five years, then transitioning to 100% net clean electricity within 10 years.

Mitigation Goals	*Resiliency Goals*

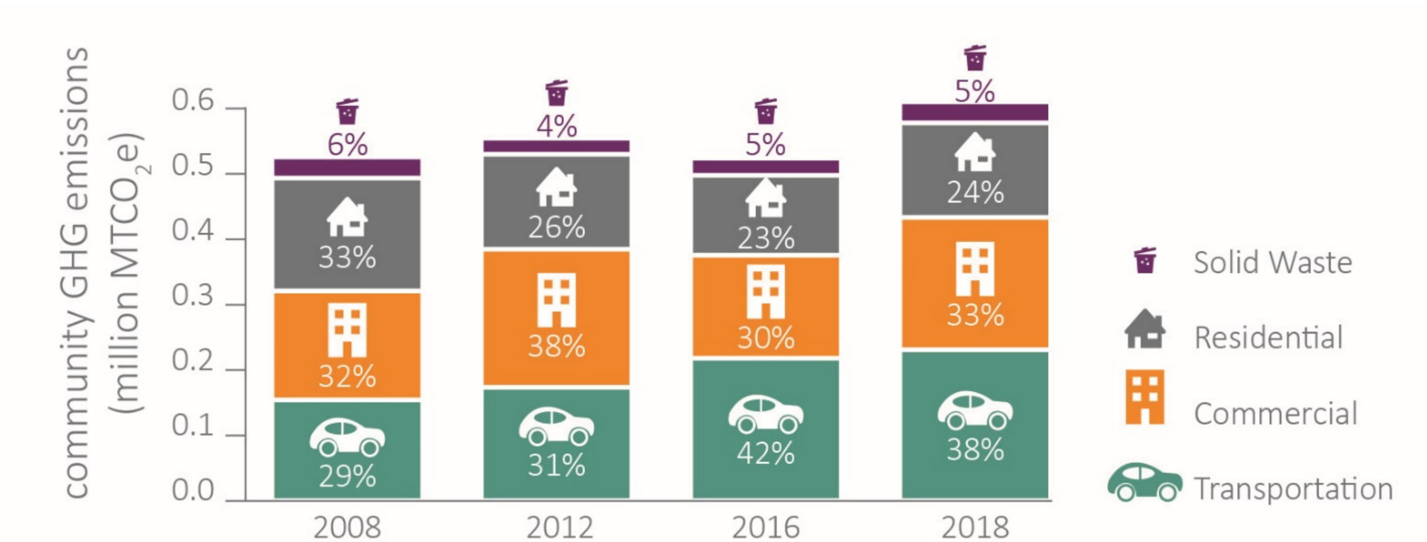




The resiliency goals address the impacts of climate change identified in the 2019 [City of Bozeman Vulnerability Assessment and Resiliency Strategy:]()

Extreme Heat	More frequent & intense	**Mountain Snowpack**	Decline in volume
Floods	More severe	**Wildfire**	More extensive, frequent, & intense
Drought	More frequent & intense	**Winter Storms**	More severe

Despite years of climate action, total community emissions have increased since 2008. This increase is closely related to the 37% population growth the community experienced in the past decade. Overall, per capita emissions have declined by 15% since 2008. Looking to the future, Bozeman will need to continue addressing per-capita emissions by prioritizing high-impact emissions reduction opportunities. Because 57% of Bozeman's 2018 emissions came from building energy use, improving building efficiency and increasing renewable energy supply are paramount to achieving the near-term emissions reduction goal. Meanwhile, transportation is the fastest growing sector of community emissions with a 50% increase since 2008, requiring a strong immediate response to offset the growth. Over time, as building-related emissions decrease, efforts to reduce transportation and waste-related emissions will grow in importance to achieve carbon neutrality.

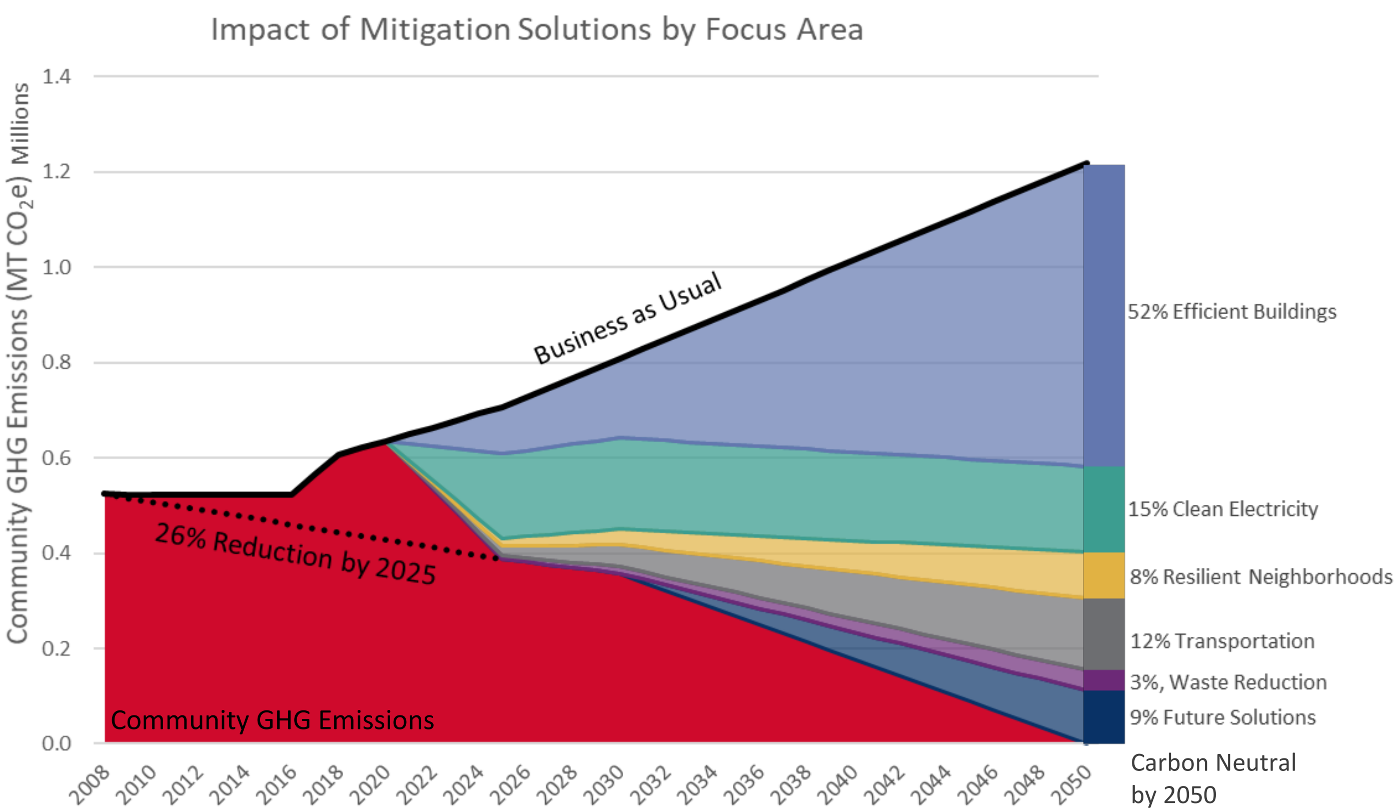


A ROBUST SET OF INNOVATIVE, ACTIONABLE SOLUTIONS

Adhering to the second guiding principle, this plan delivers a robust set of 16 innovative, actionable solutions organized across six focus areas:

Each solution includes a suite of supporting actions that ultimately help Bozeman achieve its climate vision and goals. Actions encompass a range of activities including advocacy, projects, programs, infrastructure, outreach, partnerships, policy, financial tools, studies, and more. To ensure the solutions are actionable, each action includes a detailed description, action priority level, identification of lead and implementing partners, and performance measures to keep the action on track.

Based on modeling of the emissions reduction potential for each solution, the following diagram summarizes the pathway to achieving Bozeman's GHG emissions reduction goals. The black line along the top of the graph represents actual or projected business as usual community GHG emissions and each colored wedge represents the contribution of solutions in each focus area to meeting Bozeman's GHG mitigation commitments. For more details on the individual solution contributions see the breakout in each focus area section. Note that two pathways for clean energy are explored in Focus Area 2: Responsible & Reliable Clean Energy Supply.



Projected GHG emissions and anticipated emissions reduction by focus area

WEAVING SUSTAINABILITY AND RESILIENCE INTO DECISION-MAKING PROCESSES

The third guiding principle, weaving sustainability and resilience into decision-making processes, recognizes that the solutions and their strategies are economically feasible but will require community and political will and a culture shift in ensuring climate action is infused into city services, business operations, and lifestyle choices.

Embracing this guiding principle, the plan was developed through a comprehensive engagement process. The engagement process not only served to provide an inclusive approach for developing the elements of this plan, but also to raise the public and community partners' interest, understanding, and capacity for weaving climate action into daily decision-making.

The engagement approach included workshops with an intergenerational, transdisciplinary climate team, focus area meetings, in-person and online community forums, community survey, public-facing project website, community events such as a climate lecture at Montana State University, and City staff coordination meetings. See Appendix C: Climate Team

Workshop Summaries and Appendix D: Community Forum Summaries for detailed notes and materials from these engagement efforts.

CITY LEADERSHIP AND PARTNERSHIPS ARE CRITICAL TO SUCCESS

In addition to grassroots action across all decision-making, the fourth guiding principle, pursuing partnerships with other municipalities and our utility provider, recognizes the importance of the City leading the way through its municipal operations, collaborating with other municipalities regionally and at the state level, and advancing its strategic initiatives with NorthWestern Energy. Example Priority Level 1 actions in these areas include:

City Leading the Way

- Action 1.A.1. Increase Energy Efficiency at City Facilities
- Action 2.F.1. Plan and Install Renewable Energy Projects for City Facilities
- Action 6.O.1. Invest in Landscaping and Irrigation Upgrades at City Facilities
- Action 6.P.2. Maintain and Expand the Urban Forest

Collaborations with NorthWestern Energy

- Action 2.D.2. Collaborate and Innovate Utility-Scale Renewable Energy Solutions
- Action 4.K.1. Support Community EV Roadmap Development
- Action 4.K.2. Collaborate to Install Publicly Accessible EV Infrastructure

State-Level Partnerships with Other Municipalities

- Action 1.B.2. Advocate for Adoption of State-Wide Net Zero Energy Code
- Action 2.D.3. Support Policies to Expand Renewable Energy and Just Transition Initiatives

Collaborations with Regional Municipalities and Partners

- Action 3.G.1. Continue Regional Coordination on Compact Growth and Sustainable Development
- Action 4.J.1. Prioritize Regional Multi-Modal Planning and Connectivity
- Action 4.J.4. Pursue Sustainable Transit Funding and Expansion
- Action 4.J.6. Support Regional Transit Service Coordination and Outreach

INVITING ALL BOZEMAN RESIDENTS TO JOIN US, INCLUDING CURRENT AND FUTURE LEADERS

Finally, the fifth guiding principle recognizes the shared responsibility for implementing the solutions in this plan in order to achieve the myriad emissions reduction and resiliency benefits documented throughout this plan. As the City and its partners create climate action policies, programs, and services, Bozeman residents and businesses can do their part by learning about their contribution to Bozeman's greenhouse gas emissions and becoming supportive adopters of new climate actions as they become available. Example action items that invite all Bozeman residents, businesses, and community organizations to get involved are outlined on the following pages.

Inviting Bozeman Residents to Join the Climate Effort!

Bozeman residents are encouraged to print a copy of this page and the next as a quick reference guide for how you can do your part in supporting the Climate Plan. See the more detailed action descriptions in the Climate Plan to learn more about how each action contributes to Bozeman's climate goals. Check off actions as you complete them!

Action 1.A.2.

Reduce energy usage during peak energy demand (4pm to 8pm)

Healthy, Adaptive & EFFICIENT BUILDINGS

Action 1.A.3.

Learn about and begin to make changes to your energy behaviors

Healthy, Adaptive & EFFICIENT BUILDINGS

Action 1.A.3.

Contact NorthWestern Energy, or other qualified auditor, to schedule a home energy audit and make efficiency improvements to your home

Healthy, Adaptive & EFFICIENT BUILDINGS

Action 1.B.2.

Advocate for more stringent state-wide energy and water efficiency regulations

Healthy, Adaptive & EFFICIENT BUILDINGS

Action 2.F.4.

Explore opportunities to install on-site renewable energy and storage on your property

Responsible & Reliable CLEAN ENERGY SUPPLY

Action 3.G.3.

Plan a neighborhood activity to help build social connections and improve community cohesion

Vibrant & Resilient NEIGHBORHOODS

Action 3.H.3.

Get to know your neighbors and swap contact information for times of need or emergency

Vibrant & Resilient NEIGHBORHOODS

Action 3.H.3.

Review City maps to understand if you are in a location that is vulnerable to flooding, fires, or other hazards and develop an emergency plan if a hazard event occurs

Vibrant & Resilient NEIGHBORHOODS

Actions 4.J.2. & 4.J.6.

Walk, bike, carpool, or take transit to destinations instead of driving alone

Diverse & Accessible TRANSPORTATION OPTIONS

Action 4.K.1.

Limit idling and combine trips when using a vehicle for transportation

Diverse & Accessible
TRANSPORTATION OPTIONS

Action 4.K.6.

Consider investing in an electric vehicle for your next vehicle purchase

Diverse & Accessible
TRANSPORTATION OPTIONS

Action 4.L.1.

Find alternatives to air travel, avoid binge flying, and/or purchase offsets for your next airline trip

Diverse & Accessible
TRANSPORTATION OPTIONS

Action 5.M.1.

Review your waste and consumption practices and look for opportunities to reduce, reuse, or share products

Comprehensive & Sustainable
WASTE REDUCTION

Action 6.N.2.

Volunteer at or donate to a local food bank

Regenerative Greenspace, Food Systems &
NATURAL ENVIRONMENT

Action 6.N.3.

Learn to garden and grow your own food

Regenerative Greenspace, Food Systems &
NATURAL ENVIRONMENT

Action 6.P.2.

Plant and maintain a tree

Regenerative Greenspace, Food Systems &
NATURAL ENVIRONMENT

Action 6.O.2.

Update irrigation equipment and landscaping to use less water

Regenerative Greenspace, Food Systems &
NATURAL ENVIRONMENT

Action 6.P.4.

Reduce pesticide and herbicide use

Regenerative Greenspace, Food Systems &
NATURAL ENVIRONMENT

Inviting Bozeman Businesses and Community Organizations to Join the Climate Effort!

- ☐ Reduce energy usage during peak energy demand (4pm to 8pm) (Action 1.A.2.)
- ☐ Become a City of Bozeman Energy Project Partner (Action 1.A.3.)
- ☐ Contact NorthWestern Energy to schedule an energy appraisal and implement appraisal recommendations (Action 1.A.3.)
- ☐ Monitor and benchmark your building's energy performance (Action 1.A.4.)
- ☐ Explore opportunities to install on-site renewable energy and storage on your property (Action 2.F.4.)
- ☐ Review City maps to understand if you are in a location that is vulnerable to flooding, fires, or other hazards and develop an emergency and continuity of operations plan if a hazard event occurs (Action 3.H.3.)
- ☐ Engage your employees or constituents in emergency preparedness planning, drills, and protocols (Action 3.H.3.)
- ☐ Provide options and incentives for employee telecommuting and alternatives to single-occupancy vehicle travel (e.g., bike to work days, preferred parking spots, carpool matching, bicycle racks, wellness programs, etc.) (Actions 4.J.2. and 4.J.5.)
- ☐ Install electric vehicle charging infrastructure for fleet, employee, and potentially public use (Action 4.K.2.)
- ☐ Convert fleet vehicles and equipment to electric or alternative fuel models (Action 4.K.3.)
- ☐ Establish and enforce employee idling policies when using personal or fleet vehicles for business use (Action 4.K.5.).
- ☐ Limit non-essential airline travel and/or purchase carbon offsets for airline trips (Action 4.L.1.)
- ☐ Review your supply chain and consumption practices and look for opportunities to use less packaging, reuse or recycle materials, and compost organic waste (Action 5.M.1.)
- ☐ Provide markets for recycled products by supporting suppliers and businesses that use recycled materials (Action 5.M.1.)
- ☐ Reuse or donate used equipment and goods (Action 5.M.1.)
- ☐ Donate unused food and right-size large catering orders (Action 5.M.1.)
- ☐ Plant and maintain trees (Action 6.P.2.)
- ☐ Purchase products that support growth of the local food system (Action 6.N.3.).
- ☐ Update irrigation equipment and landscaping to use less water (Action 6.O.2.)
- ☐ Reduce pesticide and herbicide use (Action 6.P.4.)

Solution and Action Summary Table

Focus Area	*Action*	*Priority Level*
Focus Area 1. Healthy, Adaptive & Efficient Buildings	**Solution A. Improve Efficiency of Existing Buildings**	
	1.A.1. Increase Energy Efficiency at City Facilities	1
	1.A.2. Use Data and Price Signals to Advance Energy Efficiency	1
	1.A.3. Expand Energy Efficiency Information and Resources for Private Property	1
	1.A.4. Establish an Energy and Water Benchmarking Standard for Commercial Buildings	1
	1.A.5. Require Home Energy Labeling at Time of Listing	2
	1.A.6. Promote Energy Efficiency Financing and Investment	2
	1.A.7. Create a Rental Registry Program to Advance Renter Safety and Energy Efficiency	3
	Solution B. Achieve Net Zero Energy New Construction	
	1.B.1. Support High Performance Building Resources and Training for the Development Community	1
	1.B.2. Advocate for Adoption of State-Wide Net Zero Energy Code	1
	1.B.3. Encourage High Performance Construction for All Publicly Funded Buildings	2
	1.B.4. Analyze and Support Opportunities for District Energy	2
	1.B.5. Offer a Voluntary Pathway & Incentives for Above-Code Construction	3
	Solution C. Electrify Buildings	
	1.C.1. Advance Electrification Upgrades and Conversion Projects for City Facilities	2
	1.C.2. Include an Electrification Component for Above-Code Construction	3
	1.C.3. Support Outreach and Incentives for Electric Appliances and Equipment	3
Focus Area 2. Responsible & Reliable Clean Energy Supply	**Solution D. Increase Utility Clean Energy Mix**	
	2.D.1. Complete a 100% Net Clean Electricity Community Feasibility Study	2
	2.D.2. Collaborative and Innovate Utility-Scale Solutions with Utility Provider	1
	2.D.3. Support Policies to Expand Renewable Energy and Just Transition Initiatives	1
	2.D.4. Encourage Philosophical Shift for Utility Provider	1
	Solution E. Develop and Promote Utility Green Power Programs	
	2.E.1. Advance Green Tariff Program Development and Participation	1
	Solution F. Increase Community-Based Distributed Renewable Energy Generation	
	2.F.1. Plan and Install Renewable Energy Projects for City Facilities	1
	2.F.2. Streamline Solar Permitting and Adopt Solar-Ready Code Provisions	1
	2.F.3. Advance Distributed Solar Policies	2
	2.F.4. Promote Education and Incentives for Distributed Renewable Energy and Storage	3

Focus Area	*Action*	*Priority Level*
Focus Area 3. Vibrant & Resilient Neighborhoods	**Solution G. Facilitate Compact Development Patterns**	
	3.G.1. Continue Regional Coordination on Compact Growth and Sustainable Development	1
	3.G.2. Review Development Code to Enhance Compact and Sustainable Development	1
	3.G.3. Develop Sustainable Neighborhoods Outreach	2
	Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards	
	3.H.1. Plan for Resilience Hubs at Critical Facilities	1
	3.H.2. Advance Resilience in Development Code and Development Review	1
	3.H.3. Support Business and Residential Preparedness Outreach	2
	3.H.4. Incorporate Resilience into Infrastructure Plans	2
	Solution I. Enhance Social Infrastructure and Community Preparedness	
	3.I.1. Support Community and Neighborhood Resilience Programming	1
Focus Area 4. Diverse & Accessible Transportation Options	Solution J. Increase Walking, Bicycling, Carpooling and Use of Transit	
	4.J.1. Prioritize Regional Multi-Modal Planning and Connectivity	1
	4.J.2. Pursue Innovative Funding for Pedestrian and Bicycle Connections and Network	1
	4.J.3. Improve Maintenance of Multi-Modal Infrastructure	1
	4.J.4. Pursue Sustainable Transit Funding and Expansion	1
	4.J.5. Support Employee Trip Reduction Programs and Transportation Demand Management	1
	4.J.6. Support Regional Transit Service Coordination and Outreach	1
	4.J.7. Leverage Parking Policies to Encourage Other Modes of Transportation	2
	4.J.8. Develop Bike and Car Share Programs	3
	Solution K. Decrease Direct Vehicle Emissions	
	4.K.1. Support Community EV Roadmap Development	1
	4.K.2. Collaborate to Install Publicly Accessible EV Infrastructure	1
	4.K.3. City Fleet and Transit EV Investment	2
	4.K.4. Advocate for EV Utility Rates, Incentives, Infrastructure, and Efficiency Standards	2
	4.K.5. Limit Wasteful Vehicle Emissions	2
	4.K.6. Support EV Group Buy and Outreach	3
	Solution L. Improve Air Travel Efficiency	
	4.L.1. Build Awareness of Air Travel Impacts and Alternatives	2
	4.L.2. Advocate for Increased Air Travel Efficiency	3

Focus Area	*Action*	*Priority Level*
Focus Area 5. Comprehensive & Sustainable Waste Reduction	**Solution M. Move Toward a Circular Economy and Zero Waste Community**	
	5.M.1. Actively Promote Source Reduction, Recycling, and Repair	1
	5.M.2. Expand Composting Services and Collection	1
	5.M.3. Improve Waste Policies, Services, and Operations	2
	5.M.4. Support Construction Waste Diversion	2
	5.M.5. Encourage the Development of Material Markets	3
	5.M.6. Develop Plans for Green Purchasing and Zero Waste Events for City Operations	3
Focus Area 6. Regenerative Greenspace, Food Systems & Natural Environment	**Solution N. Cultivate a Robust Local Food System**	
	6.N.1. Support the Formation of a Local Food Council	1
	6.N.2. Help Develop a Food System Assessment and Security Plan	2
	6.N.3. Encourage Local Agriculture and Preservation of Working Lands	2
	6.N.4. Support Local Food Production, Processing, and Distribution	3
	Solution O. Manage and Conserve Water Resources	
	6.O.1. Invest in Landscaping and Irrigation Upgrades at City Facilities	1
	6.O.2. Build on the Success of Water Conservation Education and Incentives	2
	6.O.3. Evaluate Additional Water Conservation Code and Water Rate Structure Adjustments	2
	Solution P. Manage Land and Resources to Sequester Carbon	
	6.P.1. Protect Local Wetlands for Flood Resilience and Water Quality	1
	6.P.2. Maintain and Expand the Urban Forest	1
	6.P.3. Enhance Greenspace and Carbon Sequestration for New Development	1
	6.P.4. Provide Outreach on Water Pollution Prevention and Carbon Sequestration Strategies	2

CONTENTS

FIGURES

TABLES

CHAPTER 1:

INTRODUCTION

CHAPTER 1: INTRODUCTION

In 2017, the City of Bozeman passed Resolution 4866 to join the Climate Mayors, a partnership of over 400 United States mayors committing to uphold the goals of the Paris Climate Agreement through local action. This Climate Plan updates the City's previous emissions reduction goals, outlined in the 2011 Community Climate Action Plan (CAP), to reflect this commitment. This plan also updates the mitigation strategies from the 2011 CAP to reflect the latest available climate science, new technologies, and evolving community priorities. Anticipating future changes and the need to adapt to the risks associated with changing local climate conditions, the Climate Plan also includes strategies to improve Bozeman's climate resiliency.

For more than a decade, Bozeman has made strong commitments to climate action and demonstrated significant progress in establishing and working toward its climate goals. Noteworthy achievements include launching and growing the Bozeman Energy Project for businesses, energy efficiency upgrades to municipal facilities, updates to the Unified Development Code, a food composting pilot program, installation of a 385 kW solar array at the Water Reclamation Facility through a partnership with NorthWestern Energy, and fare-free transit rides. This plan aims to leverage and grow these emissions mitigation activities, while also infusing new ideas and opportunities to reduce emissions and prepare the Bozeman community to be more resilient in anticipation of future climate change impacts.

This plan is a comprehensive and complex policy document, touching many different facets of daily life in the Bozeman community and greater region. As such, it is essential to focus on issues beyond emissions reduction and resiliency benefits and consider the related equity and human health and well-being facets of each solution and action. Implementation of this plan will undoubtedly create myriad co-benefits as well as impacts – some known and some yet to be discovered. Highlighting these considerations and tradeoffs in this plan is a first step in exploring and understanding these issues, though these concepts will need to remain important lenses and perspectives throughout plan implementation.

The Bozeman community faced many serious challenges and noteworthy events during the development of this Climate Plan, including the COVID-19 pandemic, wildfire, economic disruptions, community protests, and municipal leadership changes. While not all are climate-related, all are linked to community resiliency and Bozeman's ability to respond, recover, and adapt. The intent of this Climate Plan is to usher in a new era of community collaboration and innovation not only to address future climate-related goals, but also to recover and rebound together.

ALIGNING WITH OTHER EFFORTS

The Climate Plan is designed to align, support, and strengthen other City of Bozeman and regional efforts. It does not replace other plans; instead, it helps to create linkages between those plans and climate impacts, while also providing specificity in terms of implementation roles and priorities. Appendix B: Existing Plan Summary provides an overview of various plans and policies that relate closely to this effort.

Perhaps most notably, the 2018 City of Bozeman Strategic Plan establishes the overarching direction and strategic priorities for the City of Bozeman.

The 2020 strategic priorities include:

- Affordable Housing
- Annexation Analysis
- Climate Action
- Community Outreach
- Parks & Trails District
- Planning & Land Use
- Property Tax Relief / Tax Fairness

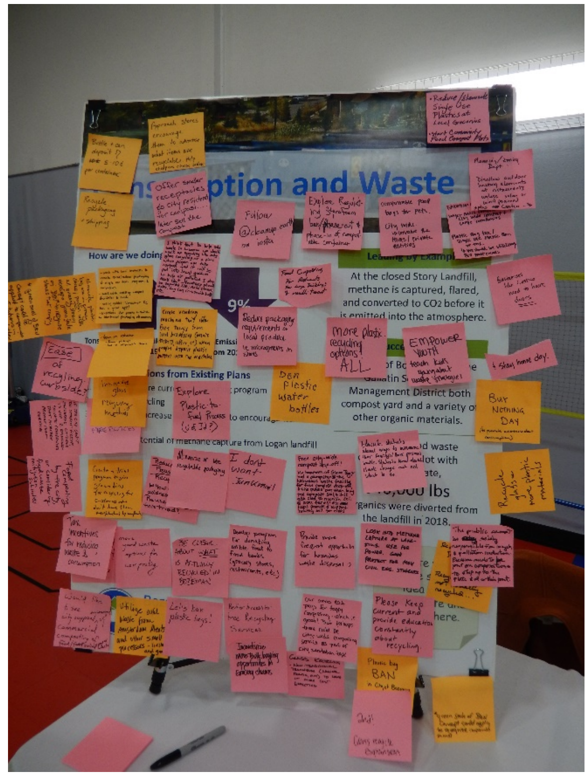
This plan directly addresses the priority of Climate Action, and also supports the strategic priorities of affordable housing, community outreach, and planning and land use.

Beyond the Strategic Plan, other closely linked efforts that helped shape the content of this Climate Plan include:

- Bozeman Community Plan Update (2020)
- Montana Climate Solutions Plan (2020)
- City of Bozeman Climate Vulnerability Assessment and Resilience Strategy (2019)
- NorthWestern Energy Electric Supply Resource Procurement Plan (2019)

Moving forward as other City of Bozeman plans and policies are updated, it is anticipated that this plan can serve as a resource to help infuse climate considerations and actions into those documents.

PLAN DEVELOPMENT PROCESS

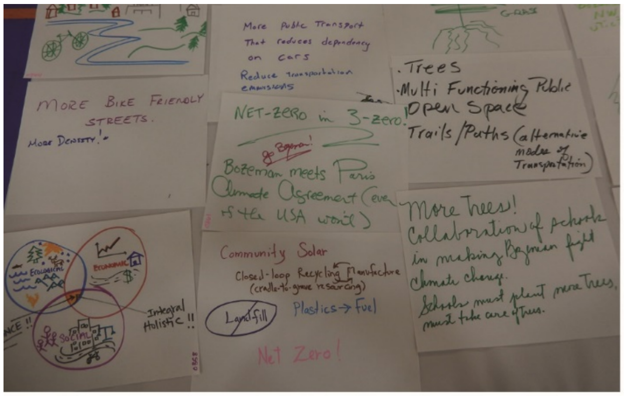


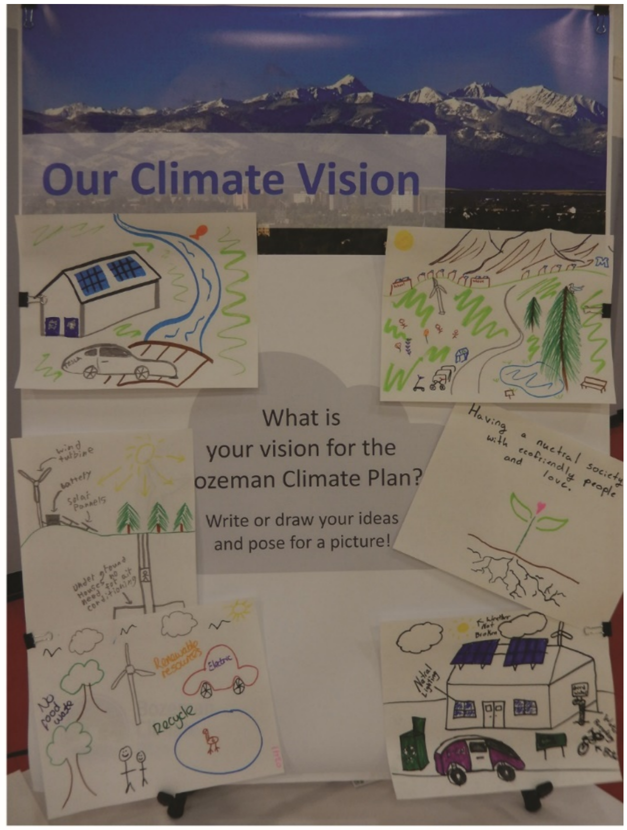
In June 2019, the City of Bozeman initiated the Bozeman Climate Plan process. To help facilitate diverse public involvement, the City Manager invited a core group of approximately 30 Bozeman representatives to serve as Climate Team members to help guide the development of the Climate Plan (see [Resolution 5077)](). Based on the Mayor's encouragement, the Climate Team includes four youth representatives from Montana State University and Bozeman High School and several early-career professionals to ensure that the City would hear from those who will be most impacted by climate change. See the Acknowledgements for a list of all Climate Team members.

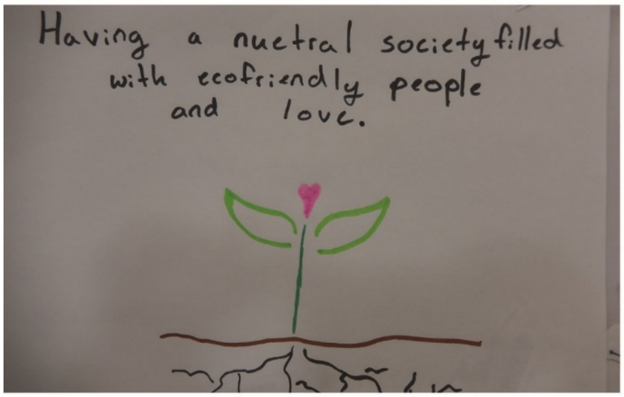
The project management team planned engagement strategies for community members, the Climate Team, City Departments, and the City Commission. A summary of engagement activities to support the plan development is illustrated in the following graphic.

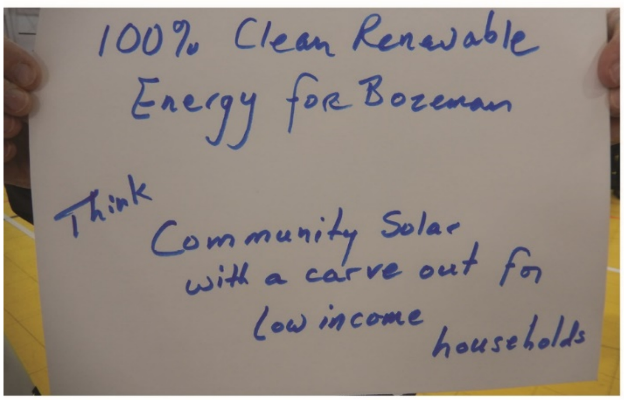
Note that following the COVID-19 pandemic declaration in March of 2020, the project approach and timeline was adapted to accommodate remote public engagement. Surveys and remote workshops were employed in lieu of a condensed in-person workshop.

Figure 1. Climate Plan development process timeline

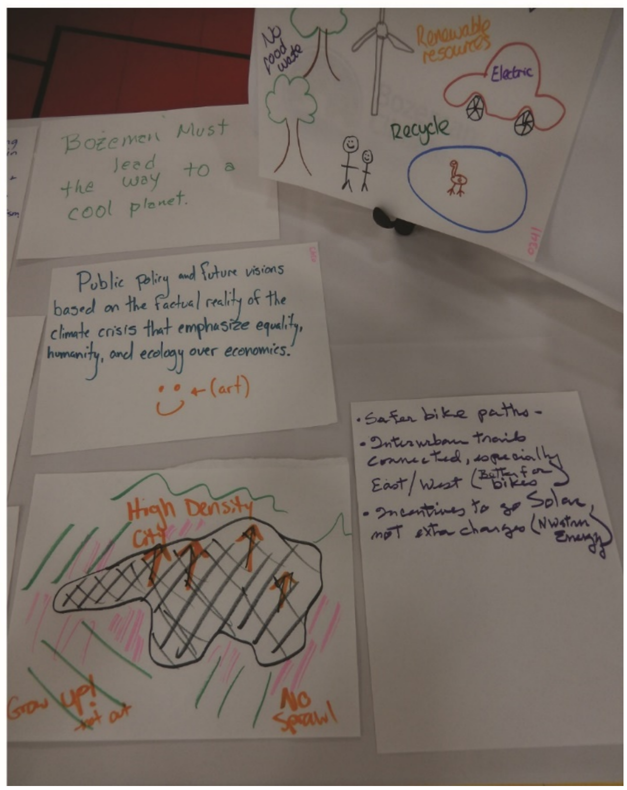

More Public Transport That reduces dependency on cars
MORE BIKE FRIENDLY STREETS.
More Density!
NET-ZERO in 3-zero!
Bozeman meets Paris Climate Agreement (even if the USA won't)
Community Solar
Landfill
Plastics -> Fuel
Net Zero!
Integral Holistic!!
.Trees
·Multi Functioning Public Open Space
Trails/Paths (alternative modes of Transportation)
More Trees!
Collaboration of schools in making Bozeman fight climate change.
Schools must plant more trees, must take care of trees.


Our Climate Vision
What is
your vision for the
ozeman Climate Plan?
Write or draw your ideas
and pose for a picture!
Having a nuetral society filled with ecofriendly people and love.
Renewable resources
Electric
Recycle
No food waste


Having a nuetral society filled
with ecofriendly people
and love.


100% Clean Renewable
Energy for Bozeman
Think
Community Solar
with a carve out for
low income households


M


No food waste
Renewable resources
Electric
Recycle
Bozeman Must lead the way to a cool planet.
Public Policy and future visions based on the factual reality of the climate crisis that emphasize equality, humanity, and ecology over economics.
(art)
High Density City
Grow up! not out
No Sprawl!
·Safer bike paths -
·Interurban trails connected, especially East/West (Better for bikes)
·Incentives to go Solar, not extra charges (NWestern Energy)

BOZEMAN'S CLIMATE VISION

A vision is the guiding force that orients the overarching direction for a plan. The Bozeman Climate Plan vision is organized into two components: a position statement and guiding principles. The position statement establishes the foundation upon which this plan was built, while the guiding principles form the pillars that hold up each component of this plan. Together, the position statement and guiding principles answer the question: how will the City of Bozeman address the imminent and far reaching impacts of climate change in a manner that lifts and strengthens the entire community?

Position Statement

Through leadership and collaboration, the City of Bozeman will advance innovative solutions to cultivate a more equitable and resilient low-carbon community for current and future generations.

Guiding Principles

Bozeman will be a leader in addressing climate change by:

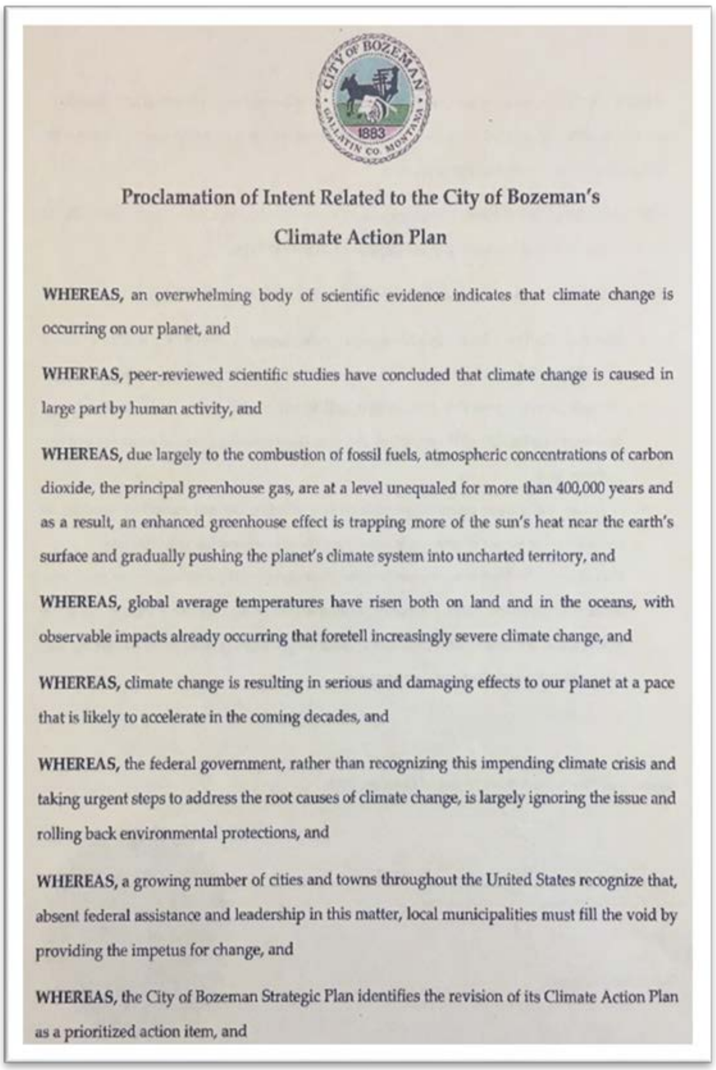
- Adopting bold targets for emissions reduction and renewable energy.
- Weaving sustainability and resilience into decision-making processes.
- Pursuing partnerships with other municipalities and our utility provider.
- Seeking innovative, actionable solutions to mitigate climate change.
- Inviting all Bozeman residents to join us, including current and future leaders.

Figure 2: Vision words for the Climate Plan generated by the Climate Team

BOZEMAN'S CLIMATE GOALS

This Climate Plan establishes an assortment of goals for community greenhouse gas emissions reductions and climate resiliency. These climate goals are based on past City of Bozeman commitments, state-wide pledges, and the necessity established by current climate science, as summarized below.

- The 2011 Community Climate Action Plan established a goal of "reducing community-wide greenhouse gas emissions 10% below 2008 levels by 2025."
- The 2011 Community Climate Action Plan established a goal of reducing per capita emissions to 10 metric tons of greenhouse gas emissions, not to exceed 695,000 metric tons, by 2020.
- In July 2019, Montana Governor Steve Bullock issued an Executive Order that establishes an interim goal of "net greenhouse gas neutrality for average annual electric loads in the state by no later than 2035."
- The United Nations Environment Program (UNEP) 2019 Emissions Gap Report released on November 26, 2019 indicates that we must reduce global greenhouse gas emissions 7.6% each year between 2020 and 2030, and achieve carbon neutrality by 2050, to avoid more than 1.5°C (2.7°F of warming).



Proclamation of Intent Related to the City of Bozeman's Climate Action Plan

WHEREAS, an overwhelming body of scientific evidence indicates that climate change is occurring on our planet, and

WHEREAS, peer-reviewed scientific studies have concluded that climate change is caused in large part by human activity, and

WHEREAS, due largely to the combustion of fossil fuels, atmospheric concentrations of carbon dioxide, the principal greenhouse gas, are at a level unequaled for more than 400,000 years and as a result, an enhanced greenhouse effect is trapping more of the sun's heat near the earth's surface and gradually pushing the planet's climate system into uncharted territory, and

WHEREAS, global average temperatures have risen both on land and in the oceans, with observable impacts already occurring that foretell increasingly severe climate change, and

WHEREAS, climate change is resulting in serious and damaging effects to our planet at a pace that is likely to accelerate in the coming decades, and

WHEREAS, the federal government, rather than recognizing this impending climate crisis and taking urgent steps to address the root causes of climate change, is largely ignoring the issue and rolling back environmental protections, and

WHEREAS, a growing number of cities and towns throughout the United States recognize that, absent federal assistance and leadership in this matter, local municipalities must fill the void by providing the impetus for change, and

WHEREAS, the City of Bozeman Strategic Plan identifies the revision of its Climate Action Plan as a prioritized action item, and

WHEREAS, the City of Bozeman has convened a diverse consortium of talented and passionate citizens to develop a draft Climate Action Plan to serve as a comprehensive strategy for addressing climate change in Bozeman, and

WHEREAS, the plan will identify strategies and actions to reduce our community's greenhouse gas emissions and build resiliency to the impacts of climate change.

NOW, THEREFORE, AS THE MAYOR OF BOZEMAN, I PROCLAIM THAT;

- Bozeman shall be a leader in addressing climate change by developing a climate action plan that identifies bold targets for carbon reduction and climate change mitigation, and
- We will create a policy document that will weave sustainability and climate change resilience into the City of Bozeman's decision-making processes, capital planning and city budgets, and
- Bozeman will pursue partnerships with other municipalities and our utility provider to reach a goal of 100% net renewable energy for the City of Bozeman by 2030 and
- We call upon the Montana governor's office, state agencies, the state legislature, the Public Service Commission, Gallatin County, Northwestern Energy and all Bozeman residents to join us as we seek bold, actionable climate solutions to reverse global warming and leave a better world for future generations.

Signed and Proclaimed this 22nd day of October, 2019.

Cynthia L. Andrus
Mayor
Bozeman, Montana

Figure 3: Proclamation of Intent related to the City of Bozeman's Climate Plan from October 2019

Bozeman's goals include a mix of quantitative emissions mitigation goals and qualitative community resiliency goals. Together, these goals work to position the city to achieve its climate vision.

Emissions Mitigation

Mitigation focuses on interventions to reduce or prevent emissions of greenhouse gases and lessen the human impact on the climate system.

The City of Bozeman's mitigation goals recognize that a rapid transition to non-carbon emitting clean energy is necessary to eliminate over one-third of community greenhouse gas emissions. Furthermore, clean electricity has the potential to replace gasoline and diesel, which represent an additional one-third of overall greenhouse gas emissions. The goals include a mix of near- and longer-term goals for the years 2025, 2030, and 2050.

Resiliency

Resiliency focuses on increasing the ability of the community to prepare, plan for, absorb, respond to, recover from, and adapt to the effects of climate-related shocks and chronic stressors.

This plan emphasizes the need for the City of Bozeman to incorporate climate change resiliency considerations into all City decision making. The resiliency goals center around being a community that prepares for, responds quickly to, and recovers from climate-related events and stressors through well-planned infrastructure, supportive social networks, and balanced economic prosperity.

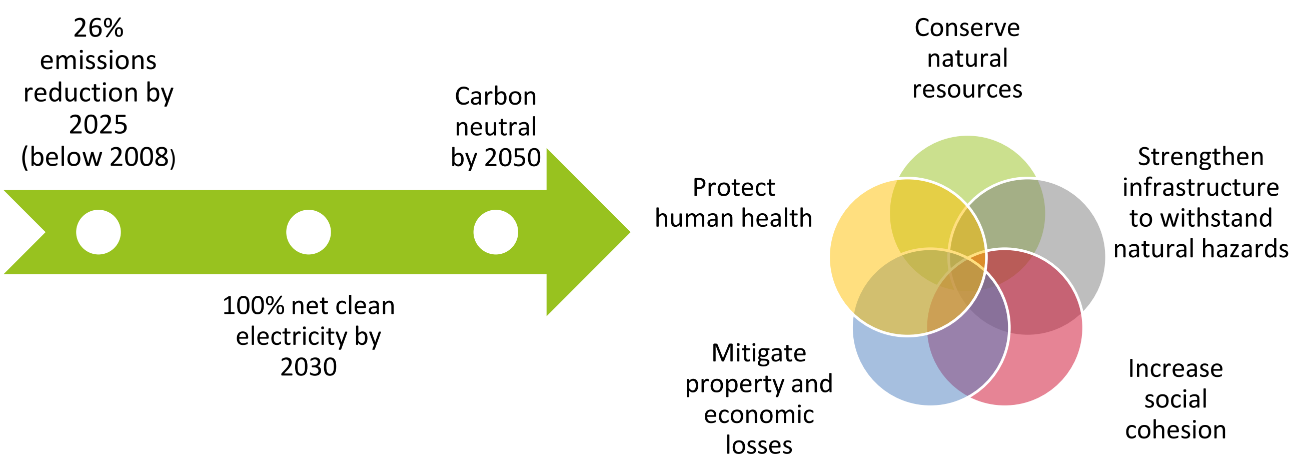


Figure 4. Goals of mitigation and resiliency

PLAN ORGANIZATION

The Climate Plan is organized to support Bozeman's climate vision and goals. The plan emphasizes **six focus areas** that help categorize the solutions and actions of the plan. Each focus area presents one or more **mitigation and/or resiliency solutions** that can support achievement of the vision and goals.

The final chapter describes how the plan can be **implemented** and updated over time to reflect new opportunities and changing community conditions. A series of appendices provide more detailed documentation and technical information that supported plan development.

Focus Areas

The focus areas serve to categorize the actions necessary for Bozeman to mitigate, and adapt to, projected climate change impacts. The focus area sections are ordered by their potential to achieve the mitigation goals, with an underlying concept of "efficiency first." By focusing on efficiency first, the Bozeman Climate Plan seeks to accelerate mitigation and resiliency efforts that will maximize the community's return on investment. The six focus areas of the Bozeman Climate Plan are:

Focus Area 1:
Healthy, Adaptive & EFFICIENT BUILDINGS

Focus Area 2:
Responsible & Reliable CLEAN ENERGY SUPPLY

Focus Area 3:
Vibrant & Resilient NEIGHBORHOODS

Focus Area 4:
Diverse & Accessible TRANSPORTATION OPTIONS

Focus Area 5:
Comprehensive & Sustainable WASTE REDUCTION

Focus Area 6:
Regenerative Greenspace, Food Systems & NATURAL ENVIRONMENT

Each focus area section includes an introduction to the theme and how it relates to Bozeman's greenhouse gas emission trends and recent efforts. Additional components detailed within each focus area section include:

- Where are we now
- Related and supporting plans, policies, and programs
- Key activities in this focus area that are necessary to achieve Bozeman's emissions reduction goals
- Reduction targets for each emissions reduction goal
- Resiliency opportunities and considerations for each climate hazard

Cross-Cutting Topics

In addition to the six focus areas, three cross-cutting topics are infused throughout the entire plan: equity, health and well-being, and City of Bozeman assets. Each component of this plan was developed within the context of these three topics, highlighting their importance and relevance to each focus area. A description of how each of these are addressed throughout the plan is provided in the Climate Solutions section that follows. Similarly, each Climate Solution includes discussion of how it supports the City of Bozeman's Strategic Plan priorities.


Equity


Human Health
& Well-Being


City Assets

Equity

Providing a more equitable future for the entire Bozeman community is a core value captured in the vision of this plan. Under an equitable climate plan, all community members will reap the benefits of mitigation and adaptation efforts. Ensuring an equitable path toward a resilient and low-carbon future will mean considering how each solution proposed in this plan might positively or negatively impact diverse sectors of the Bozeman community. For example, though pursuing distributed renewable energy (see Solution F. Increase Community-Based Distributed Renewable Energy Generation) is central to achieving mitigation goals, not every resident may be able to afford the upfront costs, and therefore cannot reap the long-term savings. Through this careful consideration, all actors involved in implementing this plan can work to maximize equitable benefits, while avoiding or mitigating negative equity impacts wherever possible.

Human Health and Well-being

This cross-cutting topic seeks to prioritize the physical, mental, and emotional well-being of Bozeman's community members in the context of climate action. Mitigation and resiliency activities often yield significant health benefits, though in some cases, there may be unintended or negative consequences associated with solutions. For example, if a more diverse transportation future includes significant on-demand transportation (e.g., Uber and Lyft), air pollution may increase where idling occurs. As with equity considerations, exploring all of the potential impacts on human health and well-being can help ensure benefits are maximized, while unintended consequences are limited or avoided.

City Assets

In 2019, the City of Bozeman completed the Vulnerability Assessment and Resiliency Strategy specific to City assets. This plan identified the City's most vulnerable assets and set forth strategies for improving resiliency in the face of five natural hazards. The recommendations from the 2019 Vulnerability Assessment and Resiliency Plan find synergies in this Climate Plan. Though every member of the Bozeman community is responsible for implementing this plan, the City will seek to lead by example, making direct and influential impact by leveraging City Assets.

Climate Solutions

Climate solutions are the big ideas that help further organize and describe how the community will make progress toward Bozeman's climate vision. Each focus area has one to three proposed solutions, identified through the first two workshops with the Climate Team. As with the focus areas, solutions are organized around the concept of improving and maximizing efficiency first, to establish a solid foundation upon which to build other approaches and innovations. For example, community efforts should focus first on improving the efficiency of existing buildings, before layering in opportunities to switch fuel sources.

Focus Area 1:
Healthy, Adaptive & EFFICIENT BUILDINGS

- Solution A. Improve Efficiency of Existing Buildings
- Solution B. Achieve Net Zero Energy New Construction
- Solution C. Electrify Buildings

Focus Area 2:
Responsible & Reliable CLEAN ENERGY SUPPLY

- Solution D. Increase Utility Clean Energy Mix
- Solution E. Develop and Promote Utility Green Power Programs
- Solution F. Increase Community-Based Distributed Renewable Energy Generation

Focus Area 3:
Vibrant & Resilient NEIGHBORHOODS

- Solution G. Facilitate Compact Development Patterns
- Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards
- Solution I. Enhance Social Infrastructure and Community Preparedness

Focus Area 4:
Diverse & Accessible TRANSPORTATION OPTIONS

- Solution J. Increase Walking, Bicycling, Carpooling, and Use of Transit
- Solution K. Decrease Direct Vehicle Emissions
- Solution L. Improve Air Travel Efficiency

Focus Area 5:
Comprehensive & Sustainable WASTE REDUCTION

- Solution M. Move Toward a Circular Economy and Zero Waste Community

Focus Area 6:
Regenerative Greenspace, Food Systems & NATURAL ENVIRONMENT

- Solution N. Cultivate a Robust Local Food System
- Solution O. Manage and Conserve Water Resources
- Solution P. Manage Land and Resources to Sequester Carbon

Solution Template Overview

Each solution begins with an introductory narrative that provides an overview of the solution's scope, the regulatory context within which the solution operates, and key areas of impact for the solution. These introductory narratives also highlight specific resiliency considerations and call out related efforts led by the City or key community partners. In addition to a general overview of scope, each solution introduction identifies the equity, health and well-being considerations, and City asset opportunities relevant for that solution. Each cross-cutting topic description is marked with an icon.

Equity considerations explore how a solution might provide equity benefits, or how it might negatively impact equity. Wherever possible, solutions should be implemented to maximize equity benefits, while avoiding unintended consequences.

Health and well-being considerations identify how a solution could positively or negatively impact the physical, mental, and emotional health and well-being of Bozeman community members.

City asset considerations identify opportunities for the City to lead by example through improvements to public facilities and infrastructure and municipal operations.

Climate Goal Contributions

Each solution is categorized as primarily a mitigation solution or primarily a resiliency solution. For solutions that provide climate mitigation, the benefits are summarized in terms of which emissions reduction goals are supported and which fuel sources are impacted. For solutions that provide resiliency benefits, the summary tables identify which resiliency goal(s) the solution helps to advance.

Emissions Reduction Benefits

Describes how the solution supports achievement of one or more of the emissions mitigation goals.

Also identifies the emissions sources impacted:

- Electricity
- Natural Gas
- Gasoline & Diesel
- Waste
- Carbon Sequestration

Resiliency Benefits

- Conserve natural resources
- Strengthen infrastructure to natural disaster
- Increase social cohesion
- Mitigate property and economic losses
- Protect human health

Related Solutions

None of the solutions in this plan stand alone. In some cases, one solution may find synergies with another or many other solutions. In other cases, a solution may be wholly dependent on the successful implementation of one or more solutions. The lists of related solutions help to identify these interdependencies, and act as a reminder that Bozeman cannot and will not achieve its vision without collaboration and partnership.

Actions

Each solution in this plan includes a suite of supporting actions that provide strategy and focus to help advance the solutions to ultimately help Bozeman achieve its climate vision and goals. Actions encompass a range of activities including advocacy, projects, programs, outreach, partnerships, policy, financial tools, studies, and more. Each action begins with a background narrative to explain the steps and activities necessary to advance the idea. A summary table presents other important details associated with each action:

Priority	1 – The action is a top priority to show municipal leadership or proof of concept; to capture the momentum of existing efforts, is necessary to achieve 2025 goals; some resources for implementation may be available; and/or may be a necessary building block for subsequent actions. 2 – The action is important to advance in the near-term, but may be dependent on other actions to begin first; may not be as time sensitive and is more closely related to long-range goals; and/or the resources for implementation need to be identified or secured. 3 – The action is a longer-term priority; it may need to begin after other actions; and/or additional resources are needed for implementation.

Lead & Implementing Partners	Achieving the goals in this plan will require a community-wide effort. The City of Bozeman can play a leadership role by investing taxpayer resources in ways that advance the goals, but will need the participation of residents, businesses and community organizations, utilities, and more to successfully implement the solutions identified. Each action identifies an implementation lead, as well as supporting partners.

Strategic Plan Alignment	The City of Bozeman Strategic Plan was adopted via Resolution 4852 to focus on specific outcomes and guide policy and funding decisions over the next 2 to 5 years. This section references actions that align with priorities from the Strategic Plan.

Performance Measures	Periodic greenhouse gas emissions inventories will continue to serve as the over-arching metric of progress towards Bozeman's goals. In addition to monitoring total greenhouse gas emissions, one or more metrics are identified to quantitatively measure and monitor the implementation status, impact, or performance of each action.

Note that all of the solutions in this plan require a coordinated and collaborative community effort to ensure successful implementation. The major groups that will play leading and supporting roles in implementing the solutions and actions within the plan are noted below and referenced throughout this document.

Government

- City of Bozeman
- Gallatin County
- State of Montana
- Adjacent Jurisdictions

Utilities

- NorthWestern Energy
- Solid Waste
- Telecommunications
- Water & Sewer

Businesses & Community Organizations

- Businesses
- Developers
- Investors
- Montana State University
- Non-profits
- Schools

Residents

- Homeowners
- Renters
- Students
- Taxpayers
- Property owners
- Homeowners associations

LEADING BY EXAMPLE

The City of Bozeman has demonstrated a commitment to reducing emissions and enhancing resiliency through various municipal projects, programs, and initiatives. Below are just a few recent and noteworthy achievements.

Focus Area 1: **Healthy, Adaptive & EFFICIENT BUILDINGS**

- From 2012-2019, municipal energy efficiency upgrades saved 3,583 MWh and $371,440.
- Bozeman Energy Project business partner savings from 2015-2019 equaled 522 MWh and $54,882.

Focus Area 2: **Responsible & Reliable CLEAN ENERGY SUPPLY**

- From 2009-2019, Solar PV on city property generated 1,390 MWh of electricity.
- The City of Bozeman partnered with NorthWestern Energy to install a 385 kW solar array at the Water Reclamation Facility, which produces enough electricity to offset the annual consumption of 60 homes.

Focus Area 3: **Vibrant & Resilient NEIGHBORHOODS**

- In April 2018, the Unified Development Code was adopted to address the growing population; it includes changes to standards such as allowing higher density requirements and smaller lot sizes.
- The Landscaping chapter in the Bozeman Municipal Code requires drought tolerant landscaping.

Focus Area 4: **Diverse & Accessible TRANSPORTATION OPTIONS**

- Streamline Transit provided 283,714 fare-free rides in 2018.
- Bozeman was designated a 2016 Silver Bike Friendly Community by the League of American Bicyclists.

Focus Area 5: **Comprehensive & Sustainable WASTE REDUCTION**

- Both the City of Bozeman and the Gallatin Solid Waste Management District compost yard waste and a variety of other organic materials.
- At the closed Story Landfill, methane is captured, flared, and converted to CO_2 before it is emitted into the atmosphere.
- Through a food waste composting pilot with Montana State, 540,000 pounds of organics were diverted from the landfill in 2018.

Focus Area 6: **Regenerative Greenspace, Food Systems & NATURAL ENVIRONMENT**

- The Story Mill Community Park is helping to build healthy streams and wetlands. The project received the 2017 Montana Wetland Stewardship Award and doubled the size of Bozeman's largest wetland to 14 acres.
- The 7th Ave. streetscape project integrated green infrastructure components, including permeable pavers and underground soil vaults to collect and absorb stormwater while supporting the maintenance of new street trees.

CHAPTER 2:

CLIMATE TRENDS & GOAL CONTRIBUTIONS

CHAPTER 2: CLIMATE TRENDS & GOAL CONTRIBUTIONS

According to the 2017 Montana Climate Assessment, climate change in Montana is predicted to lead to temperature variability, shifts in precipitation, varying risk of certain severe weather events, and changes to other features of the climate system (Whitlock et al., 2017).

Table 1 and Table 2 summarize the Montana Climate Assessment's future projections of impacts to Bozeman under two future emissions concentration scenarios (i.e., Representative Concentration Pathways (RCP)), RCP 4.5 and RCP 8.5, respectively. Under the stabilization scenario, RCP 4.5, greenhouse gas emissions peak around 2040 and then begin to decline through the end of this century as atmospheric carbon dioxide (CO_2) concentration begins to level off. Under the business-as-usual scenario, RCP 8.5, greenhouse gas emissions increase through 2100 and warming continues to rise.

The Montana Climate Assessment examined 20 global climate models and calculated the strength of consensus for each climate variable's direction of change (either increasing or decreasing). This is reported as the percentage of model agreement. Additional information on the climate modeling approach can be found in the Montana Climate Assessment.

Table 1. Under the stabilization emissions scenario (RCP 4.5), the difference, or change, projected from historical conditions (1971-2000) to mid-century (2040-2069) and end-of-century (2070-2099) thirty-year averages for Southwest Montana.

	RCP 4.5 (2040-2069)	**RCP 4.5 (2070-2099)**	**Model Agreement**
Average annual temperature	+4.5˚F	+5.6˚F	100%
Average daily summer maximum temperature	+4.5˚F	+6.5 ˚F	100%
Average number of days above 90˚ F	+25 days	+29 days	100%
Average number of freeze free days above 32˚ F	+30 days	+41 days	100%
Average annual precipitation[2]	+0.9 inch/year	+1.1 inch/year	85%[3]/90%[4]
Change in summer precipitation	-0.1 inch/month	-0.1 inch/month	65%

[2] Interannual variability projected to range from -0.5 inch/year to +1.5 inch/year (≤80% model agreement).
[3] 85% model agreement for RCP 4.5 (2040-2069).
[4] 90% model agreement for RCP 4.5 (2070-2099).

Table 2. Under the business-as-usual emissions scenario (RCP 8.5), the difference, or change, projected from historical conditions (1971-2000) to mid-century (2040-2069) and end-of-century (2070-2099) thirty-year averages for Southwest Montana.

	RCP 8.5 (2040-2069)	RCP 8.5 (2070-2099)	Model Agreement
Average annual temperature	+6.0˚F	+9.8˚F	100%
Average daily summer maximum temperature	+6.0˚F	+11.8˚F	100%
Average number of days above 90˚ F	+33 days	+54 days	100%
Average number of freeze free days above 32˚ F	+41 days	+70 days	100%
Average annual precipitation[5]	+1.2 inch/year	+1.7 inch/year	85%[6]/100%[7]
Change in summer precipitation	-0.1 inch/month	-0.1 inch/month	65%

As characterized in the modeled scenarios, average and annual seasonal temperatures in Montana have been increasing since the mid-20th century and are predicted to continue to increase through the end of the century. Likewise, in the modeled scenarios, the timing of precipitation (i.e., winter versus spring and summer) and the form in which it will occur (i.e., rain versus snow) is anticipated to shift. This combination of increasingly warmer days with variable precipitation results in interrelated, indirect local climate impacts.

For example, decreased snowpack may lead to more severe droughts in the summer and a susceptibility to wildfire risk in the watershed. This type of direct impact will have a broad range of additional, indirect effects on the local and regional economic and social systems. The heightened susceptibility to wildfire could reduce the amount and quality of water available, along with damaging ecosystems and infrastructure, potentially limiting city-wide services available to address the impacts. Therefore, the local outdoor and tourist economy could be compromised in addition to public health considerations.

Another indirect consequence of climate change is human migration caused by sea level rise and other extreme weather shocks and stressors. Sea-level rise in coastal communities in the United States is predicted to increase net-migration to Gallatin County up to an estimated 50,000 people by the end of the century, exacerbating Bozeman's existing challenge of rapid population growth (Hauer, 2017).

[5] Interannual variability projected to range from -0.4 inch/year to +1.9 inch/year (≥80% model agreement).
[6] 85% model agreement for RCP 8.5 (2040-2069).
[7] 100% model agreement for RCP 8.5 (2070-2099).

Bozeman Climate Plan

Local Climate Change Impacts

Over the past decade, the City of Bozeman has changed rapidly due to a growing population and bustling economy, leading to increased urbanization, suburbanization, and stress on the area's natural resources. Climate change is a major factor that has the potential to continue to significantly shape the community's future. Evidence of climate change is well documented throughout the United States (Climate Change Impacts in the United States: The Third National Climate Assessment, 2014) and Rocky Mountain West region. The potential climate impacts for Montana (Whitlock, Cross, Maxwell, Silverman & Wade, 2017) and Bozeman are significant.

The 2019 City of Bozeman Vulnerability Assessment and Resiliency Strategy identifies the following consequences of climate change. This plan explores solutions to mitigate the community's impacts on climate change, as well as solutions that enable the community to adapt and be resilient to these impacts.

Extreme Heat	More frequent & intense	**Mountain Snowpack**	Decline in volume
Floods	More severe	**Wildfire**	More extensive, frequent, & intense
Drought	More frequent & intense	**Winter Storms**	More severe

Snowblowing Efforts December 12, 2005

Bozeman Creek Flooding May 24, 2011

Bozeman Creek December 11, 2009

©Rachel Leathe/Chronicle

Still-standing house, surrounded by charred fields September 6, 2020

BASELINE INVENTORY & FUTURE PROJECTIONS

The City of Bozeman has been tracking community greenhouse gas (GHG) emissions since 2008 to better understand the sources of Bozeman's GHG emissions and monitor progress over time. The information is used to help inform City decision-making. This inventory adheres to the Global Protocol for Community-Scale Emissions and the U.S. Community Protocol for Accounting and Reporting of Greenhouse Gas (GHG) emissions. The City of Bozeman Sustainability Office collected data related to energy, transportation, solid waste, water, wastewater, and emissions factors. We calculated our carbon emissions using the ICLEI ClearPath platform.

For Bozeman, GHG emissions result from four primary sectors: energy use in homes, energy use in commercial buildings, fuel use in transportation, and solid waste (see Figure 5). In 2018, Bozeman's overall community emissions totaled 607,139 metric tons of carbon dioxide equivalent (MT CO_2e). Since the baseline year of 2008, total community emissions have increased 16% while population has increased 37% over the same period. Based on 2018 census data, Bozeman had 48,532 people, resulting in per capita emissions of 12.5 MT CO_2e per person. Since 2008, per capita emissions have decreased by 15%. Over half of these emissions (57%) are linked to the energy generated to power our homes, businesses, and institutions. The next largest source of emissions comes from transportation fuel sources (38%)—most transportation emissions come from gasoline (63%). The remaining emissions result from solid waste (5%) and wastewater treatment (<1%).

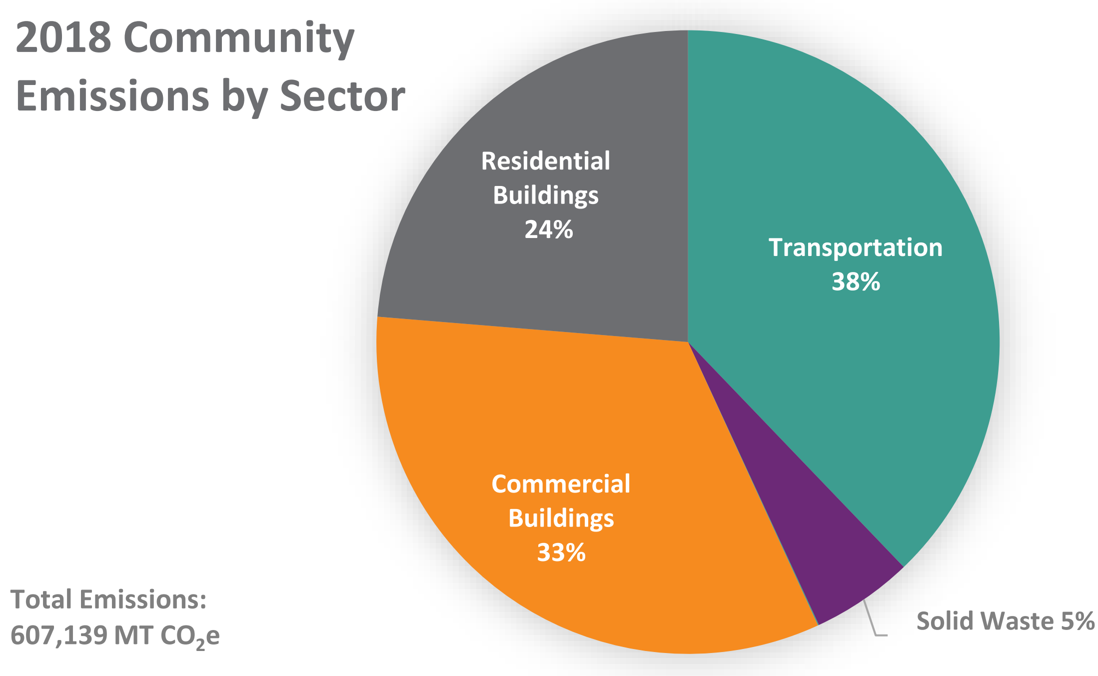


Figure 5. 2018 Bozeman community emissions by sector

The three primary greenhouse gas emissions sources at the local level are electricity, natural gas, and fuel for our vehicles (Figure 6). Electricity generation remains the largest source of greenhouse gas emissions at 29%, which is driven by the carbon intensity of coal and natural gas burned to produce the electricity that serves Bozeman. Natural gas is the second leading emissions source at 27%, which is primarily used to heat buildings. Gasoline for transportation is the third leading source of emissions totaling 24%.

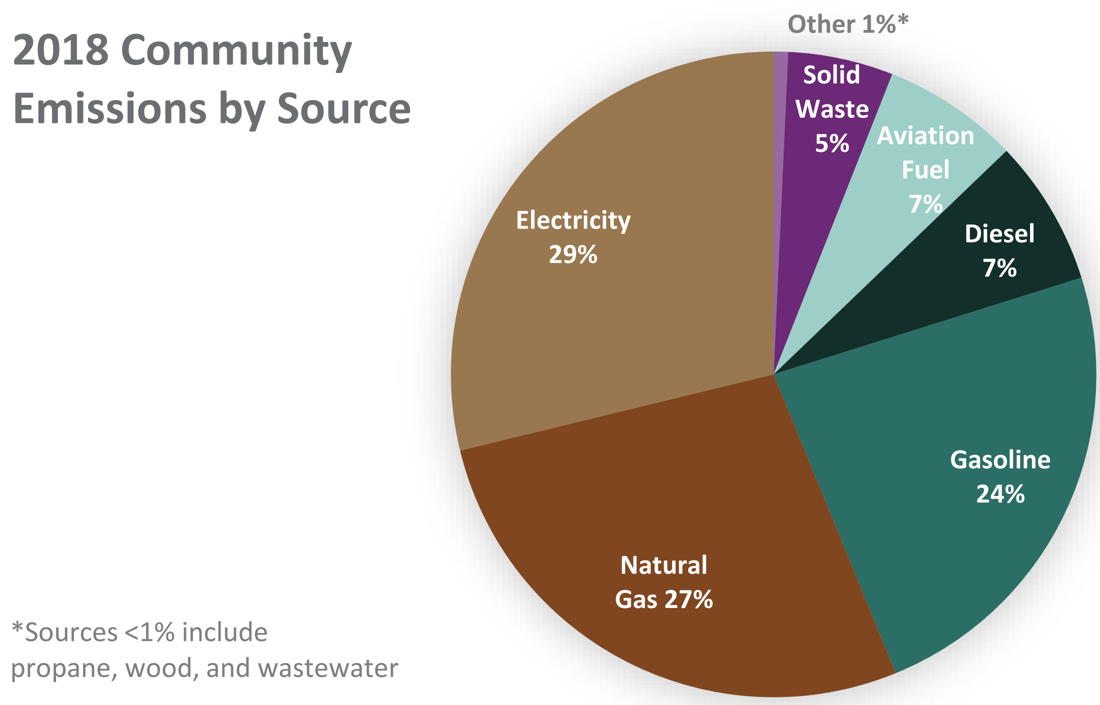


Figure 6. 2018 community emissions by source

Energy Emissions

Energy supports our modern way of life. We use it for lighting, heating, cooking, powering our devices, manufacturing, and beyond. This energy use encompasses the largest proportion (56%) of our GHG emissions. A slight majority of building energy emissions (29%) come from electricity consumption. Natural gas, which is largely used for building heating, comprises 27% of emissions (Figure 6). Commercial, institutional, and municipal buildings account for 33% of total emissions and residential buildings account for 24% of total emissions (Figure 5).

The two primary factors that influence emissions trends over time are the amount of energy used in different sectors and the composition or carbon intensity (carbon emissions per unit of energy) of the energy source. As an example, the carbon intensity of Bozeman's grid electricity has decreased by 23% since 2008, due to the addition of hydroelectricity and wind energy resources. As of 2018, approximately 60% of the power grid that delivers electricity to Bozeman was derived from low-carbon sources (46% hydro, 15% wind, and <1% solar) and 40% from non-renewable sources (20% coal and 19% natural gas). Bozeman consumed nearly 340,000 megawatt hours of electricity in 2018.

Figure 7 shows the how the emissions by sector relates to each of the emissions sources. Note that some of the smaller emissions sources have been combined in previous graphics. For example, line loss emissions have been combined with electricity and fugitive natural gas emissions have been combined with natural gas use emissions.

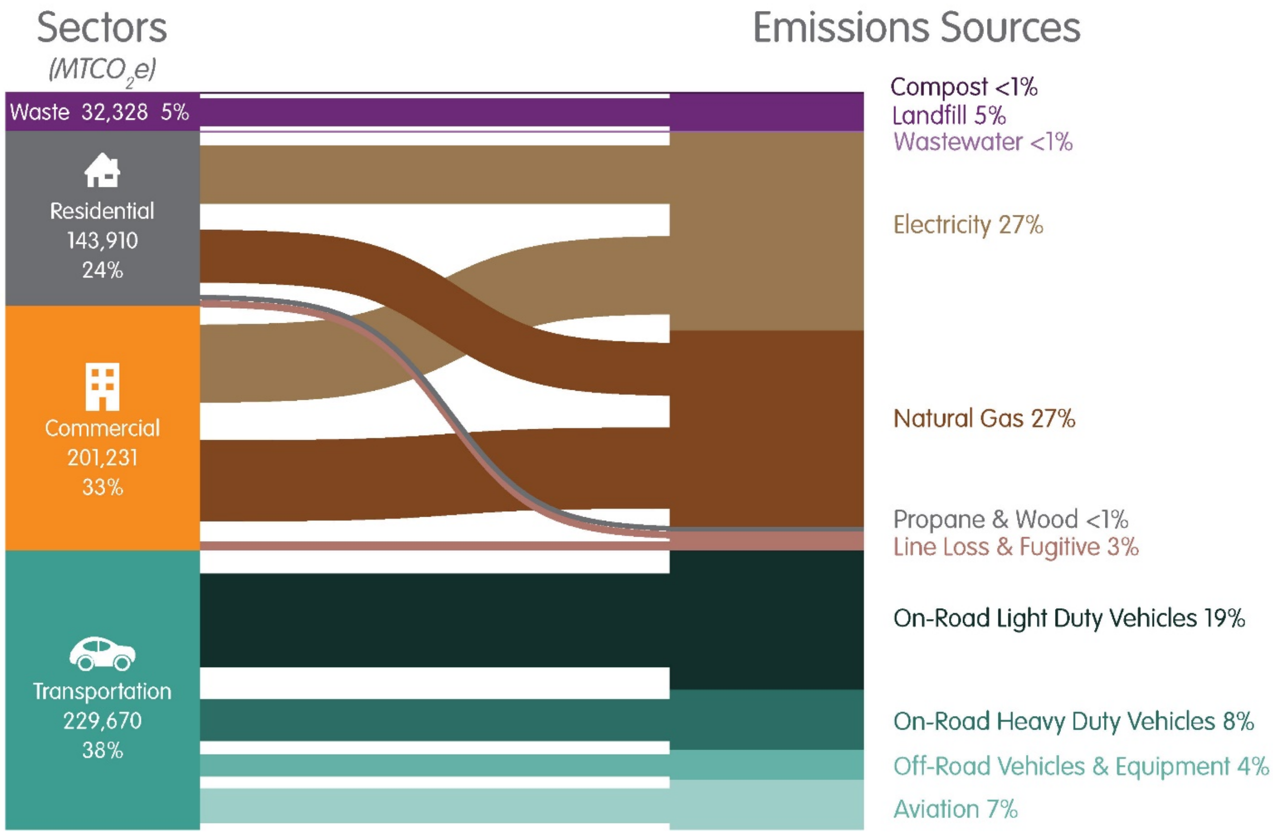


Figure 7. 2018 energy flow diagram of community emissions by sector and source

Transportation Emissions

Bozeman residents use a variety of transportation modes, but the city remains a car-centric community. According to the 2017 Bozeman Transportation Master Plan, nearly 70% of Bozeman commuters drive a personal vehicle to work. The mean commute time is 14.5 minutes, while the national average is 26 minutes.

Streamline Transit, operated by the Human Resources Development Council, is Bozeman's major public transportation service that offers fare-free paratransit, local, and commuter bus service. In fiscal year 2018, Streamline reported 283,714 fare-free rides.

In 2018, 38% of total emissions came from the transportation sector. On-road emissions from passenger cars and light duty trucks contributed 19% of total emissions. On-road emissions came from heavy duty commercial hauling, and freight vehicles accounted for 8% of total emissions. In 2018, Bozeman residents, tourists, commuters, and commercial vehicles drove over 315 million miles in city limits.

Off-road vehicles and equipment, such as lawn mowers, construction equipment, recreational vehicles, and boats, contributed approximately 4% of total emissions. Aviation emissions, estimated based on fuel use for locally originating flights from Bozeman-Yellowstone International airport, contributed approximately 7% of total emissions.

Waste Emissions

Solid waste represents about 5% of our total inventory. In fiscal year 2019, the City of Bozeman Solid Waste Division serviced approximately 11,800 residential customers with curbside garbage service and 34% of these customers use curbside recycling service. During the same period, residents helped divert over 3,000 tons of waste through the City's composting and recycling programs. Composting organic materials, like yard waste and kitchen scraps, reduces greenhouse gas emissions compared to landfilling. Composting emissions totaled less than 1% in 2018. Private haulers also serve commercial and residential customers in Bozeman, and the City does not operate an active landfill for household waste. Bozeman's waste is hauled to the Logan Landfill, which is operated by the Gallatin Solid Waste Management District; City of Bozeman waste emissions are estimated based on total municipal solid waste received from Gallatin County and Bozeman's population.

Wastewater Emissions

A small amount of methane and nitrous oxide emissions are associated with wastewater treatment processes. This generated less than 1% of total emissions in 2018.

Comparison Between Inventories

The 2018 emissions sector and source results are relatively consistent with previous inventory years (Figure 8). Total community emissions have increased by 16% (Table 3) since the baseline year of 2008 and the most recent inventory year of 2018. Notable increases in emissions, as compared to the baseline, are transportation and commercial emissions. Notable decreases are residential emissions. From 2016 to 2018, there were evident increases in the commercial, residential, and solid waste sectors.

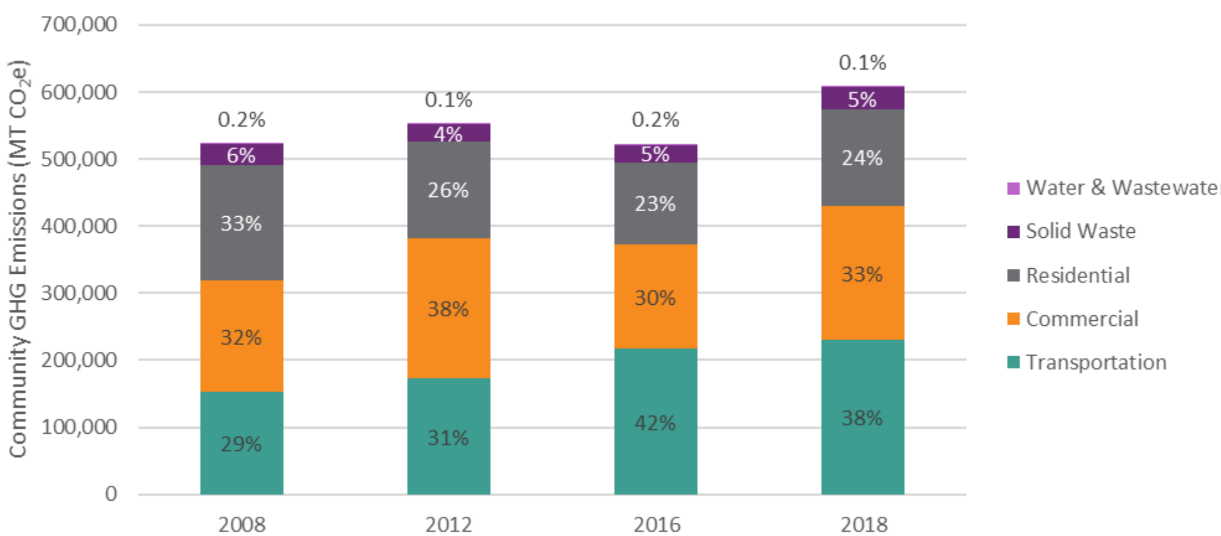


Figure 8. Comparison of annual emissions by sector

While it is critical for Bozeman to achieve absolute emissions reductions to meet the challenge of climate change, normalizing for population growth can make greenhouse gas emissions figures more relatable and help uncover efficiency trends. Table 4 provides a summary of the sectors on a per capita basis. Overall, 2018 per capita emissions have declined by 15% since 2008 with decreases in all sectors except transportation. Since 2016, per capita emissions increased by 8%, though there was a slight yet encouraging decrease in per capita transportation emissions.

Per capita emission trends help Bozeman residents better conceptualize how their local activity fits within the broader context of greenhouse gas emissions. Overall, residents of Bozeman contributed 12.5 MT CO_2e per person in 2018. To meet our 2025 goals and the intent of the Paris Climate Agreement, Bozeman residents will need to reduce emissions to between 7.2 to 4.3 MT CO_2e per person. The range reflects potential low and high growth scenarios.

Table 3. Total emissions (metric tons CO_2 equivalent) by sector and inventory year

	2008	2012	2016	2018	Change since 2008	Change since 2016
Transportation	153,211	172,391	216,608	229,670	50%	6%
Commercial	166,005	210,082	156,894	201,231	21%	28%
Residential	171,457	144,384	121,344	143,910	-16%	19%
Solid Waste	32,232	24,502	26,354	32,025	-1%	22%
Wastewater	921	757	1,204	303	-67%	-75%
Total	**523,826**	**552,116**	**522,404**	**607,139**	**16%**	**16%**

Table 4. Per capita emissions (metric tons CO_2 equivalent) by sector and year

	2008	2012	2016	2018	Change since 2008	Change since 2016
Transportation	4.3	4.6	4.8	4.7	10%	-1%
Commercial	4.7	5.6	3.5	4.1	-11%	20%
Residential	4.8	3.8	2.7	3.0	-39%	11%
Solid Waste	0.9	0.7	0.6	0.7	-27%	13%
Wastewater	0.0	0.0	0.0	0.0	-76%	-77%
Total	**14.7**	**14.7**	**11.5**	**12.5**	**-15%**	**8%**

Factors Influencing Emissions

Both commercial and residential natural gas consumption increased in 2018 and appears to be a primary reason for the emissions increase in 2018. The winter of 2018 was relatively cold with several days of record low temperatures, which was coupled with particularly low natural gas prices. This may have encouraged residents and businesses to turn up the thermostat. However, normalizing for both population and weather shows that energy use per person per degree days (the total number of degrees above or below 65° F during a given period) increased in 2018 relative to the inventory years of 2012 and 2016 (Figure 9). While we cannot provide evidence that cold weather necessitated more heating and natural gas use, we can presume that the perception of extreme cold may have been a factor.

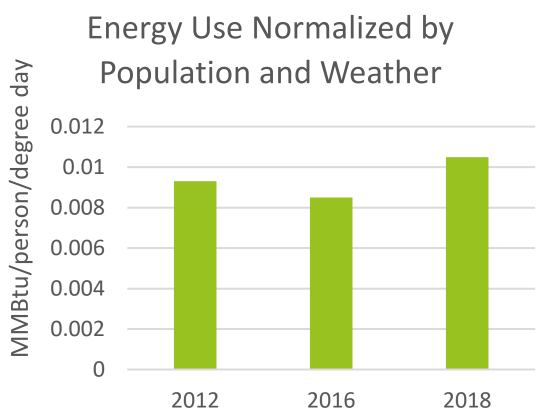


Figure 9. Energy use by inventory year normalized for population and weather

The community indicator of growth is an important trend to track in relation to greenhouse gas emissions. In the decade between our baseline inventory of 2008 to 2018, population increased by 37%. The annualized growth rate between 2010 and 2018 was 3.4% (American Community Survey). Enrollment at Montana State University has increased to 16,900 students with 3,000 employees. Employment growth in Gallatin County has grown at a rate of 4% per year since 2010, with 80% of this growth occurring in Bozeman.

Key Findings from 2018 Inventory

The 2018 City of Bozeman community GHG inventory showed an emissions value of 607,139 metric tons of carbon dioxide equivalent (MT CO_2e), representing a 16% increase in total emissions from the 2008 baseline and previous inventory year of 2016. Per capita emissions were 12.5 MT CO_2e, which is an increase from 2016, but still 15% below the 2008 baseline. Population growth and an increase in natural gas use in the residential and commercial sectors were significant contributors to the increase in emissions.

The majority of Bozeman's local greenhouse gas emissions can be attributed to residential and commercial energy and transportation. Within these sectors, the three leading sources of emissions are electricity, natural gas, and light-duty gasoline vehicles.

Projected Future Emissions

Based on the 2016 baseline inventory, the community GHG emissions were projected to 2050. This projection is used to understand the magnitude of emissions reductions required to meet the community's climate mitigation goals. Bozeman is a rapidly growing community, with 20% population growth between 2012 and 2016. Because of this tremendous growth and the unknowns with regard to climate migration and the potential for future pandemic migration, there is considerable uncertainty around the growth rate looking out to 2050.

Population growth will have a significant impact on the projected GHG emissions for the community. To best understand the range of emissions likely, population estimates from various sources, including the Demographic and Real Estate Market Assessment from 2018, were compiled as shown in Figure 10. Using this array of population estimates, minimum and maximum growth scenarios were estimated to encompass the majority of the data points.

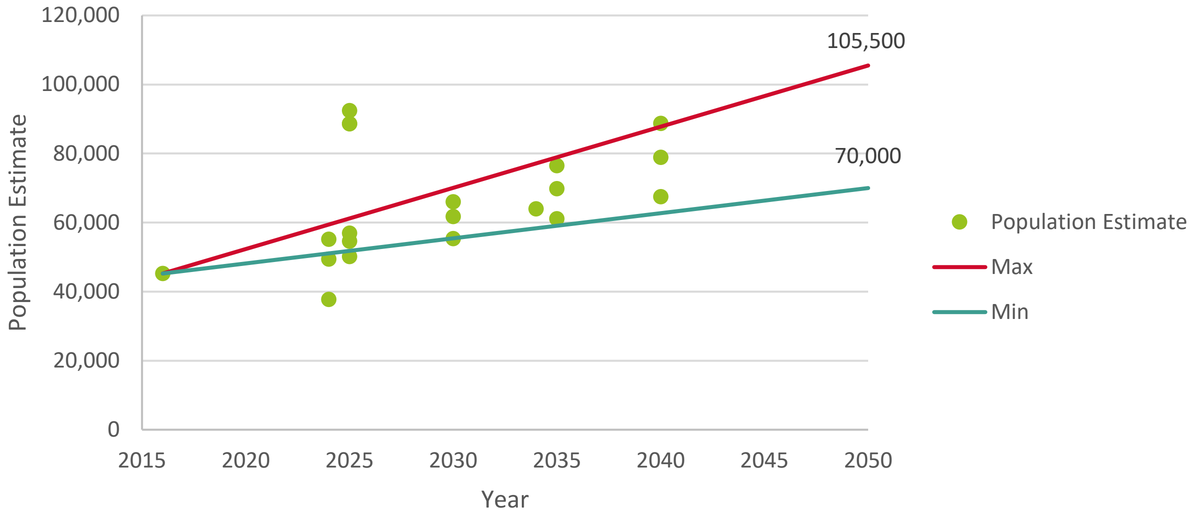


Figure 10. Review of City of Bozeman population growth estimates

To translate this likely population growth to expected GHG emissions, it was assumed that in the business-as-usual scenario, the patterns of use for energy, energy sources, and equipment efficiency remain unchanged over this time, keeping the per capita emissions constant. As shown in Figure 11, this leads to significant uncertainty in the estimated GHG emissions in 2050. For the purposes of this report, we looked at a high growth rate scenario to ensure the community is able to achieve its GHG reduction goals for any of the likely population growth forecasts.

Figure 11. Range of projected GHG emissions with business as usual

EMISSIONS MITIGATION GOAL PROGRESS & KEY MILESTONES

To establish a pathway to achieve Bozeman's climate mitigation goals, it is necessary to look at both the potential magnitude of impact of each solution and the sequencing of solutions. In terms of sequencing, this plan focuses on applying efficiency opportunities to reduce overall energy needs first, followed by renewable energy generation, and then switching to alternative fuel sources. While these solutions work in tandem, reducing overall energy needs first will help ensure that renewable energy and fuel-switching efforts are appropriately scaled.

Efficiency Solutions

Solutions that reduce the community's overall energy demand through improving efficiency of existing buildings, vehicles, equipment, or thoughtfully planning community growth should be implemented first. By reducing the community's energy demand, the amount of energy that must be produced through renewable energy generation is reduced. This improves the feasibility of conversion to 100% renewable energy as well as reduces the investment in infrastructure required to meet the community's energy needs. Solutions considered to be efficiency solutions are:

- Solution A. Improve Efficiency of Existing Buildings
- Solution B. Achieve Net Zero Energy New Construction
- Solution G. Facilitate Compact Development Patterns
- Solution J. Increase Walking, Bicycling, Carpooling, and Use of Transit
- Solution M. Move Toward a Circular Economy and Zero Waste Community

Renewable Energy Generation

These solutions focus on meeting the community's energy needs through clean energy sources and include:

- Solution D. Increase Utility Clean Energy Mix
- Solution E. Develop and Promote Utility Green Power Programs
- Solution F. Increase Community-Based Distributed Renewable Energy Generation

All three of these solutions work together to meet the community's clean energy needs. With efficiency of scale, the utility can generally provide the lowest cost clean energy generation. Over time, this will provide the largest source of clean energy for the community. It does take time for the utility to convert its generation to clean sources while ensuring that utility rates do not cause undue burden on vulnerable populations. Community participation in green power programs can help demonstrate the City's desire for more renewable energy and enable the utility to install clean generation more quickly than would otherwise be feasible. Distributed generation provides the smallest portion of electricity to the community but is an important tool for improving resiliency. These systems can provide back-up power to critical buildings in the case of a power outage or natural disaster.

Fuel Switching

For some GHG emissions sources, and with today's technology, there are no clean alternatives for these fuels. In this case, the source of power for these uses can be switched to electricity which is generated through clean sources to eliminate these emissions. Fuel switching solutions include:

- Solution C. Electrify Buildings
- Solution K. Decrease Direct Vehicle Emissions

Potential Emissions Reduction by Solution

To understand the range of emissions reduction potential from each solution, the expected reduction potential from each strategy through existing Bozeman efforts was compared to top performing communities from across the country. Stakeholder input was used to refine this range of possible emissions to reflect the probable potential reduction for Bozeman.

In Figure 12, the colored bars indicate the range of potential impact for each solution in Bozeman. The low end of that range is what the emissions reduction might be if existing initiatives continue at current levels of engagement and the high end is the maximum emissions reduction that could be expected from that strategy. The grey bars extending beyond the expected range shows the expected emissions from the best-case scenario, which may not be practical or feasible in Bozeman based on local factors or conditions (e.g., state regulations, or growth forecasts) or current technology. This analysis is used to break the solutions into categories based on their expected impact on Bozeman's GHG emissions as shown below.

Please note that this analysis is used to inform scenario modeling and does not consider interaction between strategies. For detailed information on this analysis and scenario modeling, please reference Appendix A. This analysis assumes the maximum growth scenario discussed in the baseline inventory section.

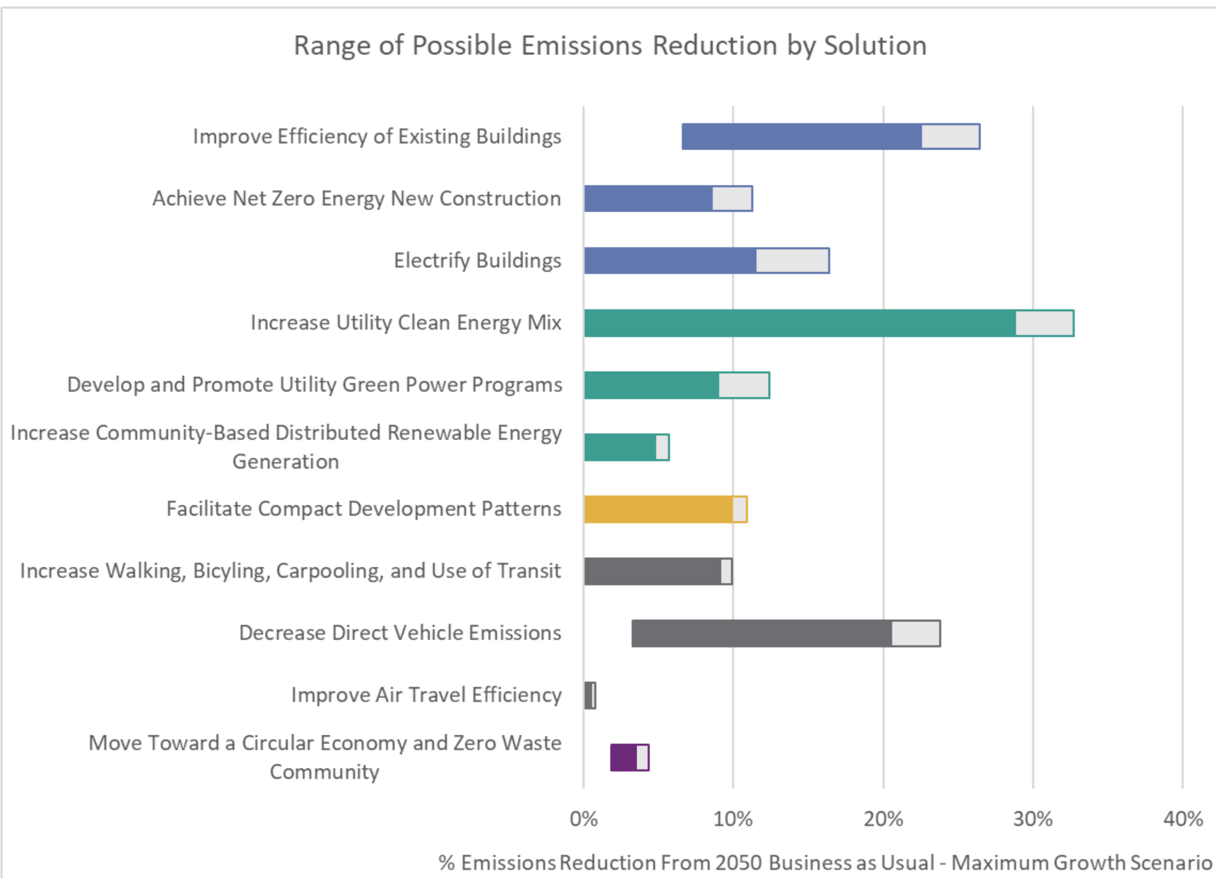


Figure 12. Emissions reduction potential by solution

Scenario Modeling

The range of potential impact for each solution shown in Figure 13 as well as the expected rate and timing of implementation were used to model the most likely scenarios for Bozeman to meet each of solution's GHG mitigation goals. The two graphs below show the business as usual emissions projections on the left and the estimated emissions reduction by focus area in each goal year on the right. Note that the future solutions section in the 2050 emissions reduction shows that there are some emissions sources for which there are not currently good solutions to address, such as aviation fuel use and heavy-duty vehicles. It is expected that between now and 2050 there will be new technologies or policies that can be used to address these emissions sources.

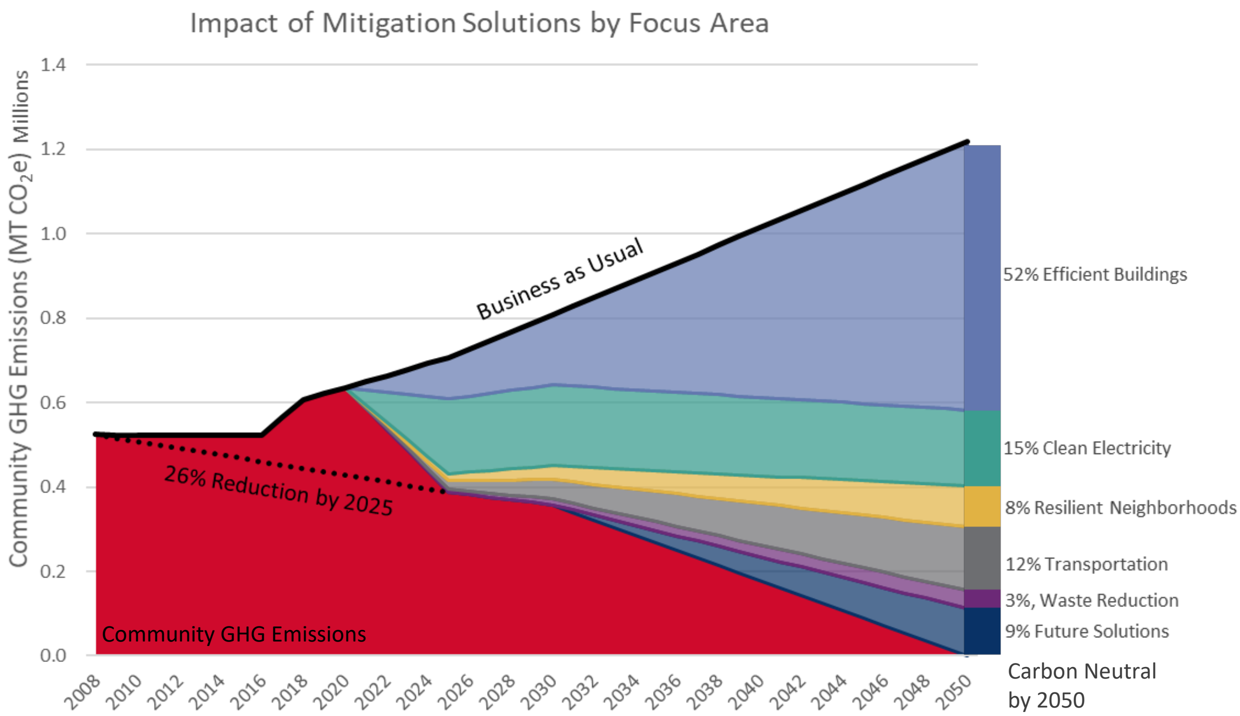


Figure 13. Projected GHG emissions and anticipated emissions reduction by focus area

For more detailed information on the contributions of each solution, see the corresponding chapter. The details of each goal scenario are shown in Appendix A: Analysis Summary, Table 13, and key takeaways for each goal year are outlined in the following sections.

26% reduction in GHG emissions by 2025 (below 2008)

To meet the community's commitment to uphold the Paris agreement, swift action will be required on key mitigation strategies that can be implemented over the next 5 years, including encouraging NorthWestern Energy to advance new utility-scale clean energy and to develop a green tariff program that together provide 204 million kWh, or 40% of the projected community's electricity use, above and beyond the 280 million kWh clean electricity delivered to the City via the utility's current generation mix. These strategies are supported by improving efficiency in new and existing buildings, as well as distributed renewable electricity generation (16 million kWh). Energy efficiency improvements in existing buildings and rooftop or community solar have the potential to be a major contributor to meeting Bozeman's sustainability goals, but the capital improvements necessary cannot be implemented as quickly as other strategies. The community should target about 2% energy savings from existing building efficiency programs each year.

There will also be lesser contributions from a variety of other solutions. While these solutions are not as readily scalable to meet the 2025 goal, it is important to start working on these solutions to set the community up for success on future goals.

2025 Climate Goal

26% reduction in GHG emissions by 2025 (compared to 2008)

2025 Goal: Reduce Emissions to 388,000 MT CO_2e

- Electricity Savings: 126,000,000 kWh
- Additional Clean Electricity Production: 220,000,000 kWh
- Natural Gas Savings: 6,900,000 therms
- Gasoline Savings: 4,400,000 gal
- Other Emissions Reductions: 8,100 MT CO_2e

Total GHG Emissions Reduction: 319,000 MT CO_2e*

2008 Community GHG Emissions: 524,000 MT CO_2e

* As compared to the 2025 business-as-usual projection

Primary mitigation solutions

- Increase Utility Clean Energy Mix
- Develop and Promote Utility Green Power Programs

Secondary mitigation solutions

- Improve Efficiency of Existing Buildings
- Achieve Net Zero Energy New Construction

Supporting or long-term mitigation solutions

- Increase Community-Based Distributed Renewable Energy
- Facilitate Compact Development Patterns
- Increase Walking, Bicycling, Carpooling and Use of Transit
- Decrease Direct Vehicle Emissions
- Limit Emissions from Air Travel
- Move Toward a Circular Economy and Zero Waste Community

100% net clean electricity by 2030

The next milestone set by the City is to supply the community electricity needs with 100% net clean electricity by 2030. To meet this milestone, NorthWestern Energy will need to continue to meet its emissions reduction targets and the community will need to continue to advocate to NorthWestern Energy to increase the percentage of utility-scale clean energy and to develop and grow a green tariff program. To meet the City's goals, the total grid delivered clean energy must be at least 484 million kWh in 2025 and 478 million kWh by 2030. These values include the 280 million kWh clean electricity delivered to the City via the Utility's current generation mix. The energy efficiency improvements for existing buildings need to continue at the rate of 2% energy savings across sectors each year.

High performing new construction is also important to ensure that new growth is as efficient as possible. Housing and commercial building space projections from the Bozeman Community Plan estimate 12,700 new homes (an average of 450 new homes per year), and 6.3 million square feet of non-residential construction demand through the year 2045.

Transportation emissions reduction measures required to keep the community on track to meet their 2050 emissions reduction goals will increase the demand for renewable electricity as more of the community fleet is electrified. This additional electrical load has been accounted for in the required renewable electricity generation calculations.

2030 Climate Goal

100% net clean electricity by 2030

2030 Electricity Emissions Goal: Net Zero MT CO_2e

- Business-as-Usual Electricity Use: 712,000,000 kWh
- Additional electricity use from electric vehicles: 19,000,000 kWh
- Electricity Efficiency Savings: 213,000,000 kWh
- Clean Energy Production
  - Baseline: 280,000,000 kWh
  - Additional: 238,000,000 kWh

Net Carbon-based Electricity Use: 0 kWh

Primary mitigation solutions

- Improve Efficiency of Existing Buildings
- Increase Utility Clean Energy Mix
- Develop and Promote Utility Green Power Programs

Secondary mitigation solutions

- none

Supporting or long-term mitigation solutions

- Achieve Net Zero New Construction
- Increase Community-Based Distributed Renewable Energy

Carbon Neutral by 2050

To meet the community's commitment to be carbon neutral by 2050, the community will need to transition natural gas used in buildings for heating and cooking to electric appliances and 100% renewable electricity. Likewise, a nearly complete transition of personal passenger and fleet vehicles to electric vehicles will be required between 2030 and 2050. Even so, scenario modeling indicates certain harder to reach portions of the GHG emissions inventory that will require some combination of future technologies, carbon sequestration opportunities, or carbon offsets to achieve the goal to become a carbon neutral community by 2050. These harder to reach areas include emissions from airline fuel use, heavy duty trucks and equipment not currently conducive to electrification, and energy storage.

2050 Climate Goal

Carbon neutral by 2050

2050 Emissions Goal: Net Zero MT CO_2e

- Electricity Savings: 750,000,000 kWh
- Additional Clean Energy Production: 386,000,000 kWh
- Natural Gas Savings: 50,400,000 therms
- Gasoline Savings: 32,000,000 gal
- Other Emissions Reductions: 48,200 MT CO_2e
- Future Solutions: 113,000 MT CO_2e

Total GHG Emissions Reduction: 1,218,000 MT CO_2e

Business-as-Usual: 1,218,000 MTCO_2e

Primary mitigation solutions

- Improve Efficiency of Existing Buildings
- Increase Utility Clean Energy Mix
- Decrease Direct Vehicle Emissions

Secondary mitigation solutions

- Achieve Net Zero New Construction
- Electrify Buildings
- Facilitate Compact Development Patterns
- Increase Walking, Bicycling, Carpooling and Use of Transit

Supporting mitigation solutions

- Develop and Promote Utility Green Power Programs
- Increase Community-Based Distributed Renewable Energy
- Limit Emissions from Air Travel
- Move Toward a Circular Economy and Zero Waste Community

RESILIENCY GOAL CONTRIBUTIONS

This table illustrates which solutions connect with and help advance the community's resiliency goals. Shaded cells indicate that the solution provides direct benefits to support the goal.

	RESILIENCY GOALS				
	Conserve natural resources	Harden infrastructure to natural disaster	Increase social cohesion	Mitigate property and economic losses	Protect human health
Solution A. Improve Efficiency of Existing Buildings	■	■			■
Solution B. Achieve Net Zero Energy New Construction	■	■			■
Solution C. Electrify Buildings	■				■
Solution D. Increase Utility Clean Energy Mix	■	■			■
Solution E. Develop and Promote Utility Green Power Programs	■				■
Solution F. Increase Community-Based Distributed Renewable Energy Generation	■	■		■	■
Solution G. Facilitate Compact Development Patterns	■		■	■	■
Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards		■		■	■
Solution I. Enhance Social Infrastructure and Community Preparedness			■	■	■
Solution J. Increase Walking, Bicycling, Carpooling and Use of Transit	■		■		■
Solution K. Decrease Direct Vehicle Emissions	■				■
Solution L. Improve Air Travel Efficiency	■				■
Solution M. Move Toward a Circular Economy and Zero Waste Community	■		■		■
Solution N. Cultivate a Robust Local Food System	■		■	■	■
Solution O. Manage and Conserve Water Resources	■	■		■	■
Solution P. Manage Land and Resources to Sequester Carbon	■	■		■	■

Bozeman Climate Plan

CHAPTER 3:

Focus Areas, Solutions & Actions

CHAPTER 3: FOCUS AREAS, SOLUTIONS & ACTIONS

Focus Area 1:

Healthy, Adaptive & EFFICIENT BUILDINGS

More than half of Bozeman's greenhouse gas emissions come from energy use in buildings. Specifically, building energy use comprised 57% of emissions in 2018, with 33% from Bozeman's commercial buildings, and another 24% from residential buildings. While efficiency of the residential and commercial sectors is improving over time through more efficient building codes, more can be done to decrease emissions from existing buildings across the community. Improvements to building envelopes and energy systems can provide myriad benefits, such as enhancing building performance, stabilizing utility bills, improving comfort and safety of building occupants, and adapting to future climate conditions.

As a rapidly growing community, new construction threatens progress made on reducing overall building-related greenhouse gas emissions if not built efficiently. According to the 2018 Demographic and Real Estate Market Assessment, there is an estimated demand for 12,700 new housing units and 6.3 million square feet of new commercial space in Bozeman through 2045. By ensuring that new buildings are constructed efficiently and ready to adapt to changing technologies, Bozeman can help make sure that new buildings are helping the community achieve its emissions reduction and resiliency goals.

RELATED PLANS & STUDIES

- Bozeman Community Plan
- NorthWestern Energy Electric Supply Resource Procurement Plan
- Climate Vulnerability Assessment and Resilience Strategy
- Downtown Bozeman Improvement Plan
- Community Greenhouse Gas Emissions Report
- Municipal Greenhouse Gas Emissions Report

These solutions emphasize reductions in energy-related emissions from existing and future buildings by prioritizing energy efficiency as a primary objective, supported by converting natural gas consumption of buildings to electricity where feasible and appropriate. To meet the GHG emissions goals, the community will need to:

- **Solution A. Improve the Efficiency of Existing Buildings**
  Maintain a concerted effort to improve efficiency of existing buildings over the 30-year planning horizon, including deep energy retrofits. The energy savings target should be 2% year over year energy savings from existing buildings.

- **Solution B. Achieve Net Zero Energy New Construction**
  Ramp up to 100% of new buildings being net zero energy by 2030.

- **Solution C. Electrify Buildings**
  Focus on electrification of building heating and cooking loads in tandem with pursuing renewable energy targets. Future technologies will be required to eliminate any natural gas loads that cannot be electrified with current technologies.

Detailed energy savings targets for each goal year are shown in Table 5.

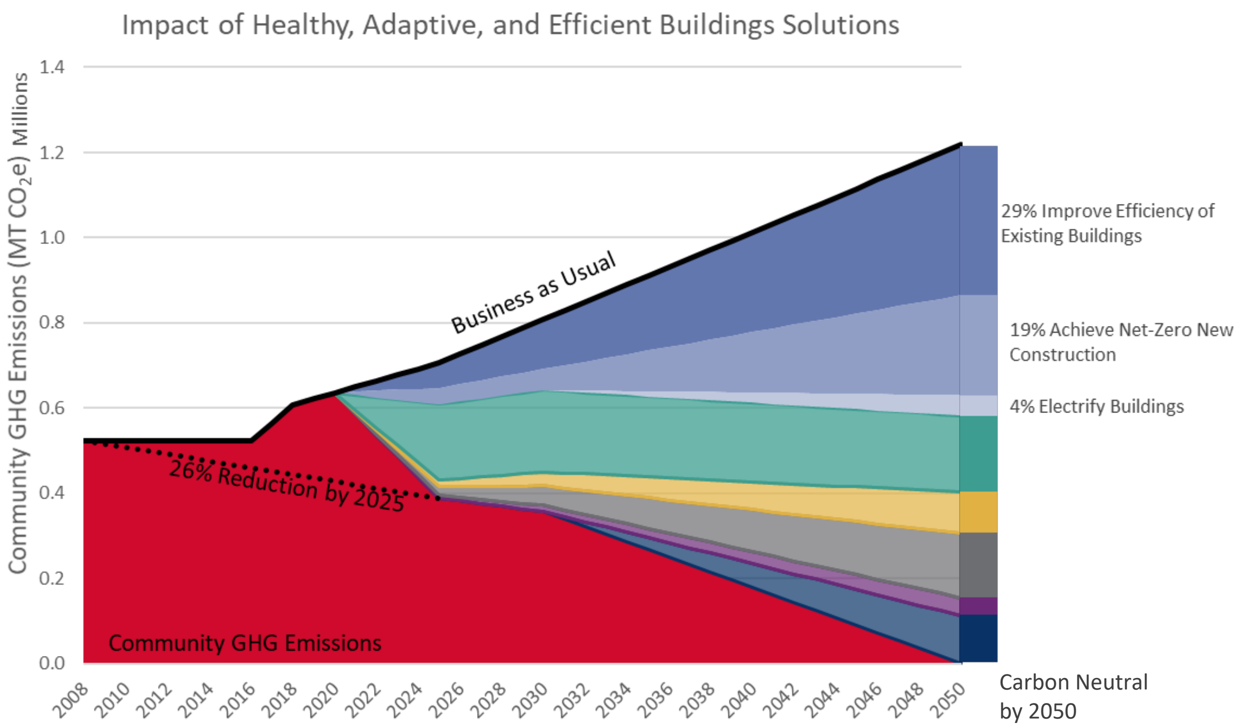
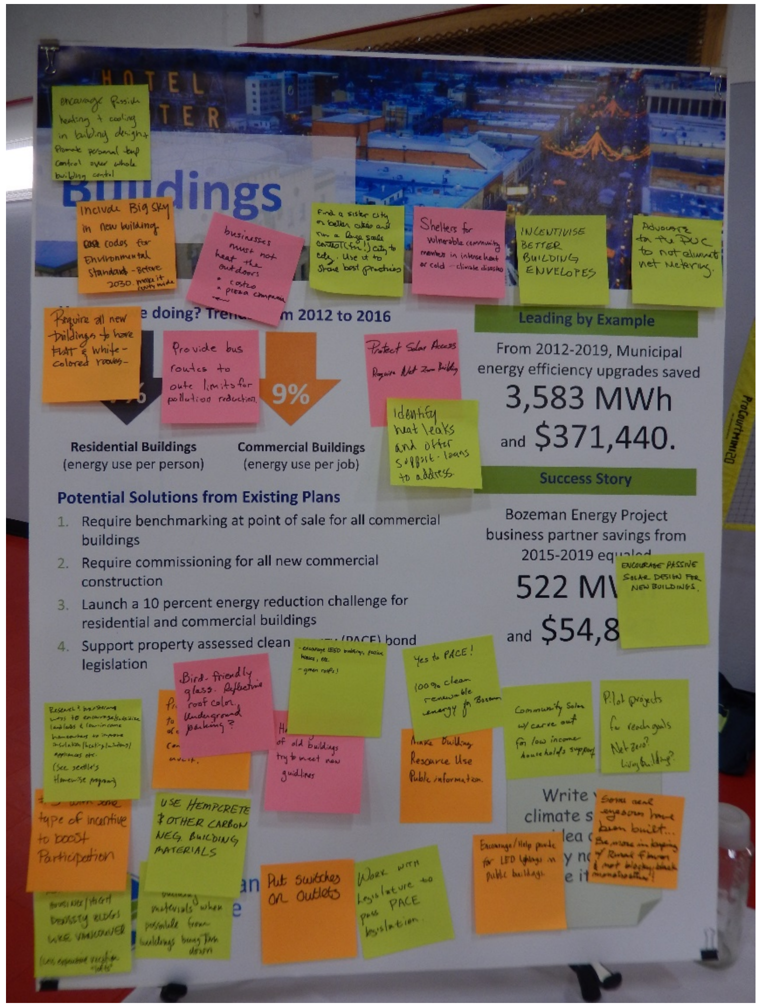


Figure 14. Projected building GHG reductions by solution

Table 5. Projected buildings solution mitigation targets

Solution	*Metric (annual savings)*	*Target* 2025 *Paris Accord*	*Target* 2030 *100% Net Clean Electricity*	*Target* 2050 *Carbon Neutral*
Solution A. Improve Efficiency of Existing Buildings	Electricity (kWh)	75 million	149 million	447 million
	Natural Gas (therms)	4.2 million	8.4 million	25 million
Solution B. Achieve Net Zero Energy New Construction	Electricity (kWh)	51 million	64 million	303 million
	Natural Gas (therms)	2.7 million	3.4 million	16.2 million
Solution C. Electrify Buildings	Electricity (kWh)	0	0	-269 million
	Natural Gas (therms)	0	0	9.2 million

Solution A. Improve Efficiency of Existing Buildings



This solution emphasizes investment in existing buildings to improve energy and water efficiency, as well as building comfort and performance. An energy or water audit, especially one with thermal imaging, can be a great first step to help identify and prioritize efficiency improvement opportunities. Some efficiency improvements may be simple do-it-yourself quick fixes, while others may require more costly investments and renovations.

Many efficiency projects generate a positive cash flow and have a simple payback of less than five years. Numerous technologies exist to help building owners and occupants monitor utility consumption and automate building systems. Energy efficiency investments and technologies present tremendous opportunities for workforce development and job training, further bolstering the economy.

Various tax credit, rebate, and incentive programs exist at the local, utility, state, and federal levels to support efficiency improvements. However, the landscape of financial resources and incentives can be confusing for customers to navigate and access. Bozeman's electric and natural gas utility provider, NorthWestern Energy, offers some limited energy audit/appraisal programs for their residential and business customers. Utility rate structures and time-of-use pricing (paying a different price per kWh depending on when the energy is consumed) can be another financial mechanism to discourage high utility consumption at certain times or thresholds.

From an equity perspective, it is essential to recognize that many utility customers experience cost burden (i.e., have difficulty paying bills) and bitter cold Montana winters can make heating a lifesaving necessity. While such customers may implement efficient practices out of necessity, they may be unable to make costly efficiency investments, or may encounter challenges with time-of-use utility rate structures. Another important equity consideration is the fact that the costs and benefits of efficiency improvements are not always fairly distributed between owners and tenants.

Health and well-being benefits associated with enhanced building performance can include improved indoor air quality and occupant comfort. All energy efficiency improvements should be coupled with ventilation measures to avoid negative impacts on indoor air quality.

The City of Bozeman can lead by example by planning, financing, and implementing energy and water efficiency improvements across municipal facilities and operations. The City can also continue to lead education and outreach efforts to help share efficiency information and resources with the community.

This solution primarily addresses Bozeman's greenhouse gas mitigation goals, while also providing supporting resiliency benefits.

Emissions Reduction Benefits

This is a secondary mitigation solution to achieve the 2025 goal and primary solution to achieve the 2030 and 2050 goals. Emission source(s) impacted include:

- Electricity
- Natural Gas

Resiliency Benefits

- Conserve natural resources
- Strengthen infrastructure to natural disaster
- Protect human health

Related Solutions

- Solution E. Develop and Promote Utility Green Power Programs
- Solution F. Increase Community-Based Distributed Renewable Energy Generation
- Solution O. Manage and Conserve Water Resources

Action 1.A.1. Increase Energy Efficiency at City Facilities

To advance our goal of 26% reduction in greenhouse gas emissions (compared to 2008 levels) by 2025, the City will identify and implement efficiency improvements in all City facilities through comprehensive energy audits and energy efficiency upgrades. Energy Service Performance Contracts (ESPC) may be used to help finance and install energy conservation and renewable energy projects at City-owned buildings and properties.

The City has already completed energy audits and Facility Condition Inventories, as well as an ESPC completed in 2015 that included an LED retrofit of 190 city-owned and metered streetlights and an LED upgrade in the Bridger Parking Garage. Individual solar photovoltaic installations, lighting upgrades, and other energy projects continue to be initiated by multiple City departments as opportunities arise, but a comprehensive ESPC approach across the entire City will help maximize efficiency, speed project deployment, and link efficiency, production, and energy storage in new ways.

While retrofitting involves replacing outdated equipment, retro-commissioning focuses on improving the efficiency of what is already in place. Using City building benchmarking (measuring energy performance against past usage or other buildings' energy usage) and energy consumption trends, the City will fine-tune existing buildings and systems in order to make them operate optimally and more efficiently through scheduling, sequencing, controls programming, optimizing set points, and equipment maintenance. Retro-commissioning can save energy and money, while also extending the service life of equipment.

In preparation for our carbon neutrality goal, the City will adopt ambitious high-performance net zero energy construction standards for all new or major renovations for City facilities. This standard should incorporate life cycle analysis (LCA) during the development or redevelopment of all municipal facilities, such as the standards established in the building life-cycle impact reduction credit available through the Leadership in Energy and Environmental Design (LEED) building rating system.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman**	• 4.3 Strategic Infrastructure Choices • 6.3 Climate Action	• Establish building energy benchmarking • Improve building energy use intensity

Action 1.A.2. Use Data and Price Signals to Advance Energy Efficiency

Customer access to data can be a powerful tool to drive energy conservation and reduce peak demand. Advanced Metering Infrastructure (AMI) refers to meters, communication networks, and data management systems that collect, transmit, and record electricity consumption data in daily or shorter intervals. More timely and granular data provides operational benefits to utilities, and can be used to engage, motivate, and reward customers.

AMI data alone does not necessarily lead to energy savings. Data paired with customer engagement tools, time-of-use (TOU) pricing, and programs with incentives that motivate and support customers are needed to change when and how customers use energy. A 2017 study of 50 utilities with TOU pricing mechanisms found an average peak demand reduction of 16% and an average reduction in overall consumption of over 2%. Utilities that offered TOU pricing plus peak-time rebates realized an overall consumption reduction of over 7% (Gold, 2020).

NorthWestern Energy is in the process of deploying AMI in its South Dakota service area and plans to extend this service to its Montana service area beginning in 2021. The City of Bozeman will coordinate with NorthWestern Energy and encourage early deployment of AMI for Bozeman residents, as well as encourage NorthWestern Energy to couple AMI with real-time resource consumption and generation source information for their customers. Reducing peak demand should be a priority in order to address NorthWestern Energy's peak load challenges referenced in the 2019 Electricity Supply Resource Procurement Plan and help eliminate the need for costly investment in additional fossil fuel generation.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **NorthWestern Energy** • City of Bozeman • Potential third-party data analyst (e.g., WattTime)	• 4.3 Strategic Infrastructure Choices • 6.3 Climate Action	• % installation of advanced metering infrastructure • Establishment of a time-of-use rate designed to offset utility peak demand • % reduction of peak energy demand

Action 1.A.3. Expand Energy Efficiency Information and Resources for Private Property

In support of Advanced Metering Infrastructure and time-of-use rates, the utility, City, and numerous community partners can provide general outreach, social marketing, and education regarding energy appraisals, energy and water efficiency rebates, energy efficiency tips, and strategies to avoid peak energy demand. The city and its partners can create a conversation around the critical role of smaller homes and apartments, as well as the size, number, and efficiency of appliances, in reducing overall energy use. Targeted outreach to low-income residents, renters, and residents living in mobile homes and older homes represents an important segment of the Bozeman community and each population requires customized recommendations. Possible opportunities to explore include working with Montana State University to develop a curriculum to engage and inform students at they move off campus or training student "energy squads" to assist with basic rental retrofits. Using technology such as aerial thermal imaging could map high energy use hotspots and energy efficiency opportunity zones.

Improving the conversion rate from commercial and residential energy appraisals to execution and completion of energy upgrades is a wide-spread challenge. The City and NorthWestern Energy should partner on innovative strategies to improve this conversion rate. One successful model involves partnering with an energy service provider to provide turn-key service from the energy appraisal to on-bill financing to contractor services for energy upgrades. With partners such as the Montana State University Extension Montana Weatherization Center, Bozeman could pilot an energy service provider training and certification program.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • **NorthWestern Energy** • Human Resource Development Council • Montana State University • Montana Weatherization Training Center • Southwest Montana Building Industry Association • Montana Department of Environmental Quality Energy Office	• 4.5 Housing and Transportation Choices • 6.3 Climate Action	• Reduced peak energy demand • Average energy use per household • Increase audit to retrofit conversion rate

Action 1.A.4. Establish an Energy and Water Benchmarking Standard for Commercial Buildings

With 33% of Bozeman's greenhouse gas emissions generated from commercial buildings, it remains a priority to advance efficiency in this sector. Building energy benchmarking tracks a building's energy performance over time and allows for transparent comparison of performance relative to other similar buildings or relative to modeled simulations of reference buildings built to a specific energy code. This comparison helps identify opportunities for technological and operational energy efficiency improvements. Benchmarking is useful for local governments to facilitate energy and water use accounting, comparing a building's energy and water use to similar buildings to assess opportunities for improvement, and quantifying or verifying energy savings. Across many commercial building markets, the practice has become standard operating procedure as energy costs and associated sustainability issues have raised awareness around the importance of energy management. Benchmarking can help ensure that building systems, such as mechanical, electrical, heating, ventilation, and air conditioning, are operating at optimal efficiency as intended by building architects and engineers. Retro-commissioning can also help extend the life of existing systems, defer expensive upgrades, and ensure timely identification of energy and water efficiency opportunities. Disclosure and data transparency enable future policy creation and enhance decision-making from both public and private entities. Social marketing, recognition, and incentives are all tools that would help accelerate benchmarking and retro-commissioning.

Using Department of Energy or Environmental Protection Agency benchmarking tools, the City should partner with NorthWestern Energy, the MT Energy Office, Montana State University, the Montana Weatherization Center, commercial and multi-family building owners and tenants, the Bozeman Chamber of Commerce, the outdoor industry, financial institutions, and others to develop a "better buildings challenge" focused on energy and water efficiency policy to require annual benchmarking and energy use disclosure for large commercial buildings. The program would initially be piloted for government and institutional buildings.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • NorthWestern Energy • MT Energy Office • Montana State University • Montana Weatherization Center • Commercial and Multi-family building owners and tenants • Chamber of Commerce • Bozeman Outdoor Industry • Financial Institutions	• 6.1 Clean Water Supplies • 6.3 Climate Action	• % of commercial buildings complying with benchmarking policy • % of reporting buildings that achieve an ENERGY STAR score >75

Action 1.A.5. Require Home Energy and Water Use Labeling at Time of Listing

Energy and water bills are a significant household expense, and yet prospective home buyers and renters are typically unable to factor this information into their decision making. Utility data is rarely provided unless a prospective buyer specifically requests it. A home energy label provides information about a property's energy consumption and costs, plus recommendations for cost-effective energy saving improvements. Like a snack's nutrition label or a car's MPG rating, it reveals valuable and comparable information not otherwise apparent to consumers.

Cities across the U.S. are working to adopt energy labeling and disclosure policies for residential properties to better inform major consumer investment decisions and provide a critical foundation for driving energy efficiency upgrades that improve our building stock and reduce greenhouse gas emissions. Residential energy use accounts for 24% of the community's 2018 greenhouse gas emissions. The City will explore a policy for Bozeman under which home sellers would disclose home energy labels to real estate agents and potential buyers through the Multiple Listing Service (MLS).

Figure 15. Example from the Dept. of Energy's Home Energy Score Program

The Department of Energy's Home Energy Score and the U.S. EPA WaterSense Labeled Home are practical tools for implementing home energy and water labeling programs. Such programs improve consumer protection through information transparency and create data to measure progress toward climate and water efficiency goals. Home sellers can increase the market value of their home while buyers and renters can better compare options in terms of energy performance and costs and consider financing options for upgrades. New jobs in energy assessments and contracting are created in the local economy. According to the Department of Energy's Better Building's Initiative, energy efficient certified homes sell faster and for 4 to 6% more.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Gallatin Association of Realtors • Southwest Montana Building Industry Association • NorthWestern Energy • Chamber of Commerce • Montana State University-Montana Weatherization Center • Financial Institutions	• 1.4 Business and Institutional Partnerships • 4.5 Housing and Transportation Choices • 6.3 Climate Action	• Implementation of energy use disclosure policy • Number of days on market for high performance properties

Action 1.A.6. Promote Energy Efficiency Financing and Investment

Today local governments are on the front lines when it comes to handling the COVID-19 pandemic, growing civil unrest, and severe weather events. Local leadership is necessary to meet local needs, including the task of recovery and job creation. State and local actors around the country are doing their part to help spur investment and job creation with public and private investment in clean infrastructure and energy efficiency.

The availability of low-cost financing is a critical factor for widespread deployment of energy efficiency and renewable energy. Green Banks are one way to help secure low-cost capital for clean energy projects, such as solar PV or energy efficiency upgrades, at favorable rates and terms. Green Bank products may be targeted at end users such as a home, business, or other finance providers, such as retail and investment banks. Examples include residential solar loan and lease programs, credit support mechanisms for energy efficiency and solar, and commercial Property Assessed Clean Energy (PACE) programs that support a variety of energy conservation measures. Through such products, Green Banks can help secure private capital by multiple diverse means. According to the National Renewable Energy Lab, as of 2016, the most advanced Green Banks in the country in Connecticut and New York have collectively invested $575 billion in total clean energy investment. These investments reportedly spur private sector investment into projects by three to six times the amount of public sector dollars.

Creating and marketing a specific local program to encourage investments in residential and commercial properties will provide an incentive for homeowners and business owners to make the capital investments in their properties necessary to help achieve our building energy reduction goals. City Economic Development and Sustainability Divisions can collaborate with area lenders via Request for Proposals or by setting loan rates and standards to create a low-interest loan program (fixed or adjustable rate) for building energy efficiency, electrification, or installation of solar or geothermal systems. Once a brand name has been developed, marketing the program can be co-funded and executed by the City of Bozeman and participating institutions.

State and local governments have established Green Banks and low-interest loan programs under a variety of different structures, legislative directives, and funding sources. The City of Bozeman will work with area financial institutions to develop a county-wide low-interest loan program to facilitate financing of energy efficiency and renewable energy systems. Through local and regional economic development entities, the City will build awareness and leverage partnerships to establish a statewide Montana Green Bank. Advancement of supporting tools such as utility on-bill financing, allocation of Universal System Benefit funds, and the authorization of commercial and residential PACE will aid in the establishment of a Montana Green Bank and county-wide low-interest loan program.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measure*
Level 2	• **City of Bozeman** • **Local Financial Institutions** • **NorthWestern Energy** • Prospera Business Network • MT Building Industry Association • Montana Legislature	• 2.1 Business Growth • 6.3 Climate Action	• Establishment of a county-wide low-interest loan program • Establishment of a Montana Green Bank

Action 1.A.7. Create a Rental Registry Program to Advance Renter Safety and Energy and Water Efficiency

With over 19,000 housing units in Bozeman, only 44% are owner occupied. Where tenants are responsible for energy and utility operating costs in addition to the monthly rent, owners have little incentive to invest in energy and water efficiency capital improvements because the tenant will be the sole beneficiary of the reduced operating costs. Where the tenant's monthly rent includes a predetermined amount to cover energy and utility operating costs, tenants have little incentive to save energy and water because the owner receives the benefits of those efforts. This split incentive between landlords and tenants makes this a particularly difficult sector in which to implement energy and water efficiency. With a consistently low rental vacancy rate in Bozeman, the market is unlikely to create the circumstances in which information, such as an audit or Home Energy Score, will lead to energy upgrades in rental properties.

The City will work with partners to develop consumer resources and develop a voluntary pathway to gradually phase in a residential renter safety and energy and water efficiency program that requires rental properties to meet minimum energy efficiency standards to receive a rental license. Such a program should be designed within a licensing program that complies with the limitations of the Landlord Tenant Act. Program specifications should be coordinated with existing and new utility incentives for lighting upgrades, water heaters and other appliances, plumbing fixtures, roof insulation, air sealing, and HVAC inspection and maintenance. Ideally, the program should be coupled with financing options, such as on-bill financing, residential Property Assessed Clean Energy, or Green Bank financing.

Performing an energy and water upgrade on a rental property (if required) can result in additional value streams for the owner including increased rent, higher tenant retention, and higher resale value. For the tenants, they are likely to experience safer, more comfortable living conditions and more affordable utility costs.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 3	• **City of Bozeman** • **Montana State University** • MSU Off-Campus Living Housing Program • Human Resource Development Council • Montana State University • Montana Weatherization Training Center • Montana Landlords Association • Financial Institutions	• 1.4 Business and Institutional Partnerships • 4.5 Housing and Transportation Choices • 6.3 Climate Action	• % of participating rental properties in outreach program • % of participating rental properties completing energy upgrades

Solution B. Achieve Net Zero Energy New Construction

This solution focuses on the design and construction of high-performance buildings. A Net Zero Energy (NZE) building is one with zero net energy consumption, typically on an annual basis. In other words, the total amount of energy used by the building is equal to (or even less than) the amount of on-site renewable energy generated. The most efficient and cost-effective pathway to achieve NZE is through new construction; however, major renovation projects can present NZE opportunities as well. Even in cases where on-site renewable energy is not feasible, Net Zero Energy Ready (NZER) or high-performance construction helps advance the intent of this solution by reducing overall energy demand and consumption.

Since Bozeman is a rapidly growing community, new building efficiency will play an important role in meeting the community's climate goals. The magnitude of impact on emissions reductions will be heavily dependent on community growth rate, but this solution is important to couple with energy efficiency in existing buildings so that emissions from all buildings (existing and future) are addressed.

The State of Montana is expected to adopt the 2018 International Building Code's International Energy Efficiency Code (with amendments) by the end of 2020. All new construction must comply with those code requirements and communities cannot adopt code requirements that are more stringent than the state adopted code. NZE construction is considered an "above code" opportunity, which can be encouraged with voluntary incentives, but not required. There are myriad resources available to inform NZE construction and many approaches align with third-party building certification programs, such as the U.S. Department of Energy's Zero Energy Ready certification program and Leadership in Energy and Environmental Design (LEED) certification. There are various ways to incentivize NZE construction, including permit fee reductions, expediting permitting processes, utility programs and rebates, tax credits, and low-interest loans. Property Assessed Clean Energy (PACE) financing is not currently available as a financing mechanism in Montana but may be considered for commercial buildings in future State legislative sessions.

Constructing new buildings to higher efficiency standards requires upfront capital investments from the developer, but the investments are repaid in reduced utility costs over the life of the home or building. A 2019 study by the Rocky Mountain Institute found the estimated incremental cost for a Net Zero Ready home in Bozeman was 2.2% with a payback of 4.6 years for homes with electric heating systems and 7.6 years for homes with natural gas heating systems (Rocky Mountain Institute, 2019). Similar assumptions likely apply to commercial buildings, but building loads vary more widely by building use type.

Equity

Benefits of NZE construction include but are not limited to efficient use of energy, low and consistent energy bills, improved occupant comfort, less reliability on the electric grid, increased resale values, and tenant attraction. High efficiency new construction often results in improved indoor air quality when the building is properly ventilated.

Human Health & Well-Being

Some of the challenges of NZE construction, such as upfront costs and finding skilled designers and builders, can be overcome with measures such as rebates, incentives, and training and education programs. Additional equity considerations include the fact that new construction of high-performing buildings may be out of reach for lower-income community members. Similarly, the potential costs to retrofit existing buildings to NZE levels need to be carefully considered.

City Assets

In terms of municipal assets, the City of Bozeman can lead by example by constructing new facilities to NZE standards and training the development community in these design and construction practices.

This solution primarily addresses Bozeman's greenhouse gas mitigation goals, while also providing supporting resiliency benefits.

Emissions Reduction Benefits

This is a supporting mitigation solution to achieve the 2025 and 2030 goals and secondary solution to achieve the 2050 goal. Emission sources impacted include:

- Electricity
- Natural Gas

Resiliency Benefits

- Conserve natural resources
- Strengthen infrastructure
- Protect human health

Related Solutions

- Solution C. Electrify Buildings
- Solution E. Develop and Promote Utility Green Power Programs
- Solution F. Increase Community-Based Distributed Renewable Energy Generation
- Solution G. Facilitate Compact Development Patterns
- Solution K. Decrease Direct Vehicle Emissions
- Solution O. Manage and Conserve Water Resources

Action 1.B.1 Support High Performance Building Resources and Training for the Development Community

While state code determines minimum building standards for all construction, the development community in Bozeman is largely responsible for whether buildings are built beyond the minimum standard. By providing high-performance building resources and training, the City can engage the development community in shifting from current baseline construction practices to more efficient buildings.

There are widespread resources available for the design and construction of high-performance buildings. The City of Bozeman will partner to develop and provide guidance for developers regarding energy and water efficient design (e.g., passive design, energy efficient building materials, low-carbon building materials, water efficient plumbing fixtures, etc.) and develop a pilot program to train builders and contractors on net zero energy construction best practices.

The City will partner with development community stakeholders such as homeowners' associations to identify opportunities to facilitate net zero energy and water efficient building practices through the development of model covenants. Through educational partnerships, the City will encourage owners of high-performance and net zero energy buildings to share and market best practices.

Given the incremental costs associated with high-performance buildings, financing must be a consideration. This will be achieved by partnering with local banks to identify opportunities for providing lower interest loans and energy efficient mortgages for high performance buildings (see also 1.A.6). High performance and net zero buildings are more durable and resilient, reducing risk for lenders. Additionally, promoting targeted financing will create value and recognition of energy efficiency. The City can also identify partnerships with building material manufacturers to encourage the production, marketing, and sale of low-carbon and energy efficient building materials.

Within the context of building energy performance, the city will continue to consider building siting and transportation requirements as important factors in building energy efficiency. The City will continue to encourage covered bicycle parking, locker facilities for commuters, integration with transit routes, or redevelopment of an existing site that does not require addition street construction.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • **Montana State University** • Montana Weatherization Training Center • Gallatin College • Southwest Montana Building Industry Association • Montana Code Collaborative • Montana Dept. of Environmental Quality • Montana Renewable Energy Association	• 1.3 Public Agencies Collaboration • 4.5 Housing and Transportation Choices • 6.3 Climate Action	• Number of trainings hosted • Number of participants

Action 1.B.2. Advocate for Adoption of State-Wide Net Zero Energy Code

While the City of Bozeman cannot require anything more or less stringent than what the state of Montana adopts, it can advocate for state level policies and regulations that increase building efficiency.

The City will collaborate with other municipalities in the state to advocate for adoption of a statewide voluntary net zero energy stretch building code. Having a statewide stretch code would provide consistency across communities and ensure builders don't have to navigate multiple versions of stretch code requirements.

The City will advocate for adoption of any necessary health and safety measures (e.g., proper ventilation), incorporation of solar-ready and electric vehicle-ready construction requirements into state building and energy codes, and more stringent statewide energy and water efficiency regulations. The City will encourage manufacturers to disclose the carbon intensity of building materials.

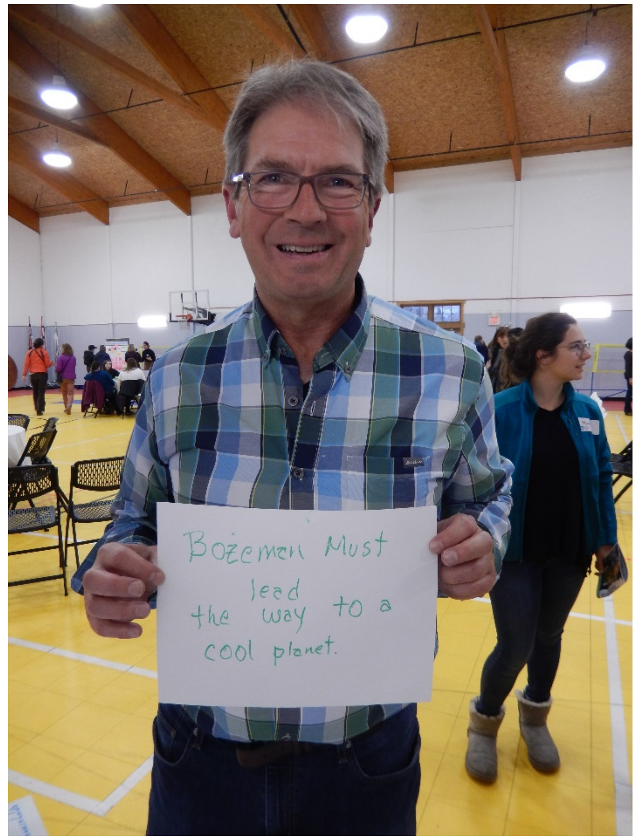
Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • Montana State University • Montana Weatherization Training Center • Southwest Montana Building Industry Association • Montana Code Collaborative • Montana Dept. of Environmental Quality • Montana Dept. of Labor & Industry	• 4.2 High Quality Urban Approach • 4.5 Housing and Transportation Choices • 6.3 Climate Action	• Establishment of a state-wide above-code building standard • Improved energy efficiency with each code adoption cycle

Action 1.B.3. Encourage High Performance Construction for All Publicly Funded Buildings

The City cannot mandate policies for non-municipal facilities that receive public funding, but it can encourage and/or incentivize high performance construction.

The City will partner with other agencies, such as the Bozeman School District and Montana State University, to identify opportunities and mechanisms to upgrade current facilities and construct new facilities to incorporate water efficiency and net zero energy standards. The City will encourage cost-effective high-performance construction for other publicly funded projects, such as affordable housing.

The City of Bozeman has direct control over the performance of municipal buildings and will adopt ambitious high-performance NZE building practices and maximize water efficiency for City-owned facilities where feasible (see Action 1.A.1).



Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Montana State University • Human Resources Development Council • Community Affordable Housing Advisory Board	• 4.2 High Quality Urban Approach • 4.5 Housing and Transportation Choices • 6.3 Climate Action	• Establishment of high performance building ranking criteria in workforce housing application • Number of above-code affordable housing units

Action 1.B.4. Analyze and Support Opportunities for District Energy

Heating, cooling, and water heating account for roughly half of global energy consumption (IEA, 2011). District heating energy systems pipe steam, hot water, or cold water from a centralized plant through a system of pipes to multiple buildings within a network and can be applied on the scale of neighborhoods or even a city. District heating is an integrated solution that can support energy efficiency, building electrification, climate resilience, renewable energy storage, support a green economy, and deep decarbonization of cities. District heating systems can also produce electricity locally. Renewable energy can provide high levels of affordable heat and cooling when incorporated into district energy systems through economies of scale and diversity of supply. District energy plus renewable energy can help communities achieve 100% renewable energy goals.

The City of Bozeman will develop a plan for City and State support of district energy through avenues such as district energy in energy efficiency building standards, state tax incentives, Combined Heat and Power (CHP) and renewable heat incentives, mapping of potential resources, land use planning, and permitting for heating and cooling infrastructure. The City will explore funding and collaboration models to study district heating in more detail, including Montana State University's advanced district energy case study (NREL, 2020), and coordinate with NorthWestern Energy and other community partners. The City will highlight replicable business models and build the business case around public-private partnerships.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Montana State University • Health Care Institutions • Development Community • Bozeman Public Schools	• 2.2 Infrastructure Investments • 4.3 Strategic Infrastructure Choices • 6.3 Climate Action	• Development of a plan to support district energy systems

Action 1.B.5. Offer a Voluntary Pathway & Incentives for Above-Code Construction

Building codes are the minimum standard for what is legally allowed to be built, yet the practice of building "above code" is relatively uncommon. As noted earlier in this chapter, the City of Bozeman is not permitted to require anything more stringent than the state-adopted code.

Creating a specific designation for superior-performing buildings, the City can encourage the development community to strive to achieve above-code construction outcomes. By setting the standards for the award designation, and creating a positive-sounding title for the achievement of the standards, a "Bozeman Net Zero Hero" building can be marketed by the builder as a model for others to follow.

The City has limited tools to incentivize above-code construction practices. Through a combination of Planned Unit Development point accumulation, fee waivers and other incentives, developers would be encouraged to make specific choices that help the City achieve its stated goals. Development code relaxations and trade-offs must be carefully considered to avoid undesirable outcomes to the built environment. Such a program would evaluate a number of factors – from neighborhood orientation to solar installation to net-zero construction – and award the appropriate amount of incentives for achieving specific benchmarks. Sequencing of a city incentive relative to the developer's stated deliverables will require further evaluation.

The City of Bozeman will create a recognition program that bestows a specific designation (e.g., "Bozeman Net Zero Hero") on buildings that meet net-zero or other objective standard of energy performance and market the program to the development and real estate community as a way of distinguishing superior performing buildings.

Further, the City will develop an incentive program for new construction (commercial and residential) that encourages all buildings to be designed and constructed to:

- Minimize energy and water use with ultra-low Energy Use Intensity (EUI) or Home Energy Score
- Maximize solar energy system electricity production to achieve Net Zero Energy (NZE) and electric vehicle ready
- Encourages passive design and optimizes neighborhood orientation
- Discourages the use of outdoor gas/electric heaters

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 3	• **City of Bozeman** • Montana State University • Montana Weatherization Training Center • Gallatin College • Southwest Montana Building Industry Association • Montana Code Collaborative • Montana Dept. of Environmental Quality	• 4.2 High Quality Urban Approach • 4.5 Housing and Transportation Choices • 6.3 Climate Action	• Establishment of a recognition program • Establishment of a voluntary above-code building standard • % of building permit applicants voluntarily participating

Solution C. Electrify Buildings

This solution emphasizes the conversion of heating equipment and cooking appliances in buildings from natural gas or propane to electricity. By shifting to electricity, it allows for energy needs to be met through renewable energy, thus reducing overall emissions from building energy consumption. This solution is vital to addressing building natural gas use and is an important element to achieve carbon neutrality by 2050.

According to the 2018 American Community Survey 5-year estimates for Bozeman, approximately 10,848 homes (59%) heat their properties with utility gas, 6,786 (37%) heat their homes with electricity, and only 382 homes (roughly 2%) use propane or fuel oil for heating fuel. While there are various appliances and equipment systems that could be converted to electricity, recommended opportunities to prioritize for conversion are propane or heating oil systems and all-electric new construction. Other system and appliance conversions should be considered on a case-by-case basis as not all conversions will be cost-effective. There is currently a higher fuel cost for electricity, yet electric heating is typically more efficient than natural gas.

As with building energy efficiency measures, conversion to or installation of building heating and cooking systems that use electricity rather than natural gas will require capital investment by the building owner or developer. A 2018 study found that electrification in residential homes for retrofits was not cost-effective. The exception to this conclusion is that "electrification is cost-effective for customers switching away from propane or heating oil, for those gas customers who would otherwise need to replace both a furnace and air conditioner simultaneously, for customers who bundle rooftop solar with electrification, and for most new home construction" (Billimoria, Guccione, Henchen, & Louis-Prescott, 2018). For this reason and to provide the greatest GHG emissions reductions, robust implementation of this solution is recommended after 2030 when the community reaches its 100% renewable electricity goal. At that time, it is expected that the economics will be more favorable as a 20 to 38% price reduction for air-source heat pumps and 42 to 48% price reduction for heat pump water heaters is expected by 2050 (National Renewable Energy Labratory, 2017). Again, we expect similar economics to apply to commercial buildings, but due to the variation in building load, the economics in commercial spaces is more difficult to model.

Equity

From an equity perspective, some households and businesses may not be able to afford the capital investment required for equipment and building retrofits. Furthermore, the last remaining users of natural gas, which may include older homes, commercial buildings, and lower-income households, should not be left with the financial burden of electrification. This means that careful attention must be paid so that they are not stuck with increasing natural gas rates as the number of natural gas customers decline.

Human Health & Well-Being

Health and well-being considerations associated with electrification of buildings include improved safety with the reduction in pilot lights needed for natural gas equipment. As building energy consumption shifts to electricity provided by renewable sources, air quality improvements will likely ensue. Residential electrification may increase demand for wood burning stove and compromise local air quality during the winter.

City Assets

As with improving efficiency and implementing Net Zero Energy practices at City of Bozeman facilities, there are opportunities for the City to lead by example and test out and pursue beneficial electrification projects at existing and future City facilities, especially those facilities with high natural gas or propane consumption.

This solution primarily addresses Bozeman's greenhouse gas mitigation goals, while also providing some supporting resiliency benefits.

Emissions Reduction Benefits

This is a secondary mitigation solution to achieve the 2050 goal.

Emission sources impacted include:

- Electricity
- Natural Gas

Resiliency Benefits

- Conserve natural resources
- Protect human health

Related Solutions

- Solution B. Achieve Net Zero Energy New Construction
- Solution D. Increase Utility Clean Energy Mix
- Solution F. Increase Community-Based Distributed Renewable Energy Generation

Action 1.C.1. Advance Electrification Upgrades and Conversion Projects for City Facilities

The City has direct influence over is its own municipal facilities. Reducing and eliminating fossil fuel use in City-owned facilities is one way the City can lead by example with the Bozeman Climate Plan.

The City will convert select City facilities and equipment to all or mostly electric, with the goal of full electric conversion as technologies advance. The City will also require all-electric new construction for new City facilities where feasible.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Rocky Mountain Institute • Montana State University • NorthWestern Energy	• 4.3 Strategic Infrastructure Choices • 4.5 Housing and Transportation Choices • 6.3 Climate Action	• Number of electrification conversion projects • % reduction in natural gas use

Action 1.C.2. Include an Electrification Component for Above-Code Construction

Building on the goals in Action 1.B.2: Offer a Voluntary Pathway & Incentives for Above-Code Construction, the City will conduct a feasibility study to explore the implication, costs, and benefits of electrification for different types of buildings in Bozeman, partner with NorthWestern Energy to encourage all-electric new construction, and advocate for all-electric new construction requirements in the stretch code and future building codes. The City will evaluate its engineering standards and NorthWestern Energy connection standards to determine if new developments are discouraged or prohibited from installing electric only service, including consideration for ground source energy or district energy technology requirements.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 3	• **City of Bozeman** • NorthWestern Energy • Montana State University • Montana Weatherization Training Center • Montana Codes Collaborative • Montana Dept. of Environmental Quality • Montana Dept. of Labor & Industry • Southwest Montana Building Industry Association • Human Resources Development Council	• 4.2 High Quality Urban Approach • 4.5 Housing and Transportation Choices • 6.3 Climate Action	• Adoption of an all-electric new construction code • % increase of homes with electric heating and appliances

Action 1.C.3. Support Outreach and Incentives for Electric Appliances and Equipment

While the City of Bozeman cannot require all-electric appliances or equipment in buildings, it has the ability to educate the community on the health and safety benefits of and encourage installation of all-electric appliances and equipment in homes and businesses.

The City will develop educational materials to encourage customers to select electric equipment or appliances where alternatives to gas exist and will incentivize private conversion of natural gas appliances to electric through avenues such as a bulk purchasing program. The City will partner with homeowners' associations to explore opportunities to electrify equipment and appliances across neighborhoods. The City will establish bulk purchase and retrofit programs to help natural gas users transition to all-electric appliances and equipment. With a robust educational program in place, the City will be positioned to advocate for appliance electrification within the state building code and explore other available pathways to accelerate the conversion to all-electric appliances and equipment.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 3	• **City of Bozeman** • Montana State University • Montana Weatherization Training Center • Human Resources Development Council • Community Affordable Housing Advisory Board	• 1.2 Community Engagement • 4.5 Housing and Transportation Choices • 6.3 Climate Action	• % Increase of buildings with electric heating and appliances

Increasing Resiliency to Climate Hazards

The following opportunities and considerations summarize how the solutions in this chapter can help improve resiliency to future climate hazards.

Opportunities	Considerations
Extreme Heat	
• Reducing energy demands during hot days can reduce risk of energy shortages. • Efficient buildings cost less to cool and can reduce energy cost burden.	• Electrification could reduce system diversity (by eliminating natural gas), increasing vulnerability if system is overloaded during a period of high demand, thus amplifying the need for energy storage and microgrids.
Flooding	
• Reduced risk of damage to natural gas infrastructure if building stock was fully electrified. • Net-zero energy buildings may avoid service interruptions if powered by onsite renewable energy.	• Siting of new construction away from flood prone areas is critical.
Drought & Mountain Snowpack	
• High efficiency building design often includes water conservation measures in addition to energy efficiency.	• Electrifying buildings will increase demand for electricity, which is currently associated with water consumption.
Wildfire	
• Efficient buildings are better sealed and better ventilated, yielding better air quality in the event of a wildfire.	• Electrified buildings could be more vulnerable to power disruptions if electricity infrastructure is damaged. • Tightly sealed buildings with poor ventilation could worsen indoor air quality.
Winter Storms	
• Efficient buildings cost less to heat and can reduce energy-burden for low-income customers.	• Electrifying existing building systems could increase utility costs depending on timing and utility rate structures. • Electrification could reduce system diversity (by eliminating natural gas), increasing vulnerability if system is overloaded during a period of high demand. • Currently, there may be limitations to cost-effective electric heating.

Focus Area 2: Responsible & Reliable CLEAN ENERGY SUPPLY

NorthWestern Energy is the primary energy utility serving the Bozeman community, providing electricity and natural gas to residential, commercial, and some industrial customers. Some community members also heat or power their homes and facilities through other energy sources or technology, including but not limited to propane, wood, solar, and geothermal. As noted in Chapter 3, more than half of Bozeman's GHG emissions come from energy use in buildings. In 2018 emissions from electricity use alone accounted for 29% of community GHG emissions. The solutions in this focus area complement the solutions in Focus Area 1: Healthy, Adaptive & Efficient buildings that emphasize energy efficiency and conversion to electric equipment and solutions. They focus on lowering energy-related emissions by addressing the energy supplied from the grid to utility customers, as well as expanding distributed generation opportunities.

RELATED PLANS & STUDIES

- Bozeman Community Plan
- NorthWestern Energy Electric Supply Resource Procurement Plan
- Climate Vulnerability Assessment and Resilience Strategy
- Community Greenhouse Gas Emissions Report
- Municipal Greenhouse Gas Emissions

In addition to emphasizing a shift to cleaner energy sources over time, these solutions advance other important topics and issues such as energy efficiency, access and affordability, electric grid stability and reliability, a just and equitable transition of the traditional energy economy, and environmental and air quality.

These solutions are focused on the generation of the required renewable electricity to serve the community's energy needs. To meet Bozeman's climate goals, the community will need to:

- **Solution D. Increase Utility Clean Energy Mix**
  Strongly encourage NorthWestern Energy to meet Bozeman's goal of 100% net clean electricity by 2030.
- **Solution E. Develop and Promote Utility Green Power Programs**
  Collaborate with NorthWestern Energy and stakeholders to introduce a green tariff. This is a key strategy to meet the City's aggressive short-term goals.
- **Solution F. Increase Community-Based Distributed Renewable Energy Generation**
  Work across the community to increase the adoption of distributed renewable energy on public and private properties.

As illustrated in Tables 6 and 7, Bozeman will need 503 million kWh of clean electricity in 2025, 519 million kWh in 2030, and 666 million kWh in 2050 to meet the community's greenhouse gas emissions mitigation goals. These electricity consumption projections include the increased electricity use from projected growth, the anticipated electricity use reduction from implementing efficiency (Solutions A and B), as well as the increased electricity use from implementing building and vehicle electrification (Solutions C and K).

The projected electricity use will come from a combination of utility-scale clean energy generation (Solutions D and E) as well as local distributed generation (Solution F). There are two pathways for renewable energy generation presented in the following subsections. The first represents the no change pathway where NorthWestern Energy does not add any additional clean energy generation, and the second represents the pathway where NorthWestern Energy uses additional clean energy generation to meet and exceed their stated carbon intensity reduction goals and reduced total emissions.

In each case, utility-scale green power programs (Solution E) are used to procure the necessary renewable energy to allow the City to meet their stated goals. The amount of distributed generation (Solution F) does not change between pathways and is set to show consistent annual increase in generation capacity to meet Bozeman's target in 2050. For more information on calculation methodology, see Appendix A.

Photo: Onsite Energy

No Additional Utility Renewable Energy Generation

This analysis illustrates the pathway if NorthWestern Energy does not add any additional renewable energy generation (pathway A).

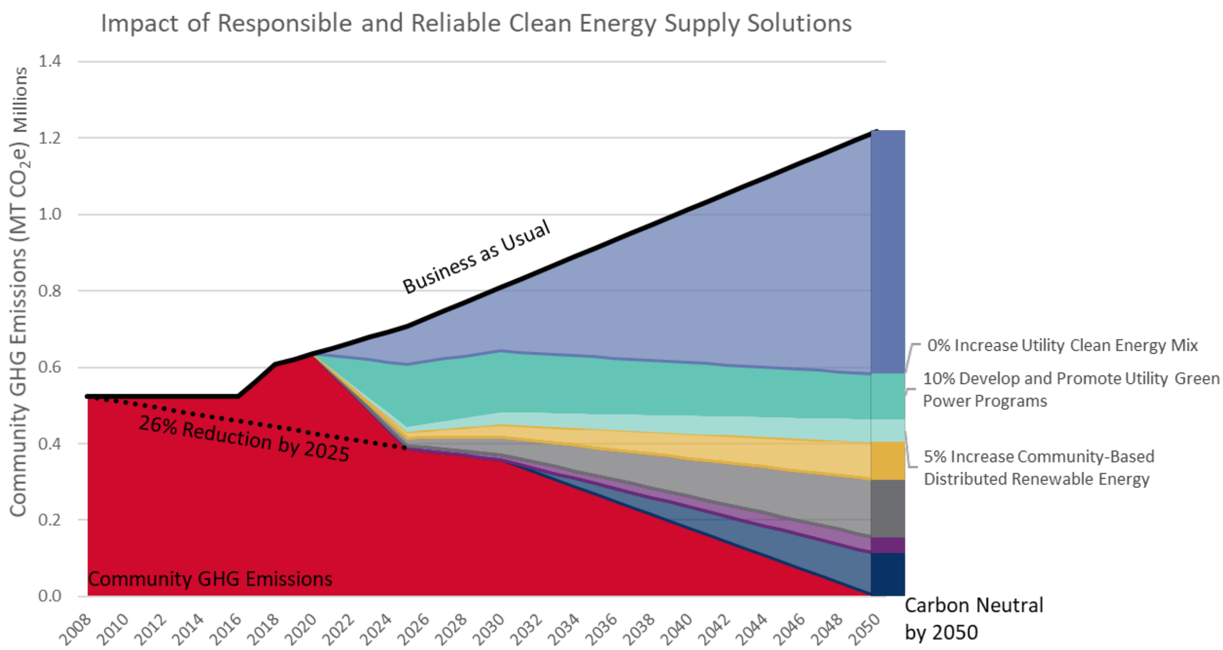


Figure 16. Projected clean energy GHG reductions by solution (pathway A)

Table 6. Projected clean energy supply solution mitigation targets (pathway A)

Solution	*Metric (unit)*	*Target* 2025 *Paris Accord*	*Target* 2030 *100% Net Clean Electricity*	*Target* 2050 *Carbon Neutral*
Total Projected Electricity Consumption	**Electricity Use (kWh)**	**503 million**	**519 million**	**666 million**
Baseline Clean Energy	Clean Electricity (kWh)	281 million	281 million	281 million
Solution D. Increase Utility Clean Energy Mix	Additional Clean Electricity (kWh)	0	0	0
Solution E. Develop and Promote Utility Green Power	Additional Clean Electricity (kWh)	204 million	198 million	256 million
Solution F. Increase Community-Based Distributed Renewable Energy Generation	Additional Clean Electricity (kWh)	16 million	40 million	130 million

Note: Columns may not add up to the total due to rounding.

Additional Utility Renewable Energy Generation

This analysis illustrates the pathway where NorthWestern Energy uses additional clean energy generation to meet and exceed their stated carbon intensity reduction goals[8] and reduces total emissions (pathway B).

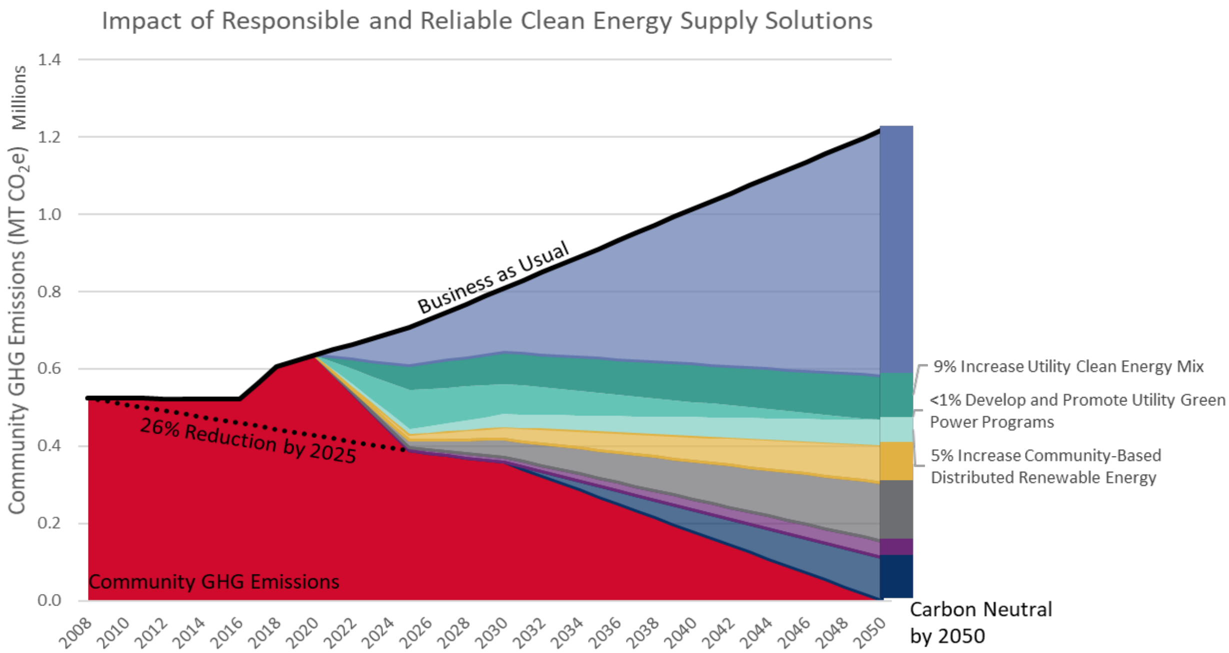


Figure 17. Projected clean energy GHG reductions by solution (pathway B)

Table 7. Projected clean energy supply solution mitigation targets (pathway B)

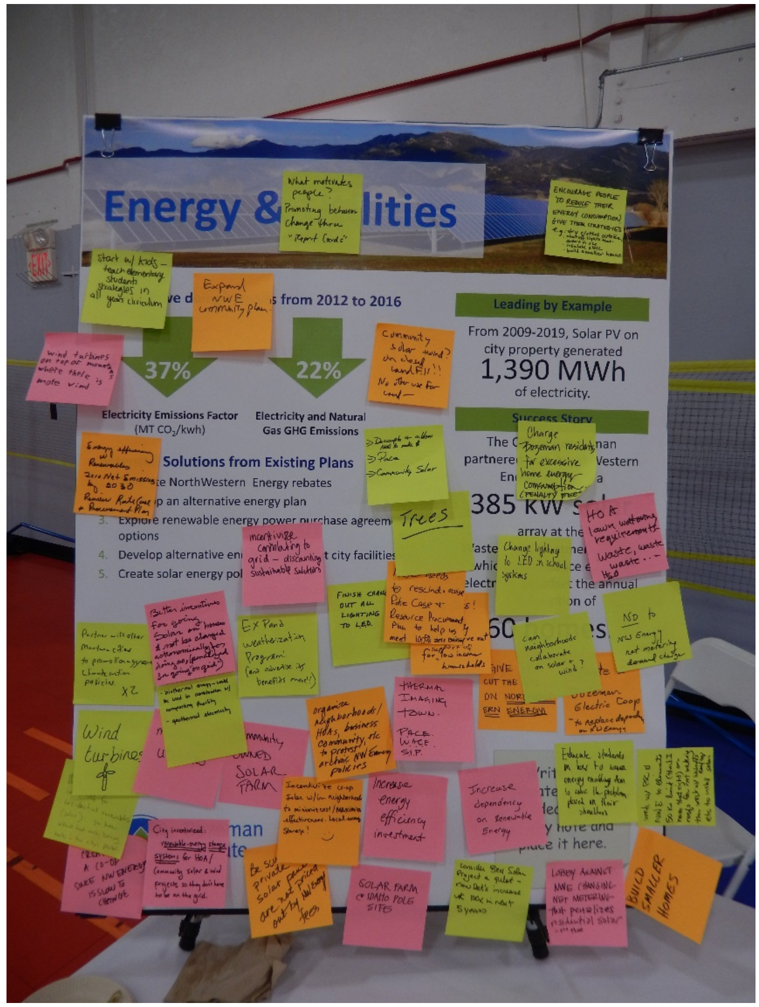
Solution	*Metric (unit)*	*Target* 2025 *Paris Accord*	*Target* 2030 *100% Net Clean Electricity*	*Target* 2050 *Carbon Neutral*
Total Projected Electricity Consumption	**Electricity Use (kWh)**	**503 million**	**519 million**	**666 million**
Baseline Clean Energy	Clean Electricity (kWh)	281 million	281 million	281 million
Solution D. Increase Utility Clean Energy Mix	Additional Clean Electricity (kWh)	80 million	100 million	306 million
Solution E. Develop and Promote Utility Green Power	Additional Clean Electricity (kWh)	124 million	98 million	0
Solution F. Increase Community-Based Distributed Renewable Energy Generation	Additional Clean Electricity (kWh)	16 million	40 million	80 million

Note: Columns may not add up to the total due to rounding.

[8] While carbon intensity is an important component of the 'math' of measuring emissions, it should not be relied upon - in and of itself - as an indication of progress; therefore, any plan that uses carbon intensity as a sole goal or metric should be taken with a grain of salt.

Solution D. Increase Utility Clean Energy Mix

This solution emphasizes the role of utilities in providing energy to customers and communities. Because of their large reach and economies of scale, utility-scale solutions are often some of the most transformational and cost-effective opportunities for communities to achieve their climate goals. A utility's resource mix typically includes an assortment of energy generation sources that are balanced to meet customer demand.



In 2018, 61% of the energy owned by NorthWestern Energy for Montana came from carbon-free sources, which includes hydropower (46%), wind (15%), and solar (<1%) (NorthWestern Energy, 2019). In 2019, NorthWestern Energy voluntarily committed to reducing the carbon intensity of its electric portfolio for Montana by 90% by 2045, from a 2010 baseline (NorthWestern Energy). If the goal is achieved with new renewable energy generation, this transition has the potential to provide significant impact and momentum, but will not advance quickly enough to achieve the City's short-term goals and my not be sufficient to achieve carbon neutrality by 2050.

Because renewable energy is not always available when customer energy demand is highest (dark, cold, still, winter evenings), solar and wind cannot be the sole solution. Clean energy portfolios should include cost-effective storage to ensure that renewable energy generation can be stored for times when generation is not available. In addition, well-designed demand response programs and time-of-use rate design can reduce peak energy demand. Customers can help reduce energy demand at peak times in evenings (usually about 4pm to 8pm). For example, delaying dishwashers until late evening and moving other non-essential energy uses to off-peak hours can reduce the peak energy need and allow more reliance on renewables rather than fossil fuel capacity resources.

Public-private partnerships have an important role to play in this solution, and innovation and the willingness of current and future partners to work together across sectors will be necessary to advance renewable energy adoption.

Equity considerations associated with significant shifts in the utility's energy generation portfolio are vast. A rapid transition could potentially cause a rate shock and increase rates, disproportionately burdening low-income households. Over the long-term, however, costs may stabilize or decrease, ultimately reducing the energy cost burden for lower-income households. More analysis is needed to accurately quantify these costs. Furthermore, the workforce associated with non-renewable energy production may experience declining employment opportunities, meaning that careful planning to provide a just transition for that workforce is essential.

Health and well-being opportunities associated with this solution include improved air quality with the reduction in burning of fossil fuels. Over time, non-renewable energy facilities may need to be decommissioned, leading to additional environmental and human health benefits, as well as employment opportunities.

Whether advancing new clean energy resources or other related efforts focused on energy efficiency and electrification, the City will advance a triple bottom line approach based on social, environmental, and economic benefits to inform utility partnership efforts.

This solution primarily addresses Bozeman's greenhouse gas mitigation goals, while also providing some supporting resiliency benefits.

Emissions Reduction Benefits

This is a primary mitigation solution to achieve the 2025, 2030, and 2050 goals.

Emission sources impacted include:

- Electricity

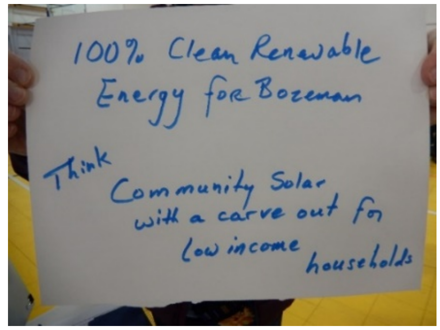
Resiliency Benefits

- Conserve natural resources
- Strengthen infrastructure to natural disaster
- Protect human health

Related Solutions

- Solution B. Achieve Net Zero Energy New Construction
- Solution C. Electrify Buildings
- Solution E. Develop and Promote Utility Green Power Programs
- Solution K. Decrease Direct Vehicle Emissions

Action 2.D.1. Complete a 100% Net Clean Electricity Community Feasibility Study



With approximately 60% of Bozeman's grid energy sourced from clean resources, Bozeman is positioned to achieve 100% net clean energy by 2030. To help map the pathway, the City will seek interlocal partnerships with interested jurisdictions to complete an independent economic analysis focused on the rate implications and environmental benefits of pursuing utility-scale clean energy solutions. The scope of the study will include an assessment of the potential cost of available policy mechanisms, including 1) the creation of a collaborative community-wide renewable energy program, modeled after Utah's Community Renewable Energy Act (HB 411), allowing communities to choose to be supplied with 100% net clean electricity with new renewable energy sourced through the utility and 2) a Community Choice Aggregation (CCA) program, which allows local governments to procure power from an alternative supplier while the utility still provides transmission, distribution, and billing services in regulated electric markets, like Montana. Both options would most likely require state enabling legislation.

Under a community-wide renewable energy program, NorthWestern Energy and interested communities would collaborate with stakeholders to help drive the path to 100% clean energy by taking advantage of low-cost renewable energy to meet growing customer demand. Under the unique Utah model, HB 411 directs the Public Service Commission to implement a 100% renewable plan for communities and their customers who opt into the program. It also provides legislative guidance for customer participation and key opt-out provisions for customers who do not wish to participate.

Under Community Choice Aggregation (CCA), local governmental entities (or a contracted third party) procure electricity on behalf of customers within a city, county, or group of jurisdictions. The CCA is responsible for choosing and procuring the mix of resources, as well as system reliability in regulated markets. Eight states have passed state CCA-enabling legislation and each state has at least one CCA that was formed when a local government body voted to aggregate its retail electricity or passed a public referendum. Most CCAs use an opt-out structure where customers are automatically enrolled unless they choose to switch to the basic utility service. CCA voluntary green power program structures vary, but many have "opt-up" products for their customers to select 100% renewable electricity (NREL, 2019).

As part of the City's due diligence to assess cost, long-term savings, and community equity considerations, the economic feasibility and policy options study will help inform the appropriate path forward to effectively meet Bozeman's clean energy goals.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Montana Cities and Counties • NorthWestern Energy	• 1.3 Public Agencies Collaboration • 6.3 Climate Action	• % increase of utility-scale renewable energy • % decrease in delivered electricity CO_2 emissions factor • Decrease in total annual utility-scale emissions

Action 2.D.2. Collaborate and Innovate Utility-Scale Solutions with NorthWestern Energy

With a strong track record of collaboration with NorthWestern Energy on local efforts such as the Bozeman Energy Project and the Bozeman Solar Project at the City's Water Reclamation Facility, the City will seek to formalize the existing long-term partnership with NorthWestern Energy with a Memorandum of Understanding (MOU). The MOU will provide a framework of joint actions and implementation timeline for the City and NorthWestern Energy to seek mutually beneficial programs and policies that will address the City's clean energy goals and the utilities' stated electric capacity shortfall.

Examples of joint programs contained within the MOU may include efforts such as the development of green power purchasing options for the City and community members, formal engagement in the development of the Electricity Supply Resource Plan, community renewable energy projects, Demand Side Management (DSM) energy efficiency programs, Advanced Metering Infrastructure (AMI) deployment, local planning for large new commercial loads, electrification coordination, and possible support for batteries and pumped storage. Through the MOU, the City will urge greater transparency from NorthWestern Energy on their reported emissions factors and carbon goal and urge the utility to establish an absolute emissions reduction goal. The City will seek input from stakeholders and outside subject matter experts on the development of this MOU and community members will have an opportunity to comment on any final agreement. The City will continue to grow interlocal partnerships with Montana cities and counties to advance common clean energy and climate goals contained in the MOU.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • **NorthWestern Energy** • Energy and environmental non-profits • Montana Dept. of Environmental Quality • Montana League of Cities & Towns	• 1.3 Public Agencies Collaboration • 6.3 Climate Action	• % increase of utility-scale renewable energy • % decrease in delivered electricity CO_2 emissions factor • Decrease in total annual utility-scale emissions

Action 2.D.3. Support Policies to Expand Renewable Energy and Just Transition Initiatives

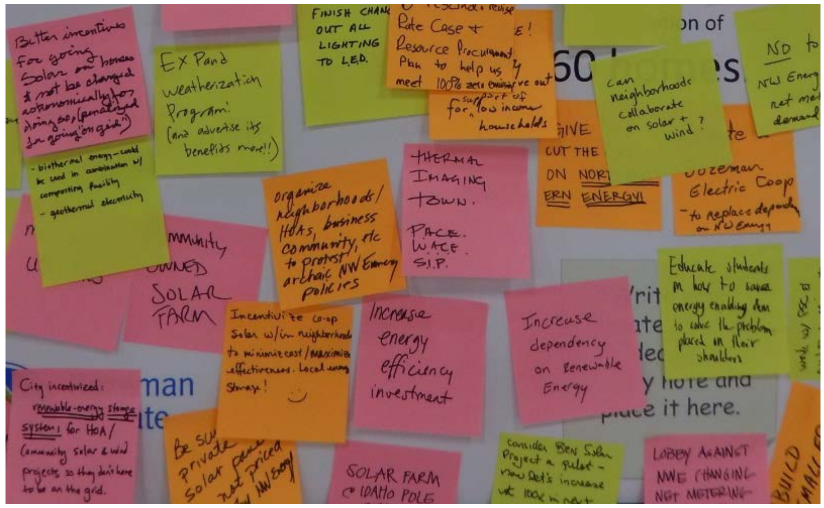
The City of Bozeman can play a proactive role in shaping Montana's energy future. Cities have the ability to build support within our existing networks or form new partnerships to help advance our climate goals. While the City has supported a clean energy agenda in the Montana Legislature for over a decade, for the first time the City's 2021 Legislative Agenda & Priorities includes clean energy among the top three priorities.

Offering testimony at the Montana Legislature and the Public Service Commission can help advance critical conversations across the state. Examples of policies aimed at advancing energy efficiency and clean energy may include time-of-use rate design, energy storage policies, utility business model reform, Property Assessed Clean Energy, oppose future efforts to acquire additional shares of Colstrip, or advancing state-wide research and development focused on the potential for geothermal, hydrogen resources, energy storage, or other disruptive technologies.

The decline of coal in Montana (U.S. Energy Information Administration, 2020, Montana State Profile and Energy Estimates) and closure of Colstrip Units 1 and 2 have helped reduce greenhouse gas emissions in the electricity sector, but more significant reductions are needed to achieve Bozeman's long-term climate goals. These emissions reductions will improve health, reduce the impacts and associated costs of climate change, and offer new economic opportunities. At the same time, these changes will cause significant economic disruption for workers and communities closely tied to fossil fuels in Montana.

A just transition focuses on policies that enable greater fairness for and input from workers and communities that are negatively affected by the transition away from fossil fuels, especially low-income communities that may have dealt with environmental injustice by living near polluting fossil fuel infrastructure. Examples of programs that could address just transition goals include economic development programs, Brownfield or Superfund programs, loan programs, clean energy tax credits, workforce development, and social safety net programs that benefit workers and communities impacted by the closure of coal mines, coal power plants, oil refineries, and associated industries. Robust cleanup efforts, such as dewatering Colstrip ash pond and implementing a "high and dry" secure storage of toxic material to protect ground water for future generations supports local environmental justice efforts. Bozeman will help advance rural priorities and just transition initiatives within the City's legislative priorities. Just transition funding should be qualified to guarantee that the funding goes toward workforce development, retraining programs, and impacted tribal entities. Locally, Bozeman may identify opportunities to support clean energy workforce development and job re-training programs through Gallatin College.

The City will proactively plan to advocate for clean energy and just transition initiatives in the Montana Legislature and Public Service Commission.



The City of Bozeman recognizes that the Intergovernmental Panel on Climate Change (IPCC) considers carbon pricing "necessary" to reach carbon reduction goals. While national carbon pricing legislation enacted by Congress would be the most efficient way to implement this market-based approach, there are also efforts underway on state and regional levels as well. The City can become a leader by engaging in discussions about creative bipartisan market-based policies that effectively reduce greenhouse gas emissions by spurring low/no carbon innovation, and work in a way that sustains local economic development and offers other benefits to Bozeman and rural Montana. A revenue neutral carbon fee and dividend, carbon adjustments and regulatory reform, including eliminating fossil fuel subsidies, offer promising market-based approaches to combatting climate change. The City of Bozeman will support state and federal legislation that embodies the principles of effectiveness and equity (an example is H.R. 763, the Energy Innovation and Carbon Dividend Act) in addressing greenhouse gas emissions reduction

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • NorthWestern Energy • Montana Legislature • Public Service Commission • Montana Consumer Counsel • Energy and environment non-profits • Bozeman School District • Gallatin College	• 1.3 Public Agencies Collaboration • 6.3 Climate Action	• % increase of utility-scale renewable energy • % decrease in delivered electricity CO_2 emissions factor • Decrease in total annual utility-scale emissions

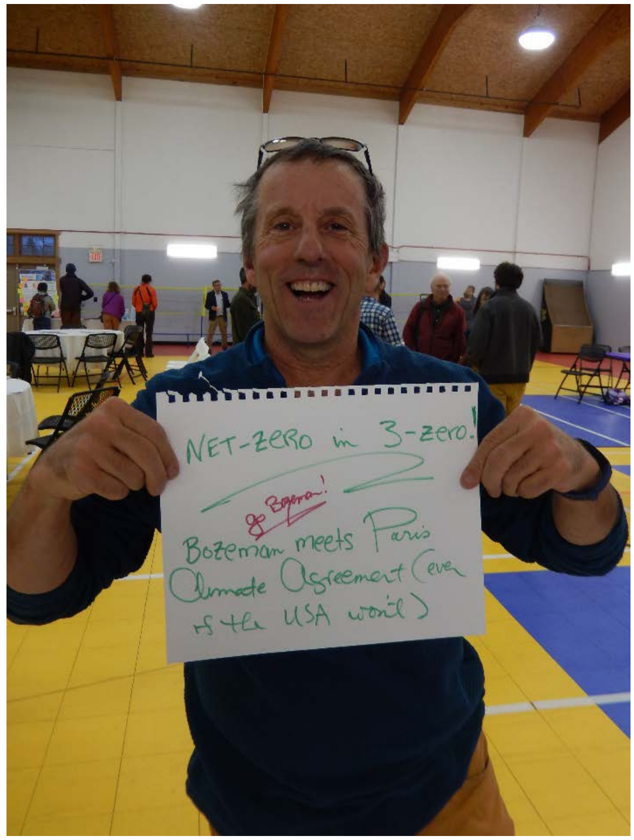
2.D.4. Encourage Philosophical Shift for Utility Provider

Bozeman's ability to meet any of its stated goals is dependent upon a rapid transition to renewable energy sources by the utility. There is no current indication that NorthWestern Energy intends to make such a transition to meet Bozeman's goals. Other utilities have recognized that their corporate mission must recognize a responsibility for future generations—and they have set aggressive goals to transition away from fossil fuels. Absent a similar governing philosophy, NorthWestern Energy will not be able to serve the specific desires of their customer base. Representing the majority of NorthWestern Energy's Montana customer base, Bozeman, Gallatin County, Missoula (city and county), Helena and other partners should make direct appeals to the Board of Directors and major shareholders of the utility to meet the needs of their customers by adopting similar transition goals comparable to other enlightened utility providers.

The City will work with partner communities to encourage NorthWestern Energy's Board of Directors and shareholders to adopt a corporate philosophy that recognizes an urgent need to shift to renewable energy sources in order to safeguard the long-term health and well-being of their customers.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • **Montana Cities and Counties** • NorthWestern Energy	• 1.3 Public Agencies Collaboration • 6.3 Climate Action	• % increase of utility-scale renewable energy • % decrease in delivered electricity CO_2 emissions factor • Decrease in total annual utility-scale emissions

Solution E. Develop and Promote Utility Green Power Programs



This solution focuses on institutional, business, and community voluntary participation in clean energy programs provided through the energy utility. Two common green power purchase program models are renewable energy certificates (RECs) and green tariffs.

Under a REC program, customers opt-in to paying a slight premium on their electric bill to support the development of renewable energy generation. In exchange, customers receive the RECs associated with electrical generation, which may be located inside or outside the utility's service area (unbundled RECs). If purchasing RECs, some buyers may want to consider buying RECs from areas with low renewable capacity or where the grid is dirtiest. Some disadvantages to RECs are that there may be potential for double counting of the clean energy attributes of renewable energy and RECs do not necessarily contribute to "additionality" or an increase in the amount of renewable energy on the local grid. NorthWestern Energy's green power purchase program is called E+ Green. It is available to the utility's electric customers. The E+ Green program currently provides RECs from the Bonneville Environmental Foundation, which has resources in Wyoming, Montana, and the Northwest. Energy purchased through the E+ Green program, or other similar REC programs, cannot be used to help Bozeman reach its' emissions reduction goals under the Global Protocol for Community-scale Greenhouse Gas Emissions.

Conversely, a green tariff is an increasingly popular optional pricing structure offered by a utility through a contractual arrangement that allows customers to purchase both the electricity from a large-scale renewable energy project and the associated RECs or environmental attributes (bundled RECs). For Bozeman, the key elements of green tariff program are the addition of clean electrons to the utility's grid and the associated RECs. Both are necessary to help reduce Bozeman's greenhouse gas emissions.

With respect to equity, this solution may be unattainable to lower-income or energy cost-burdened community members. Some entities provide scholarships or sponsorships to increase low-income access to green power purchase programs. Similarly, green tariff programs may only be available to large commercial, industrial, or municipal customers, not the entire community.

As with the solution to increase utility clean energy mix, increased investment in renewable energy technologies will likely lead to health and air quality benefits as fossil fuel consumption declines and facilities are decommissioned over time.

The City of Bozeman could advance this solution by opting into a green tariff for all facilities and electricity accounts.

This solution primarily addresses Bozeman's greenhouse gas mitigation goals, while also providing some supporting resiliency benefits.

Emissions Reduction Benefits

This is a primary mitigation solution to achieve the 2025, 2030 goals, and a supporting solution to achieve the 2050 goal.

Emission sources impacted include:

- Electricity

Resiliency Benefits

- Conserve natural resources
- Protect human health

Related Solutions

- Solution B. Achieve Net Zero Energy New Construction
- Solution D. Increase Utility Clean Energy Mix

Action 2.E.1. Advance Green Tariff Program Development and Participation

Increasing community demand for clean energy products will result in greater investment in clean energy across Montana. Developing and participating in a green tariff program by the City of Bozeman and other institutions, businesses, or residents will be necessary to ensure that Bozeman meets our 2025 emissions reduction goal.

A green tariff can be designed and branded in a variety of ways. The following green tariff programs represent cost-competitive examples that result in customer savings over longer-term contract periods:

- Puget Sound Energy's Green Direct – For commercial customers
- Xcel's Renewable*Connect - Colorado – For commercial customers
- Florida Power & Light's Solar Together - For commercial and residential customers with low-income participation option

NorthWestern Energy has agreed to develop a green tariff through a Green Power Stakeholder Advisory Committee, which includes City of Bozeman representation. The City of Bozeman, Missoula County, the City of Missoula, and the City of Helena are jointly seeking a third party consultant to advance the local governments' 100% net clean electricity goals and economic interests, including consideration for a residential program and low-income option to address equity. Bozeman's key principles for a green tariff product include: expanded choice in procuring renewable energy; cost competitiveness between the standard retail rate and the renewable energy product offering; and access to projects that are new and help drive emissions reduction beyond business as usual in NorthWestern Energy's Montana service territory.

Bozeman will continue to work with partner communities and NorthWestern Energy to develop the green tariff in 2021 and potentially support filing of a green tariff with the Montana Public Service Commission in 2022. Bozeman will continue to engage with stakeholders, institutional partners, and community members throughout the process. If successful, Bozeman will seek to procure renewable energy for city operations and actively recruit institutions, businesses, and residents to participate, as appropriate.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • **NorthWestern Energy** • **Montana Cities** • Public Service Commission • Montana Consumer Counsel • Energy & Enviro. Advocates • Business Leaders • Montana State University & Bozeman School District	• 1.3 Public Agencies Collaboration • 6.3 Climate Action	• Establishment of a voluntary Montana green tariff • Total kWh of clean energy procured for Bozeman through the green tariff

Solution F. Increase Community-Based Distributed Renewable Energy Generation

This solution supplements utility-scale clean energy generation with more localized community investments. It encourages installation of distributed renewable energy technologies, such as on-site solar, community solar gardens, methane capture, and small wind and hydro generation.

Investment in local renewable energy showcases local commitment to sustainability and climate goals. For example, from 2009 to 2019, solar photovoltaic installations on City property generated 1,390 megawatt hours of electricity.

While typically more expensive than utility-scale projects, there are myriad resources and incentives to make distributed renewable energy projects economically viable. Plus, when linked to energy storage options like batteries, microgrid technologies, and district energy, distributed renewable energy increases opportunities for resiliency in the event of a power outage or grid failure.

NorthWestern Energy customers who install solar panels, small wind, or hydro technology on their properties can request net metering, which provides a utility bill credit for any surplus energy generated. Self-generating systems of up to 50 kW in size are allowed to net meter under current Montana law.

Equity

Equity considerations associated with distributed renewable energy include access/availability to community members from various geographic and socio-economic positions. For example, some locations of the community are not well-positioned to take advantage of solar resources due to building orientation or tree coverage/other obstacles to solar access. Similarly, lower-income community members may not be able to afford investments into on-site improvements, and renters may be unable to make permanent improvements to structures. Solutions such as community solar gardens can help improve access and participation.

Human Health & Well-Being

Like the other solutions in this focus area, increased investment in renewable energy technologies will likely lead to health and air quality benefits as fossil fuel consumption declines and facilities are decommissioned over time.

City Assets

There are many opportunities for the City of Bozeman to continue to advance this solution. This includes planning for, financing, and implementing rooftop solar on municipal buildings and parking structures, making land available for community solar projects, and continuing to investigate opportunities for methane capture, small wind, and hydro generation projects at suitable City facilities.

This solution primarily addresses Bozeman's greenhouse gas mitigation goals, while also providing some supporting resiliency benefits.

Emissions Reduction Benefits

This is a supporting mitigation solution to achieve the 2025, 2030, and 2050 mitigation goals.

Emission sources impacted include:

- Electricity

Resiliency Benefits

- Conserve natural resources
- Strengthen infrastructure to natural disaster
- Mitigate property and economic losses
- Protect human health

Related Solutions

- Solution B. Achieve Net Zero Energy New Construction
- Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards

Action 2.F.1. Plan and Install Renewable Energy Projects for City Facilities

The City will continue to lead and demonstrate the value of local renewable energy within municipal operations by setting a target of achieving 100% net clean electricity consumption by 2025. The City will identify, plan, and invest in municipal solar PV, or other suitable renewable energy projects, such as district energy, geothermal and Combined Heat and Power. To build resiliency, the City will plan and invest in cost-effective energy storage projects and microgrids to enhance reliability of renewable energy systems at strategic locations.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • NorthWestern Energy	• 4.3 Strategic Infrastructure Choices • 6.3 Climate Action	• % Increase of distributed renewable energy for City facilities

Action 2.F.2. Streamline Solar Permitting and Adopt Solar-Ready Code Provisions

The City will continue to simplify and improve the permitting process for on-site renewable energy projects and begin tracking system installations to help estimate our communities' local generation potential.

Renewable energy options will be included in a voluntary above-code incentive program and the City will partner with developers and homeowner's associations to design model covenants that encourage more permissive and supportive frameworks for installing rooftop solar PV.

Further, with the adoption of the 2018 IECC and the solar-ready appendix, the City will evaluate the feasibility of adopting the appendix as a part of the local building code. This would help ensure that all new buildings are prepared to integrate cost-effective solar PV installations at a later date.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • Solar Installers • Southwest Montana Building Industry Association • Montana Code Collaborative • Montana Dept. of Environmental Quality • Montana Renewable Energy Association • NorthWestern Energy	• 2.1 Business Growth • 4.2 High-Quality Urban Approach • 6.3 Climate Action	• Adoption of the 2018 IECC solar ready appendix • Achieve SolSmart designation for solar permitting

Action 2.F.3. Advance Distributed Solar Policies

Building on the success of the Bozeman Solar Project, the City will evaluate all future options for this project with NorthWestern Energy, including co-investing to expand the project and develop a new tariff for utility-owned community solar. The City will partner to study a variety of public and private opportunity areas for community solar, evaluate rate models, and potential equity solutions to make community solar more accessible to all Bozeman residents.

In addition, the City will collaborate with renewable energy stakeholders to consider utility business model reform policies designed to remove the inherent disincentive for utilities to promote energy efficiency and distributed renewable energy generation. Given the limitations within Montana law to pursue local energy choice, the City may pursue state legislation to allow community solar, net energy metering aggregation, and net energy metering over 50 kW.

The City will continue to build the partnerships with Montana State University to test and learn from pilot renewable energy projects, including new technologies, financing models, community microgrids, and energy storage (e.g., off-duty school buses).

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • **NorthWestern Energy** • Montana Legislature • Public Service Commission • Montana Consumer Counsel • Energy and Environment Non-profits • Montana State University	• 2.1 Business Growth • 6.3 Climate Action	• Approval of a community solar tariff through the PSC • Establishment of NorthWestern Energy community solar project

Action 2.F.4. Promote Education and Incentives for Distributed Renewable Energy and Storage

The City and its partners, such as the Montana Renewable Energy Association, can work to improve access to solar energy by facilitating group purchasing of rooftop solar systems for Bozeman area residents. Allowing multiple homeowners to come together and aggregate their purchase reduces costs, making solar more affordable for many more households. The City will help develop resources and explore incentives to encourage residential and business investment in rooftop solar and energy storage, including current financing resources for renewable energy (see also supporting Action 1.A.6).

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 3	• **City of Bozeman** • NorthWestern Energy • Montana State University • Montana Weatherization Training Center • Montana Renewable Energy Association • Financial Institutions	• 1.2 Community Engagement • 6.3 Climate Action	• Establish a method of tracking distributed renewable installations • % increase in commercial and residential renewable energy installations

Increasing Resiliency to Climate Hazards

The following opportunities and considerations summarize how the solutions in this chapter can help improve resiliency to future climate hazards.

Opportunities	Considerations
Extreme Heat	
• A more diverse and distributed energy supply may have more capacity to handle wider fluctuations in demand. • Distributed energy and storage can increase the grid's capacity to accommodate spikes in demand during extreme temperatures.	• Cooling demand on hot days could exceed renewable energy capacity. • Community members may feel less motivated to practice efficiency if energy comes from "clean" sources.
Flooding	
• Increased flows associated with large rain events could support micro-hydro systems.	• Floods may pose a risk to micro-hydro or other renewable energy infrastructure.
Drought & Mountain Snowpack	
• 100% renewable mix could decrease water consumption associated with fossil fuel energy production long-term. • Increasing the generation capacity of solar and wind can help reduce reliance on hydropower.	• Drought could increase water costs, leading to high overall utility cost burden. • Drought could inhibit generation dependent on water (hydro, steam, solar heated water).
Wildfire	
• Distributed energy and storage could decrease the number of customers impacted by grid disruptions due to wildfires.	• Smoke and ash may inhibit solar energy generation. • Solar and wind infrastructure located within the wildland urban interface could be at risk for damage.
Winter Storms	
• Distributed energy and microgrids can decrease the number of customers impacted by grid disruptions due to winter storms. • Solar PV installed on steep roof pitches improves winter production and sheds snow more quickly.	• Accumulation of snow, ice, and extreme wind can inhibit solar and/or wind generation. • May be challenging to service a decentralized system in the event of a severe winter storm. • Siting of ground mounted PV should be tall enough to allow for snow accumulation from snowfall and clearing of panels.

Focus Area 3: Vibrant & Resilient NEIGHBORHOODS

Bozeman has a strong commitment to neighborhood and community planning to promote public health and safety, create a functional community that is beautiful and efficient, balance the desires of developers and community members, and support economic development. In addition to these benefits, well designed neighborhoods can help reduce the distance residents need to travel to work or to access services. This is important because it reduces transportation GHG emissions by facilitating a transition to active modes of transportation and reducing the miles traveled in personal vehicles. In 2018, transportation emissions made up 38% of community GHG emissions, and 19% of total emissions were from light-duty vehicles. Various plans, policies, and regulations exist to shape community growth, enhance existing community assets, and protect important resources. Moreover, these planning resources help establish the foundation for many other solutions and community investments such as buildings, infrastructure, transportation, and natural systems.

RELATED PLANS & STUDIES

- Unified Development Code
- 2020 Community Plan
- The Guidelines for Historic Preservation & The Neighborhood Conservation Overlay District
- Midtown Action Plan
- Triangle Community Plan

Per the 2020 Community Plan, Bozeman's Planning Area is generally the area of the City's future municipal water and sewer service boundary. It includes the City of Bozeman as well as a half-mile to two-mile area around, but outside, the City limits in the Gallatin County jurisdictional area. The Planning Area is nearly 70.8 square miles. The City's current footprint is 20.4 square miles. Since 1988, Bozeman has annexed, at landowners' request, more than 6,650 acres of land, about 10.3 square miles– more than doubling its size.

The projected land demand through 2040, based on estimated population growth, ranges from 3,820 acres to 5,716 acres, depending largely on levels of density in future residential developments. Faster rates of population growth will require additional land area. The 2020 Community Plan encourages development within the municipal boundaries where City services are available and thoughtful development in the Planning Area is guided by the Community Plan's goals and policies.

The neighborhood-focused solutions in this chapter support these existing planning resources and aim to further connect land use planning and development decision-making with climate-related considerations. They seek to reduce risk and vulnerability to natural hazards and to strengthen neighborhood capacity and resiliency. While the City of Bozeman can establish regulations and policies to inform investment and guide decision-making, private property owners have important roles to play in shaping neighborhood vibrancy and advancing resiliency.

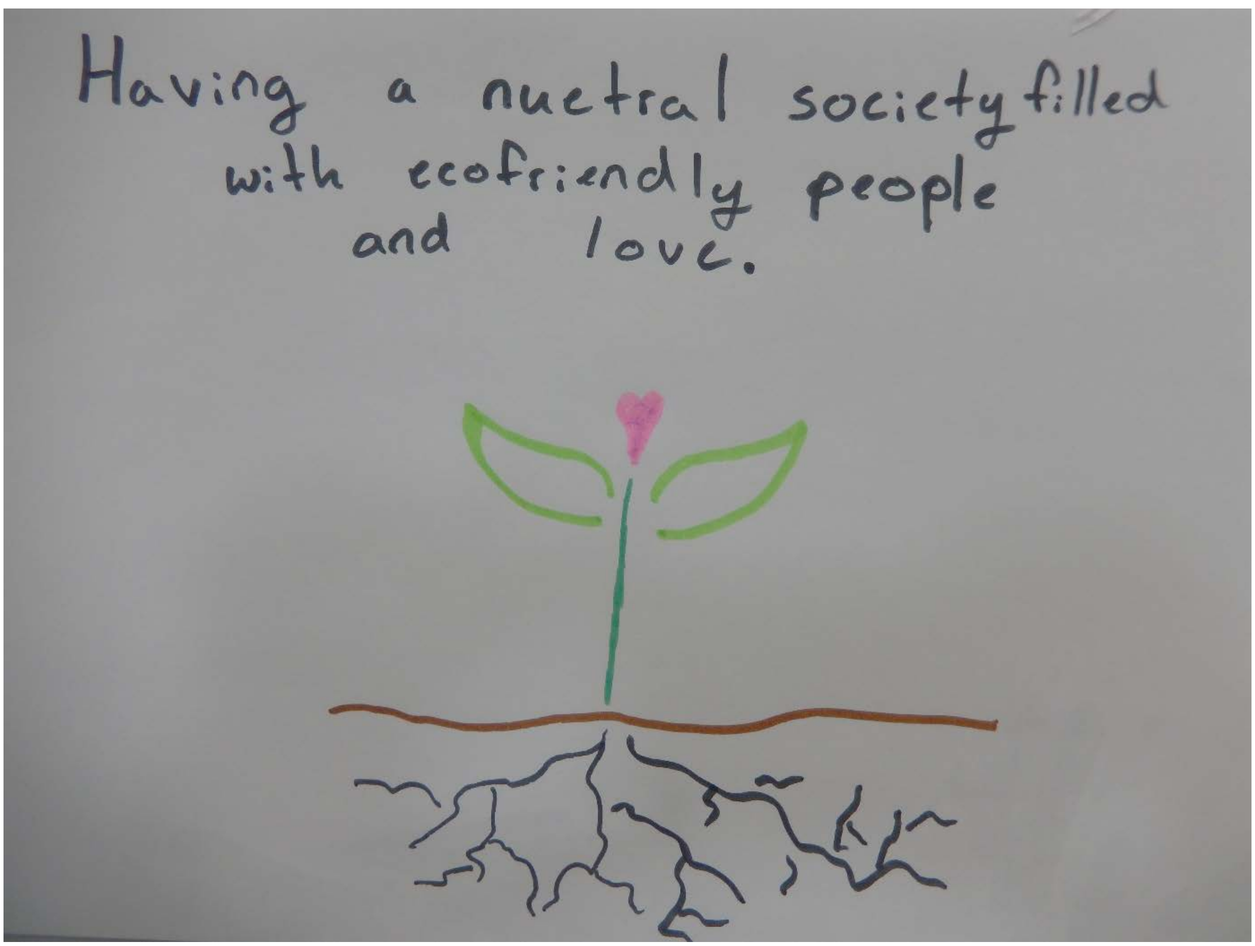
These solutions are strongly tied to improving community resiliency, and compact development is also an important long-term strategy for reducing transportation-based emissions. Compact and efficient development patterns can lead to reduced building energy consumption on a per-unit basis, though new construction will likely lead to increased energy use overall, unless net zero construction practices are employed. To support achievement of the GHG emissions and community resiliency goals, the Bozeman community will need to:

- **Solution G. Facilitate Compact Development Patterns**
  Proactively plan existing and future land uses to reduce the distance people need to travel for work and to access shopping and services.

- **Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards**
  Leverage neighborhood and infrastructure design to improve capacity to adapt to natural hazards including wildfire, flooding, drought, extreme heat, and winter storms.

- **Solution I. Enhance Social Infrastructure and Community Preparedness**
  Build social capital to improve community capacity to adapt to natural hazards and find resiliency in times of crisis.

Figure 18 shows the impact of solutions in this focus area on projected business as usual emissions and detailed fuel-saving targets for each goal year are shown in Table 8. Note that energy efficiency benefits from compact development patterns are accounted for in Solution B.



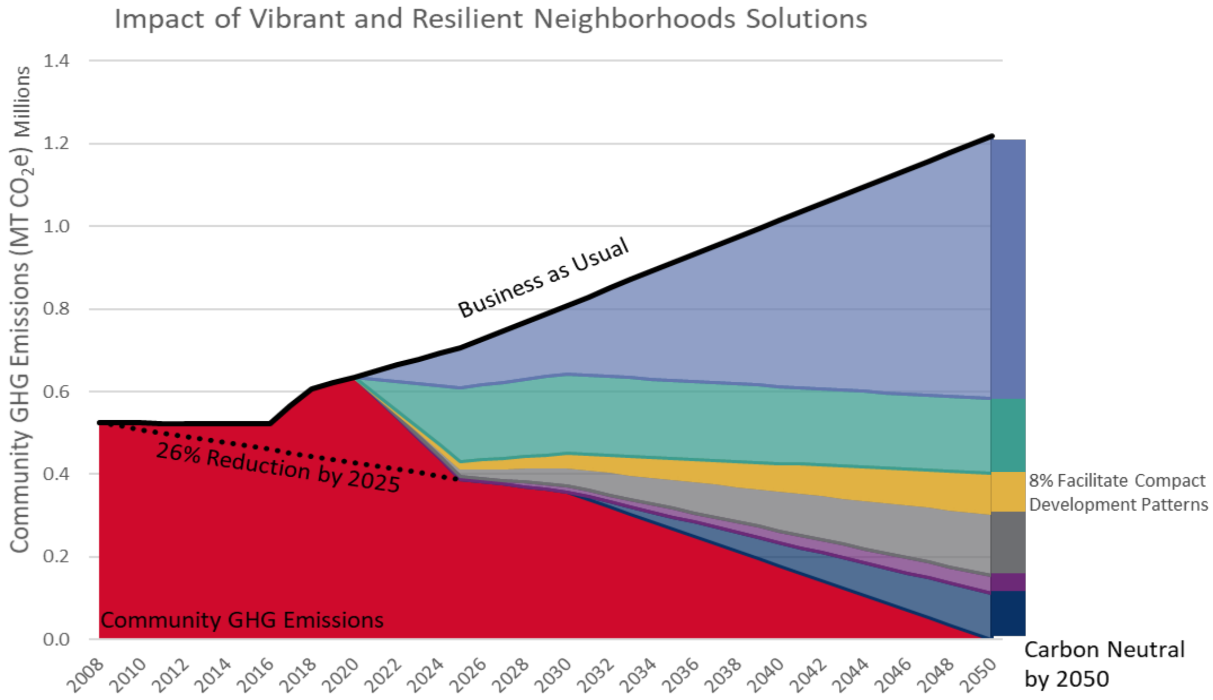
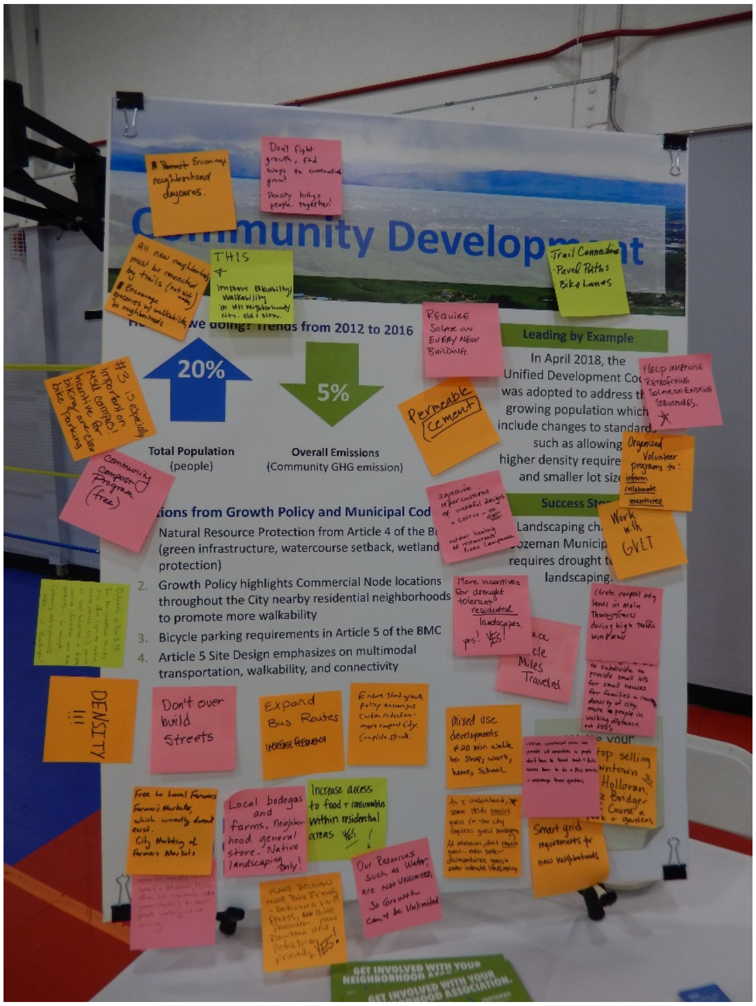


Figure 18. Projected neighborhoods GHG reductions by solution

Table 8. Projected neighborhoods solution mitigation targets

Solution	*Metric (annual savings)*	*Target* 2025 *Paris Accord*	*Target* 2030 *100% Net Clean Electricity*	*Target* 2050 *Carbon Neutral*
Solution G. Facilitate Compact Development Patterns	Gasoline (gallons)	1.9 million	3.8 million	11.2 million

Solution G. Facilitate Compact Development Patterns



This solution emphasizes the importance of proactively planning existing and future land uses to reduce the distance people need to travel for work and to access shopping and services. Compact development can be difficult to achieve in some areas of historic growth within the City, so context-sensitive infill development and redevelopment approaches are essential. For future growth and development, leveraging key planning resources such as the future land use map and Unified Development Code can help spur compact development and compatible infill and redevelopment activities.

Compact development encourages or supports smaller lot sizes, reduced setbacks, multiple units per property, and multi-story buildings in appropriate locations. However, compact development patterns do not stop at the property line. For compact development to function properly, the public right-of-way must accommodate all users, including pedestrians, bicyclists, and transit users, in addition to those that need to move around by car. In addition to the physical design, compact development must also encourage the right mix of uses. Incorporating essential land uses, such as grocery stores, medical centers, offices, and beyond, is crucial to ensure that compact development improves access to employment, goods, and services.

Though compact development can improve access to goods and services, it is important to consider: who is gaining access and who is left behind? The 2019 City of Bozeman Community Housing Needs Assessment showed that between 5,400 to 6,340 housing units are needed over the next five years to address the current housing shortfall for residents and the workforce and to keep up with job growth. Even where residential and commercial land uses are well mixed, residents may be forced to live in more affordable housing miles from where they are employed. Exacerbating this issue, improving access to goods and services can increase property values, which may make purchasing or renting a home even less attainable, pushing residents further away from these compact nodes. Fortunately, there are means of mitigating these unintended consequences. By encouraging, incentivizing, or even requiring the inclusion of affordable housing as part of compact development and redevelopment, residents typically pushed further away from activity centers can afford to live near goods and services.

In addition to improving resiliency and reducing greenhouse gas emissions, compact development can yield many co-benefits. By reducing the distance residents need to travel to access employment, goods, and services, compact development can support an uptick in the adoption of active modes of travel. In addition to reducing pollution associated with car travel, residents are able to engage in more active forms of transportation (e.g., biking and walking) and may reap the health benefits of increasing their physical activity. For many residents, compact development can also increase social cohesion by removing barriers to social interaction and increasing opportunities for gathering in shared community spaces.

The City of Bozeman can lead by example by siting and designing all future community facilities in a manner that demonstrates compact development best practices. Furthermore, the City has a distinct role to work with the development community to guide new development and redevelopment toward compact patterns.

This solution primarily addresses Bozeman's greenhouse gas mitigation goals, while also providing supporting resiliency benefits.

Emissions Reduction Benefits

This is a supporting mitigation solution to achieve the 2025 and 2030 mitigation goals, and secondary solution to achieve the 2050 mitigation goals.

Emission sources impacted include:

- Gasoline/Diesel

Resiliency Benefits

- Conserve natural resources
- Increase social cohesion
- Mitigate property and economic losses
- Protect human health

Related Solutions

- Solution B. Achieve Net Zero Energy New Construction
- Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards
- Solution I. Enhance Social Infrastructure and Community Preparedness
- Solution J. Increase Walking, Bicycling, Carpooling and Use of Transit
- Solution P. Manage Land and Resources to Sequester Carbon

Action 3.G.1. Continue Regional Coordination on Compact Growth and Sustainable Development

The current land area of the City of Bozeman is approximately 20.4 square miles. However, the City's planning area is nearly 70.8 square miles. Much of the land in Bozeman's planning area lies within Gallatin County's jurisdiction. Though the 2020 Community Plan encourages development within the municipal boundaries, growth pressures will necessitate careful coordination between the City and County to ensure sustainable development patterns as Bozeman grows into its planning area.

Current projections estimate Bozeman may need to annex between 3,820 and 5,716 acres of land by 2040. Annexing additional land will necessitate significant investment in sewer, water, and other infrastructure. By working closely with regional partners to promote compact growth, primarily within existing city limits, Bozeman can limit the total acreage of land required to accommodate new growth. Regardless, promoting compact growth within and outside of city limits will require careful consideration and implementation of resource conservation efforts and maintenance and upgrade cycles to ensure existing and new infrastructure can accommodate new growth.

Aligning around shared sources of current and future data is foundational to coordinated land use planning. For instance, the City and County may agree to use the same scenario-based software to evaluate the economic and environmental impacts of new development. In addition to sharing data sources and design standards for infrastructure, City and County Departments should develop land use and infrastructure plans to help guide projected growth locations. Finally, the inclusion of safe, accessible, and diverse transportation options is paramount to achieving the goal of compact and sustainable development. Specific to this action item, the City and County will coordinate on regional multi-modal connections to ensure a high functioning network of compact development.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • City of Belgrade • Gallatin County	• 4.1 Informed Conversation on Growth • 6.3 Climate Action	• # of education/training sessions or engagement with City and regional elected officials • # of coordinated planning efforts that promote consistency throughout the region for parks, transportation, transit service, and other community infrastructure

Action 3.G.2. Revise Development Code to Enhance Compact and Sustainable Development

At the local level, the Unified Development Code (UDC) is one of the most influential documents guiding the form and character of a community. This action is focused on ensuring the UDC, as well as other relevant City Code chapters, continue to support infill development, redevelopment, and high-quality mixed use development. As new areas are developed and existing areas redeveloped to sustainable code standards, Bozeman will continue to move toward a future where households are near the goods, services, and amenities essential to a high quality of life.

Already the City of Bozeman has taken substantial steps to facilitate compact development (e.g., by allowing Accessory Dwelling Units (ADUs) by right, by reducing minimum lot sizes by 60%, and by diversifying zoning districts to allow for a mix of uses). This action item seeks to build on activities to date.

To achieve this, the City can identify additional opportunities in the UDC to enhance compact development requirements (e.g., additional mixed use, reduced parking requirements with installed EV infrastructure, etc.). The City will explore options to encourage development above and beyond minimum requirements, such as incentive programs or performance policies. Though voluntary compliance and performance-based programs often require additional coordination with developers, if designed well, these systems can help push the envelope and ensure investment in desired amenities, such as multi-modal and transit infrastructure, low impact development, sustainable landscaping, and neighborhood gathering spaces. Neighborhood gathering spaces and other opportunities for social cohesion are a critical component of sustainable development.

Implementation activities within this action item will require coordination with utilities to better understand, accommodate, and plan for site constraints associated with compact development. This will include preparation for building and transportation electrification and ensuring adequate plans for maintaining and upgrading sewer, water, and other utilities.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • Downtown and Midtown Urban Renewal Districts	• 4.1 Informed Conversation on Growth • 4.2 High Quality Urban Approach • 4.3 Strategic Infrastructure Choices • 6.3 Climate Action	• UDC revisions that reflect density increases, such as maximum building height limits, minimum development intensity requirements, or elimination or reduction of minimum parking requirements • Number of new residential or commercial units built in existing neighborhoods or developed areas in City limits • Adopted land use management techniques and incentives that promote development within City limits

Action 3.G.3. Develop Sustainable Neighborhoods Outreach

While the City of Bozeman can establish regulations and policies to inform investment and guide decision-making, private property owners have important roles to play in shaping neighborhood vibrancy and advancing resiliency. This action focuses on empowering neighborhood partners to enact sustainable practices at the scale of the neighborhood, street, condo complex, or apartment building.

For this action, the City will partner with developers, neighborhood groups, and community members to develop and share education and engagement campaigns that allow neighborhoods to connect, participate in the sharing economy, and learn about sustainable housing options. Creating a recognition program for sustainable neighborhoods and developers will facilitate the sharing of best practices, while providing a competitive edge for developers. Homeowners' associations will be key partners in the development of shared community amenities, such as "thingery sheds" and lending libraries. The City will help promote successes and encourage the adoption of sustainable practices by creating and sharing case studies, promotional materials, and resources regarding the benefits of ADUs, co-housing and cottage developments, and local sharing economies.

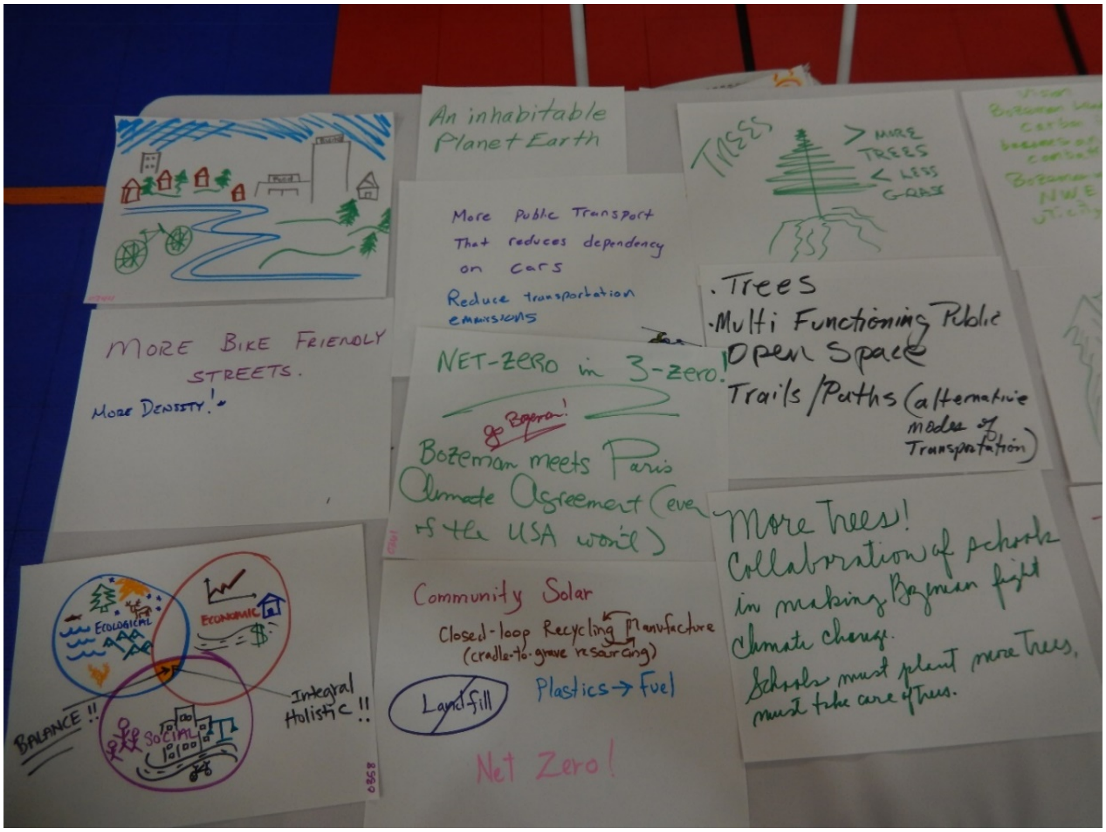
Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Inter Neighborhood Council • Bozeman Climate Partners Working Group • Homeowner's Associations • Montana State University	• 1.1 Outreach • 4.1 Informed Conversation on Growth • 6.3 Climate Action	• # of contacts with Neighborhood Associations, HOAs, and community leaders • # of trainings offered to Neighborhood Associations, neighborhoods, or community groups

Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards

This solution focuses on improving the capacity of Bozeman to adapt to natural hazards, including wildfire, flooding, drought, extreme heat, and winter storms, through the lens of neighborhood and infrastructure design. Reducing the vulnerability of neighborhoods and infrastructure can be split into two key activities: siting and design.

To the extent practicable, neighborhoods and infrastructure should be sited in safe locations (i.e., away from floodplains or wildfire activity areas). In some instances, siting neighborhoods and infrastructure in vulnerable locations may be unavoidable; in these cases, additional measures may be necessary to bolster infrastructure against hazards. Design can play an important role in reducing vulnerability against all hazards. Sites can be designed using low impact development or green infrastructure techniques to mitigate runoff and reduce the risk of flooding. On a smaller scale, buildings can be designed to withstand extreme conditions and protect inhabitants. For example, insulating walls and pipes and designing roofs to withstand heavy snows and prevent ice dams can protect inhabitants in the event of a severe winter storm. Critical infrastructure, such as roads, bridges, power lines, water treatment plants, parking structures etc., should also be designed to withstand extreme conditions, such as flooding, wildfires, high winds, and winter storms.

The building code, UDC, and other City Code chapters are foundational to resilient neighborhood and infrastructure design. For example, overlay districts can serve to restrict development in the floodway, while building codes can require the incorporation of flood-resistant building design. Through partnerships with Gallatin County Emergency Management, the City develops the appropriate plans for



managing risk in hazardous areas, including the potential relocation process for damaged property and infrastructure.

Equity

When developing in a manner that reduces the vulnerability of neighborhoods and infrastructure to hazards, it is important to consider how different socio-economic characteristics may increase vulnerability. Older adults or persons with disabilities may be less mobile and therefore more vulnerable in the event of a disaster. Alternatively, immobility may stem from a person's or family's economic status. Low-income residents may be less able to move or recover from a disaster. In many cases, low-income neighborhoods are sited in less desirable locations, such as within floodplains, which can put community members especially at risk. Often, mobile home communities are some of the most vulnerable neighborhoods, due to a combination of construction quality, siting, and limited economic mobility of residents.

Human Health & Well-Being

Resilient neighborhood and infrastructure design can dramatically improve public health and life-safety outcomes in the event of natural disasters. For instance, buildings designed to withstand flooding may reduce vulnerability to property damage while also protecting the public health and safety of residents.

Human Health & Well-Being

The City of Bozeman can lead by example by continuing to implement the [Bozeman Climate Vulnerability Assessment and Resiliency Strategy] to bolster the resiliency of identified City assets.

This solution primarily addresses Bozeman's resiliency goals, with minimal impact on the greenhouse gas mitigation goals.

Resiliency Benefits

- Strengthen infrastructure to natural disaster
- Mitigate property and economic losses
- Protect human health

Related Solutions

- Solution B. Achieve Net Zero Energy New Construction
- Solution F. Increase Community-Based Distributed Renewable Energy Generation
- Solution G. Facilitate Compact Development Patterns
- Solution I. Enhance Social Infrastructure and Community Preparedness
- Solution O. Manage and Conserve Water Resources

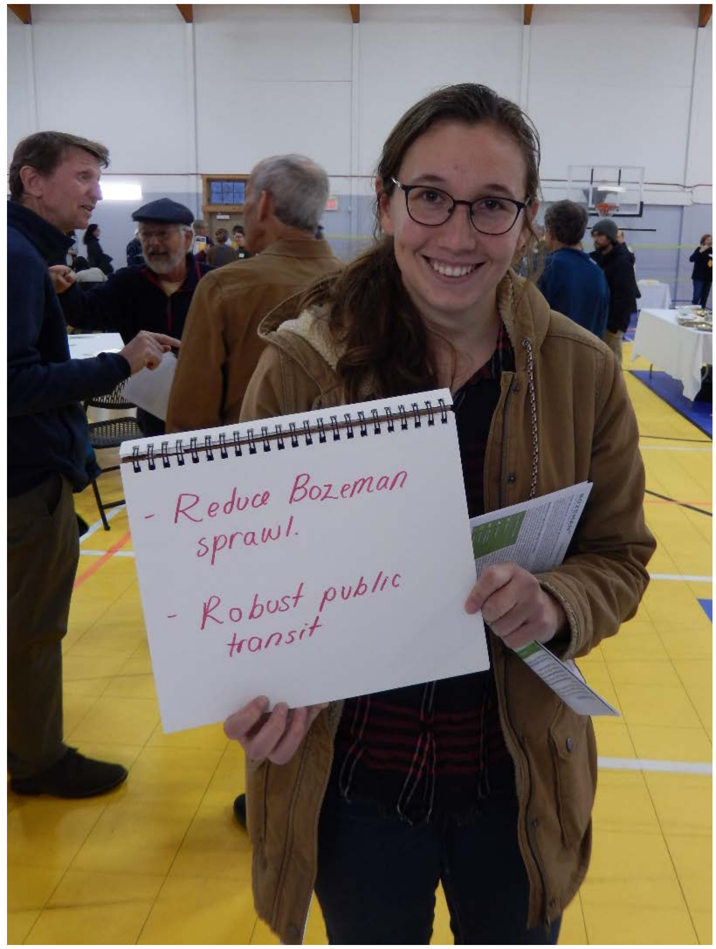
Action 3.H.1. Plan for Resilience Hubs at Critical Facilities

The 2019 Climate Vulnerability Assessment & Resiliency Strategy included a vulnerability assessment of 24 municipal buildings and facilities, organized into three primary building functions: critical city facilities, community centers, and critical infrastructure. Together, critical facilities are the structures and services necessary to meet the daily needs of the community. This includes water and wastewater treatment plants, police and emergency services departments, municipal administration, and community centers, among others. Ensuring continuous operation of these facilities and services is paramount to the safety and well-being of Bozeman community members, especially in the event of shocks and stressors. Importantly, if designed appropriately, critical facilities can serve as resilience hubs during emergencies such as extreme heat, hazardous air quality, severe winter storms, and wildfire.

This action calls for the development of one or more resiliency hub demonstration project(s) at a critical facility that showcases structural and operational resiliency, renewable energy plus storage, energy and water efficient building design, and best practices in stormwater management. This will allow these facilities to function as hubs for coordinated emergency response and recovery during periods of community stress or crisis.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • Gallatin County Emergency Management • Greater Gallatin United Way • Human Resources Development Council	• 1.3 Public Agencies Collaboration • 3.1 Informed Conversation on Growth • 6.3 Climate Action	• Establish plan for resilience hubs based on community needs • Completion of a demonstration resilience hub

Action 3.H.2. Advance Resilience in Development Code and Development Review



The UDC is central to resilient neighborhood and building design. The UDC, in conjunction with the development review process, determines both where new development is located and how it is designed. Thus, ensuring the development process discourages growth near potential hazards (e.g., floodplains, the wildland urban interface) while encouraging resilient design can have a significant impact on public health and safety.

Ensuring pathways to advance low impact development (LID) will be crucial. Low impact development is essential to effectively manage stormwater, especially as the community promotes compact design and the intensity of storm events increases. Where the building code is hindered by State-level requirements, Bozeman will explore opportunities to require or provide incentives to encourage the use of sustainable (i.e., sturdy, fire-resistant, and efficient) building materials and practices, green infrastructure, and climate-adaptive landscapes, among other best practices.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman**	• 1.1 Outreach • 3.2 Health & Safety Action • 6.3 Climate Action	• GIS data mapping for environmentally sensitive and/or hazard-prone areas • # of LID or flood control infrastructure installations (e.g., permeable pavers, bioretention facilities, rain gardens, and infiltration galleries

Action 3.H.3. Support Residential and Business Preparedness Outreach

Even as infrastructure evolves and improves, individual and neighborhood preparedness will be paramount to strengthening resiliency in the face of anticipated climate risks and hazards. This action focuses on supporting business, households, and neighborhood-scale emergency planning and preparedness.

Mapping areas of high risk and vulnerability (e.g., floodplains and wildland urban interface) is an important first step when developing a preparedness outreach campaign. Though some of these maps already exist, it will be important for the City to periodically update maps to reflect current and ever-changing conditions. Neighborhoods at Risk, developed by Headwaters Economics, is an interactive data tool that provides neighborhood-level information about potentially vulnerable people and climate change. Maps can be used to help focus on vulnerable areas of the community and to communicate the importance of preparedness. Importantly, the preparedness campaign will identify actionable steps community members can take to mitigate and adapt to potential hazards. Additional implementation activities may include developing action plans in partnership with businesses and neighborhoods to further mitigate risk and improve resiliency.

The City Stormwater Division's education and outreach program can serve as a model and touchpoint for sharing information and best practices related to residential and business preparedness related to green infrastructure, and maintenance of public and private stormwater infrastructure for flood mitigation.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Gallatin County Emergency Management • Montana State University • Greater Gallatin United Way • Human Resources Development Council • Southwest Montana Community Organizations Active in Disaster • Homeowners' Associations	• 1.1 Outreach • 3.2 Health & Safety Action • 6.3 Climate Action	• Development of outreach plan based on assessment of hazard • # of residential and business preparedness toolkits distributed • # of businesses or residents reached through outreach efforts • # of participants in the Stormwater Division's Adopt a Drain program

Action 3.H.4. Incorporate Resilience into Infrastructure Plans

The 2019 Climate Vulnerability Assessment and Resiliency Strategy includes a strategy focused on infrastructure and capital planning. The scope of the study was limited to city assets. While infrastructure and capacity outside of the city's purview was considered in the risk assessment, it was not a comprehensive evaluation of the community. The infrastructure and capital planning strategy emphasizes planning for critical facility infrastructure hardening and capital projects that limit damage to property, protect human health, and improve operational performance during and after hazard events. Supporting actions include the following:

- Continuing to build out back-up power, mobile back-up power, fuel storage, and communications redundancies for buildings and infrastructure.
- Increasing air filtration capabilities in critical facilities to cope with wildfire smoke.
- Providing covered parking for first responder vehicles and critical snow removal equipment to ensure reliable service and improve response time during extreme weather events.
- Evaluating the need for heat, air conditioning, or passive cooling within City facilities to protect the health and safety of employees and users
- Pursuing automated dam controls at Middle Creek Dam to improve operational flexibility and enhance our ability to respond to wildfire or flood events.
- Continuing to assess the feasibility of watershed or aquifer groundwater storage and recovery projects, including strategic water reserves.
- Continuing to pursue science-based forest restoration measures to protect the municipal watershed.
- Prioritize Continuity of Operations planning with Montana State University and other agencies.

This action reinforces the strategy and actions in that plan and emphasizes embedding resiliency considerations into all future capital improvement planning and design. Ultimately, this may involve establishing resiliency criteria into capital improvement prioritization processes or adopting resiliency standards for City infrastructure projects.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman**	• 3.1 Informed Conversation on Growth • 3.2 Health & Safety Action • 6.3 Climate Action	• Adoption of resiliency guidelines or prioritization criteria for infrastructure plans

Solution I. Enhance Social Infrastructure and Community Preparedness

This solution focuses on building social capital to improve the capacity of Bozeman to adapt to natural hazards and find resiliency in times of crisis. Though hard infrastructure is critical to preserve function in the face of a disaster, social infrastructure and community preparedness can be invaluable when it comes to rebuilding the community and bouncing back quickly. This means expanding the ability for community members to be self-sufficient, while simultaneously building the capacity to support each other when faced with adversity. Recent events like the COVID-19 pandemic reveal how social norms and infrastructure, like social distancing practices, shape a community's ability to adapt and recover.

A strong social infrastructure is one that connects community members to the resources they need to remain resilient in the face of adversity. Enhancing social infrastructure includes encouraging and supporting authentic connections amongst neighbors and community members to support physical, mental, and emotional needs. This connection can occur virtually, through virtual neighborhood platforms and digital communications, as well as in-person, through block parties or community volunteer events.

Building strong social infrastructure is a key component of preparedness, especially since some community members may lack the ability to be self-sufficient when facing different types of challenges or crises. A strong social fabric can help ensure all community members are included and not forgotten. Education is a crucial component of cultivating community preparedness and social infrastructure. Instilling individual and shared responsibility for preparedness can greatly reduce overall community risk, while allowing preventative and recovery resources to be funneled toward community members least able to protect themselves or respond to crises or disasters.

Enhancing social infrastructure and community preparedness for a diverse community will require accommodating diverse needs. Educational materials should be made available in multiple languages and for the hearing impaired. Meetings intended to bring together a cross-section of the community should include consideration for free transportation, access for persons with disabilities, childcare, translators, and safe spaces for the LGBTQ community.

Connecting community members to their neighbors, community partners, and City resources may reduce the isolation and vulnerability of individual persons and families by creating a network of connected resources able to lend support in the event of a crisis or disaster. Additionally, ensuring community members are fully informed and prepared before a crisis or disaster can reduce risk and vulnerability.

The City of Bozeman can lead by example by siting and constructing community centers and neighborhood gathering spaces to foster social connections and build a sense of community. Community centers should cater to Bozeman's diverse population by including programming and communications in multiple languages, and by being accessible to persons with differing abilities. Finally, community and neighborhood centers should include spaces and programming to include and support Bozeman's most vulnerable community members, including older adults, racial minorities, economically disadvantaged families, persons with disabilities, and the LGBTQ community.

This solution primarily addresses Bozeman's resiliency goals, with minimal impact on the greenhouse gas mitigation goals.

Resiliency Benefits
• Increase social cohesion • Mitigate property and economic losses • Protect human health

Related Solutions

- Solution G. Facilitate Compact Development Patterns
- Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards
- Solution M. Move Toward a Circular Economy and Zero Waste Community
- Solution N. Cultivate a Robust Local Food System

Action 3.I.1. Support Community and Neighborhood Resilience Programming

Resilience programs are place-based initiatives that empower community members to build connections, skills, and resources to be prepared, adaptable, and responsive in the face of adversity and challenges. Though grassroots efforts can be a highly effective means of change, response, and community support, they are even more impactful when well-aligned with City, regional, and State-level efforts. This action is focused on aligning neighborhood resilience programming with existing efforts such as Climate Smart Montana, the Montana Resilience Framework, and the Southwest Montana Community Organizations Active in Disaster (COAD) response, as well as expanding new resilience programming efforts.

The COVID-19 pandemic and resurgence of civil rights movements have highlighted the imperative of neighborhood-level resilience and social cohesion. Though the impacts of this pandemic have been far-reaching and severe, there are many lessons that Bozeman can carry forward to capture, build on, and leverage the change and connections formed in response to COVID-19. For instance, the City will explore the feasibility of institutionalizing mutual aid programs grown organically in response to COVID-19.

Other actions include hosting social infrastructure assessments and trainings, process debriefs and listening sessions to better understand successes and areas of opportunity in response to shocks and stressors, and community or neighborhood networking events to support social cohesion. The City will also work to connect residents with resources for mental health and personal resilience.

The City can support local resilience building activities by connecting community partners with resources, such as free meeting spaces or community-building grants to support small-scale cohesion activities and projects.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • **Greater Gallatin United Way** • Inter Neighborhood Council • Community Organizations Active in Disaster (COAD) • Bozeman Mutual Aid Network • Bozeman Community Foundation • Bozeman Help Center • HRDC and Gallatin Valley Food Bank	• 1.1 Outreach • 3.2 Health & Safety Action • 6.3 Climate Action	• Development of a resilient neighborhoods toolkit • Number of City-organized or partnered trainings or events

Increasing Resiliency to Climate Hazards

The following opportunities and considerations summarize how the solutions in this chapter can help improve resiliency to future climate hazards.

Opportunities	Considerations
Extreme Heat	
• Compact development can reduce distance needed to travel to access goods and services, which may be helpful during periods of extreme heat. • Public facilities can be designed to provide community members a refuge from extreme heat. • Development can be designed to reduce pavement and incorporate trees and green infrastructure to mitigate potential urban heat island impacts.	• Compact development without the incorporation of green infrastructure could increase in average urban temperatures
Flooding	
• Green stormwater infrastructure can slow the influx of water and reduce vulnerability to flooding. • Avoiding construction in flood prone areas will reduce risk posed to new development.	• Compact development and increased volumes of water during flooding events will increase demands on stormwater planning and infrastructure. • Much of Bozeman's land area is located in areas with a high water table.
Drought & Mountain Snowpack	
• Compact development can help reduce the water demands associated with outdoor landscaping.	• Frequent and intense periods of drought will impact the City's ability to accommodate growth, meet the needs of water users, and provide essential municipal services.
Wildfire	
• Neighborhoods near WUI areas can be designed with wildfire considerations and emergency access in mind.	• Compact developments within or near WUI areas may put large populations at risk of smoke exposure or property loss if wildfire occurs.
Winter Storms	
• Compact development decreases miles of streets that need to be plowed. • Compact development can reduce distance needed to travel to access goods and services, which may be beneficial during winter storms.	• Compact development needs to accommodate snow removal equipment and snow storage.

Focus Area 4:

Diverse & Accessible TRANSPORTATION OPTIONS

In 2018, the transportation sector accounted for 38% of total emissions in Bozeman. Emissions from this sector are on the rise with total vehicle miles traveled (VMT) per person increasing and a large share of transportation emissions coming from light-duty gas and diesel vehicles. According to the Bozeman Community Plan, 70% of Bozeman commuters drive alone, with an average commute time of 14.5 minutes. Reductions in VMT can be achieved through compact development patterns that reduce trip lengths or needs, as well as pedestrian-centered design that supports non-vehicular modes of travel such as bicycling and walking.

RELATED PLANS & STUDIES

- 2020 Bozeman Yellowstone International Airport Master Plan
- Midtown Action Plan
- Redesign Streamline 2020
- Triangle Community Plan

Telecommuting can lead to significant reductions in VMT, but when coupled with increased reliance on e-commerce activities and delivery services, those VMT reductions can be diminished. Emerging transportation and vehicle technologies such as drones, self-driving cars, and shared and autonomous fleets may also play a role in coming years in reducing trip volumes and overall VMT.

As the Bozeman community reaches a population greater than 50,000 residents (a likely milestone with the 2020 Census), federal law mandates the establishment of a Metropolitan Planning Organization (MPO). MPOs receive federal funds for carrying out transportation planning and programming in their metropolitan areas. The MPO will be a new tool to assist the greater Bozeman area in all facets of coordinated transportation planning and improvements.

As with the energy-related solutions, conservation and efficiency in the transportation system is essential to reduce emissions. Higher-efficiency transportation solutions include carpooling, hybrid electric and other fuel-efficient vehicles, and public transit. Streamline Transit, the Bozeman community bus service, provides free fixed-route bus service in and around Bozeman as well as commuter routes to Belgrade and Livingston. The highest conserving and least emitting transportation options include active and non-vehicular modes such as walking, bicycling, and other micro-mobility equipment like scooters and skateboards.

Just like the energy solutions that shift to more renewable energy sources, solutions to reduce transportation emissions also include shifting vehicles and equipment fuels from gasoline and diesel to electric and other alternative sources, such as biodiesel or hydrogen fuel cells. Solutions to address vehicle efficiency and fuel types are applicable to both personal vehicles as well as fleet vehicles and equipment.

Now the busiest airport in Montana, the Bozeman-Yellowstone International Airport is also a growing contributor to the community's transportation emissions. Few alternatives to traditional air travel currently exist, so the solutions in this chapter emphasize trip reduction and advocacy for new technologies.

Because transportation-related emissions are an increasing portion of the City's inventory, concerted efforts will need to be made to reduce overall VMT, as well as greatly enhancing the efficiency of all personal and fleet vehicles and equipment. To reduce VMT, enhance transportation efficiency, and reach GHG emissions and community resiliency goals, the Bozeman community will need to:

- **Solution J. Increase Walking, Bicycling, Carpooling and Use of Transit**
  Shift travel modes away from single-occupancy vehicles and grow the percentage of community members walking, bicycling, carpooling, avoiding or consolidating vehicle trips, and using transit.
- **Solution K. Decrease Direct Vehicle Emissions**
  Promote and transition to more efficient vehicles, vehicles that use alternative fuels, and electric vehicles.
- **Solution L. Improve Air Travel Efficiency**
  Support the development of alternatives to traditional air travel, as well as offering alternatives to air travel.

Detailed fuel and energy saving targets for each goal year are shown in Table 9: Projected transportation mitigation targets.

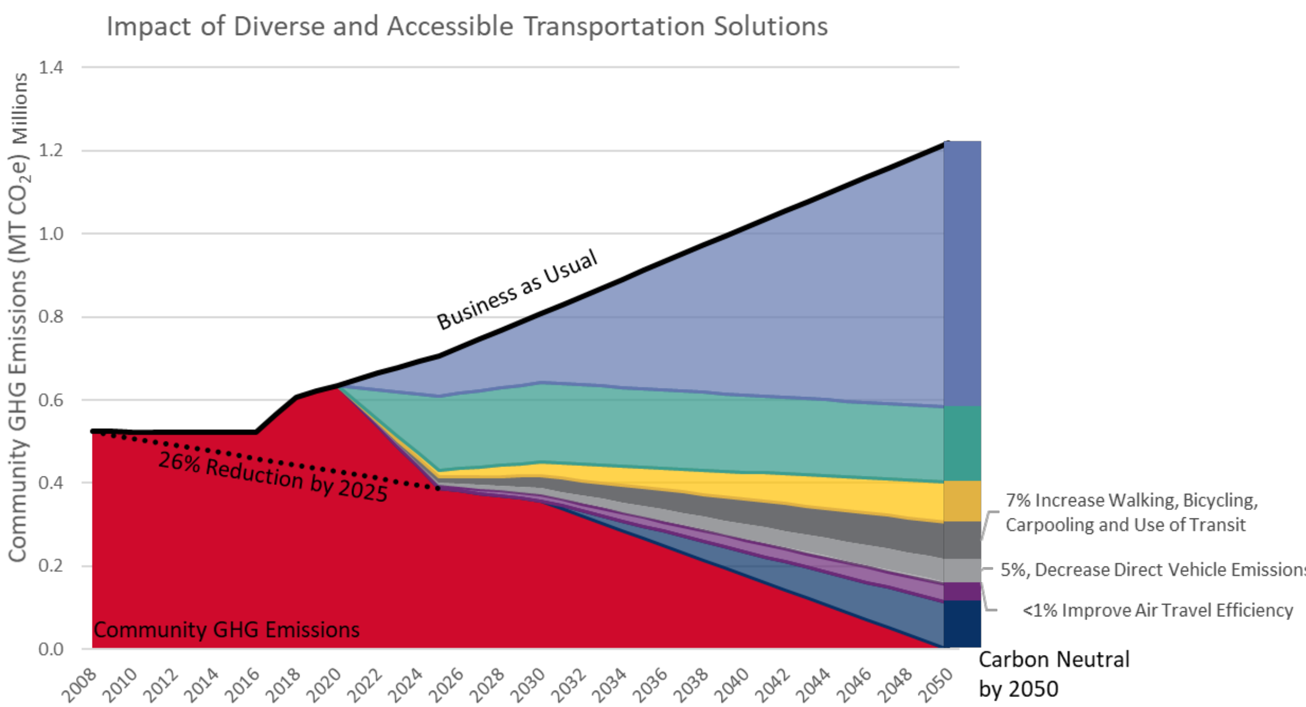


Figure 19. Projected transportation GHG emissions reduction by solution

Table 9. Projected transportation mitigation targets

Solution	*Metric (annual savings)*	*Target* 2025 *Paris Accord*	*Target* 2030 *100% Net Clean Electricity*	*Target* 2050 *Carbon Neutral*
Solution J. Increase Walking, Bicycling, Carpooling and Use of Transit	Gasoline Savings (gallon)	1.7 million	3.4 million	10.2 million
Solution K. Decrease Direct Vehicle Emissions	Gasoline Savings (gallon)	180 million	2.7 million	10.6 million
	Increased Electricity Use (kWh)	6 million	19 million	74 million
Solution L. Improve Air Travel Efficiency	GHG Emissions Reduction (MT CO_2e)	1,000	2,000	5,900

Solution J. Increase Walking, Bicycling, Carpooling and Use of Transit

This solution focuses on increasing the percentage of community members walking, bicycling, carpooling, avoiding or consolidating vehicle trips, and using transit. As noted in the Bozeman Community Plan, 16% of Bozeman commuters walk or ride a bike, 6% work from home and less than 1% use public transit. Influencing community members' mode choice can be extremely challenging, as most people choose their mode of transportation based first on convenience, and then on cost. In response to COVID-19 and resulting social distancing guidelines, working from home practices have become more prevalent in many workplaces. The widespread implementation of work from home practices, including the expansion of digital communication capabilities, may have a significant impact in the percent of employees working from home in the future.

Unfortunately, the cost of the private car and parking infrastructure is hidden and difficult to fully quantify. Many American cities have grown and developed around the easy mobility of the private car, resulting in expansive street miles with corresponding miles of water, sewer, gas, and electric infrastructure with public and private parking spaces—thus spreading infrastructure demands across a larger area and thinning the tax base needed to fund city services. Further, vehicle traffic violations and accidents require extensive parking and traffic patrol and contribute a large percent of ambulance calls, emergency room visits, and case load in the justice system. Cities may need to closely examine the full costs and benefits of all modes of transportation when making mobility infrastructure investments.

There are several mechanisms to facilitate the transition away from private cars, and especially cars moving a single driver. Building high-quality, accessible infrastructure is a primary means of encouraging active forms of transportation. Bike lanes physically separated from traffic can create a safer experience for bicyclists. Sidewalks should be wide enough to accommodate multiple users and should also be separated from vehicular traffic (i.e., setback from the road or lined with trees or planters to create a physical separation). Sidewalks that have degraded or were not constructed to meet ADA design requirements, should be rebuilt to meet these standards.

However, access to high-quality infrastructure is not enough to influence mode choice. Trip length is a major consideration when selecting a mode of transportation. Compact development can help to shorten trip lengths by reducing the distance between residents, employment opportunities, and goods and services. Parking can also play a factor in mode choice. If parking is widely available and inexpensive, community members may be incentivized to travel by car, given the extreme importance of convenience. Finally, education can play an important role in influencing mode choice. In some cases, community members may be unaware of transportation options or feel uncomfortable using those transportation options other than personal vehicles. Connecting residents with resources that will gain them confidence in the convenience, safety, and benefits of active modes of transportation and transit can help remove barriers that keep people from choosing alternatives to their cars.

The City and other community partners can help to increase the adoption of active modes of transportation by reducing financial barriers or even providing financial incentives. While transit in Bozeman is free, large businesses may provide a financial incentive for employees to use an active mode of transportation. However, infrastructure can still be very expensive and must be completed in conjunction with other measures, such as shortening trip lengths and education. Fortunately, various grant programs exist at the state and federal levels to support the development of active transportation infrastructure and education programs.

Equity

Removing barriers to active modes of transportation can enable community members to engage in more cost-effective means of travel. For example, improving access to public transit for low-income community members can reduce the transportation burden for those forced to live further from goods and services due to the cost of housing. Similarly, this solution expands mobility and accessibility for the portion of the community population that cannot drive due to physical abilities, age, and/or income.

Alternatively, some community members may maintain a need for car ownership and travel. Penalizing car trips through disincentives (i.e., restricting parking or increasing the cost of parking) may disproportionately impact residents who rely on car travel. One potential solution could be to increase the availability of affordable housing options near activity centers.

Human Health & Well-Being

In addition to reducing pollution associated with car travel, participating in active modes of transportation can improve physical and mental health. Some emerging modes of efficient transportation, such as e-bikes and e-scooters can increase mobility options for some but may discourage more active forms of transportation and, without careful planning, may present a hazard to those with certain disabilities, such as persons with visual impairments. Finally, as on-demand transportation increases in popularity for first- and last-mile connections, excessive idling and circulating may offset air quality benefits of reducing car trips.

City Assets

The City of Bozeman can lead by example by continuing to maintain its status as a Silver Bike Friendly Community by the League of American Bicyclists, implementing complete streets projects in neighborhoods underserved by active transportation infrastructure, and by continuing to use a complete streets model for the construction of new streets and street reconstruction efforts.

This solution primarily addresses Bozeman's greenhouse gas mitigation goals, while also providing supporting resiliency benefits.

Emissions Reduction Potential

This is a supporting mitigation solution to achieve the 2025, 2030, and 2050 mitigation goals.

Emission sources impacted include:

- Gasoline/Diesel

Resiliency Benefits

- Conserve natural resources
- Increase social cohesion
- Protect human health

Related Solutions

- Solution G. Facilitate Compact Development Patterns
- Solution I. Enhance Social Infrastructure and Community Preparedness
- Solution K. Decrease Direct Vehicle Emissions
- Solution L. Improve Air Travel Efficiency

Action 4.J.1. Prioritize Regional Multi-modal Planning and Connectivity

Providing high quality, safe, and accessible infrastructure to support a diversity of transportation options is a foundational step toward reducing the percent of people who drive alone. Though providing connectivity within an activity area is critical, it is also important to provide safe, convenient access to and from destinations. Thus, ensuring the adoption of multi-modal forms of transportation will require a coordinated effort amongst existing and future agencies to prioritize inclusive and diverse options for transportation.

Environmental Data that is measured, verified, and reported is an important stepping stone to lower emissions in the longer term. The 2017 Transportation Master Plan references greenhouse gas emissions and health benefits to communicate the value of active transportation. Future transportation plans will incorporate targets for total and per capita Vehicle Miles Traveled and greenhouse gas emissions reductions.

The Gallatin Triangle Planning Study and the Bozeman Transportation Master Plan establish a foundation of planning for multi-modal infrastructure. The aim of this action is to work in conjunction with other solutions, such as "Facilitate Compact Development Patterns" and "Enhance Social Infrastructure and Community Preparedness" to provide a more inclusive and systemic view of the region's transportation system. Future planning efforts should include a diversity of perspectives, including persons with disabilities, low-income households, City and County service partners, and City and County planning departments. Outcomes of future planning efforts might include a guiding policy document, which outlines a hierarchy of travel, deprioritizing travel by car, while elevating mobility for Bozeman's most vulnerable community members.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • Gallatin County • City of Belgrade • Montana Department of Transportation • Bozeman Area Bicycle Advisory Board • Pedestrian Traffic Safety Committee • Bridger Bowl • Human Resource Development Council	• 4.3 Strategic Infrastructure Choices • 4.5 Housing and Transportation Choices • 6.2 Protect Local Air Quality • 6.3 Climate Action	• % increase in walking, biking, carpool, and transit commuting • Formation of and participation in regional planning entity such as a Metropolitan Planning Organization (MPO)

Action 4.J.2. Pursue Innovative Funding for Pedestrian and Bicycle Connections and Network

This action focuses on identifying options for implementing plans and physically expanding and bolstering Bozeman's bicycle and pedestrian network. Note, the term "bicycle and pedestrian" is intended to be all encompassing and includes e-bikes, e-scooters, mobility scooters, and other forms of non-car travel. The City of Bozeman continues to invest in bike and pedestrian infrastructure through a combination of local and private funding sources. Through the City's Complete Streets principles are applied on new City projects, privately funded development, and incrementally through a series of smaller improvements and activities over time. Examples of innovative funding efforts have included bonds and urban renewal districts.

Future bicycle and pedestrian projects should respond to existing and future gap analyses, such as those conducted in the Bozeman Transportation Master Plan or through walking audits. In conjunction with adding more miles of bicycle and pedestrian infrastructure, the City will explore opportunities to bolster existing infrastructure to better serve all needs. For example, making intersection improvements, adding signs to cue motorists to share the road, or narrowing travel lanes marginally to accommodate bicycle and pedestrian infrastructure are just a few ways to transform existing infrastructure to better meet the needs of bicyclists and pedestrians. Other actions the City will consider include mapping all bike routes classified by comfort level, implementing more robust wayfinding, and identifying funding sources for protected and/or grade separated trail infrastructure.

The City will also seek opportunities to partner with developers to encourage the addition of new multi-modal amenities as part of development and redevelopment projects (see Action 3.G.2.). Updating transportation design standards and policies to better reflect all needs and abilities, including e-bikes and emerging technologies, will be an important first step to provide clear direction to developers. To prepare for autonomous and low-speed modes that may compromise pedestrian safety, planning and defining appropriate non-car travel on sidewalks, bike lanes, and shared use paths will be part of this update.

As specified in the 2017 Transportation Master Plan, ensure that the Unified Development Ordinance is amended to require sidewalk construction as a basic component of subdivisions and should be installed with the streets and utilities before individual lots are developed.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • Montana State University • Gallatin Valley Land Trust • Bzmn Area Bicycle Advisory Board • Montana Dept. of Transportation • Pedestrian Traffic Safety Committee • Gallatin Valley Bicycle Club • Bridger Bowl	• 4.5 Housing and Transportation Choices • 6.2 Protect Local Air Quality • 6.3 Climate Action • 6.5 Parks, Trails, and Open Space	• % increase in walking, biking, carpool, and transit commuting • Reduction in per capita vehicle miles traveled • Increase in network size and connection (i.e., miles of new bike lanes or sidewalks)

Action 4.J.3. Improve Maintenance of Multi-Modal Infrastructure

Once multi-modal infrastructure is developed, maintenance is a critical process for ensuring the reliability and safety of that infrastructure. For instance, prioritizing shared-use, separated paths for plowing after a winter storm will ensure a safe path of travel for bicyclists and pedestrians who prefer or rely on these paths for travel.

Additionally, proper maintenance can sometimes avoid the need to expand or extend infrastructure. For instance, intersection improvements can improve traffic flow without adding vehicle lanes, and dedicating existing lanes for carpool, vanpool, or transit can help prioritize alternative modes without the need for an expansion project. Improving lighting and wayfinding is another way the City can enhance multi-modal infrastructure to improve safety and encourage use.

As part of this action, Bozeman will seek to identify funding mechanisms, funding sources, and implementation partners to dedicate to the development and implementation of a maintenance program for multi-modal infrastructure. Private properties can play an important role in maintenance as well, by actions such as shoveling snow immediately following a winter storm or parking off the street to accommodate for snow storage while still allowing bicycle access.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measure*
Level 1	• **City of Bozeman** • Montana State University • Gallatin Valley Land Trust • Bozeman Area Bicycle Board • Pedestrian Traffic Safety Committee • Montana Department of Transportation • Bridger Bowl	• 4.5 Housing and Transportation Choices • 6.2 Protect Local Air Quality • 6.3 Climate Action • 6.5 Parks, Trails, and Open Space	• Annual miles of bike lane and shared-use path sweeping, cleaning, and plowing

Action 4.J.4. Pursue Sustainable Transit Funding and Expansion

Accessible and reliable transit options are a critical component of a high-functioning multi-modal system. This action focuses on identifying a dedicated funding source to support and expand Bozeman's existing transit system. Streamline Transit provides Bozeman community members with fixed route bus service, campus shuttles, paratransit services, and transportation to and from Bridger Bowl.

Though funding is an important driver for establishing or expanding transit systems, a community must first, or at least concurrently, work toward developing in a manner that supports transit. For instance, it may be challenging to support transit in areas with fewer than 12 to 16 people per acre. In addition to ensuring zoning districts allow the right mix of land uses and number of units, the City can work with developers to incentivize transit-oriented development (TOD) along existing or planned transit routes. TODs are typically characterized as mixed-use districts, with enough jobs, residential units, and/or attractions and amenities to support transit. Other features of TODs include walkability and bikeability, to ensure commuters can connect to and from transit stops. Many of the other actions in this solution will be critical to providing first and final mile connections to and from transit stops.

Though reliable, fixed-route transit is often the backbone of a high-functioning transit system, many communities have identified significant need for on-demand microtransit (e.g., cars and vans) and micromobility (e.g., bikesharing and e-scooters) to act as an extension of regular transportation services. Enabled by transportation technology, these flexible solutions can provide curb-to-curb or stop-to-stop service. Sometimes referred to as Mobility as a Service, through integrated technology and public and private partnerships, microtransit can be crucial for helping connect the entire community to central transit lines, especially older adults or persons with a disability.

As Bozeman's transit system grows and evolves with an emphasis on Transit Oriented Development and Mobility as a Service, the City will work with partners to identify dedicated transit funding to help ensure the maintenance and expansion of Bozeman's transit system. Funding may come from several sources, such as the formation of a Metropolitan Planning Organization (MPO), Regional Transit Authority, transportation improvement district, or other avenues such as tolls or fees. As avenues for funding are identified, the City may conduct the following feasibility studies to inform investment of transit dollars: integration of transit signal priority, expansion of Park and Ride and routes, including circulator routes for downtown and the airport.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measure*
Level 1	• **Streamline Transit** • City of Bozeman • Gallatin County • City of Belgrade • Montana State University • Montana Department of Transportation • Bridger Bowl • Human Resource Development Council	• 4.3 Strategic Infrastructure Choices • 4.5 Housing and Transportation Choices • 6.2 Protect Local Air Quality • 6.3 Climate Action	• Increased transit ridership • Reduction in per capita vehicle miles traveled • # of new transit connections

Action 4.J.5. Support Employee Trip Reduction Programs and Transportation Demand Management

The majority of road congestion in the morning and afternoon is the result of employees commuting to and from work during the same time frame. This action focuses on reducing the total number of trips taken by employees and shifting those trips away from morning and afternoon rush hours whenever possible. Reducing the number of vehicles traveling during peak times is often referred to as Transportation Demand Management (TDM). Already, the City of Bozeman has a number of TDM strategies in place as part of the Bozeman Transportation Master Plan. The City plans to add a TDM coordinator in the 2021 fiscal year to lead the implementation of these efforts.

In addition to building and maintaining a more robust multi-modal network, TDM focuses on education, outreach, and partnerships. Efforts may include hosting workshops, webinars, or temporarily closing streets to cars for "Open Streets" events to provide resources to the community on the options, benefits, and "How Tos" of multi-modal transportation. Outreach efforts will promote new efforts and tools to help navigate transportation options, such as the commuter management and ridesharing platform "Ride Amigos" launched through the Bozeman Commuter Project. Additionally, the City will partner with large employers to encourage or require the adoption of company specific performance-based TDM programs. For instance, organizations can provide a transportation incentive for employees that walk, bike, carpool or use transit to get to work.

In addition to helping employees shift how they get to work, TDM strategies will help shift when employees go to work. Companies can lead by example by allowing flexible work schedules or instating more robust work from home policies. In response to the COVID-19 pandemic, many companies have been forced to adapt work flows to allow employees to work from home. Continuing to require and/or allow employees to work from home could have a significant and positive impact on transportation demand.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **Streamline Transit** • **City of Bozeman** • Gallatin County • City of Belgrade • Montana State University • Montana Department of Transportation • Bridger Bowl • Major employers (e.g., Bozeman Health, Oracle, BSD7, Zoot, Simms) • Human Resource Development Council	• 4.3 Strategic Infrastructure Choices • 4.5 Housing and Transportation Choices • 6.2 Protect Local Air Quality • 6.3 Climate Action	• Increased transit ridership • Reduction in per capita vehicle miles traveled

Action 4.J.6. Support Regional Transit Service Coordination and Outreach

Beyond local transit service, connections to the greater region are a significant need in order to reduce vehicle trips and mitigate traffic congestion from tourism and population growth. While Streamline Transit provides Bozeman community members with fixed route bus service, campus shuttles, paratransit services, and transportation to and from Bridger Bowl, more can be done to connect Bozeman to other communities and regional destinations.

This action focuses on working with private and public transport operators to expand transit service between major regional cities and visitor destinations. It also includes exploring the feasibility of a train system to connect regional destinations, including showing support for restoring the North Coast Hiawatha Route passenger rail.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • Regional Cities • Streamline Transportation • Sky Line Transportation • Montana State University • Chamber of Commerce	• 2.2 Infrastructure Investments • 6.2 Protect Local Air Quality • 6.3 Climate Action	• Expansion of transit service within the region

Action 4.J.7. Leverage Parking Policies to Encourage Other Modes of Transportation

Parking infrastructure and policy play a significant role in vehicle use, vehicle miles traveled (VMT), and emissions. An abundance of free parking encourages and enables the decision to drive versus walking, biking, carpooling, or using transit.

Public and private parking inventory, Unified Development Code parking requirements, and parking management all influence mode choice. This action reinforces coordination with other transportation studies and plans to ensure that the community’s parking paradigm and policies strengthen and support planned transit, bicycle, and pedestrian infrastructure improvements.

Where expanded access to multi-modal transportation options are available, the City will pursue updates to the Unified Development Code that eliminate minimum parking requirements in commercial districts and affordable housing projects and reduce parking minimums elsewhere (see Community Plan M-1.12). Further, the City will proactively coordinate with Montana State University to ensure complementary parking management strategies that encourage multi-modal transportation options.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Parking Commission • Montana State University • Chamber of Commerce	• 2.2 Infrastructure Investments • 6.2 Protect Local Air Quality • 6.3 Climate Action	• Increased transit ridership • Reduction in per capita vehicle miles traveled

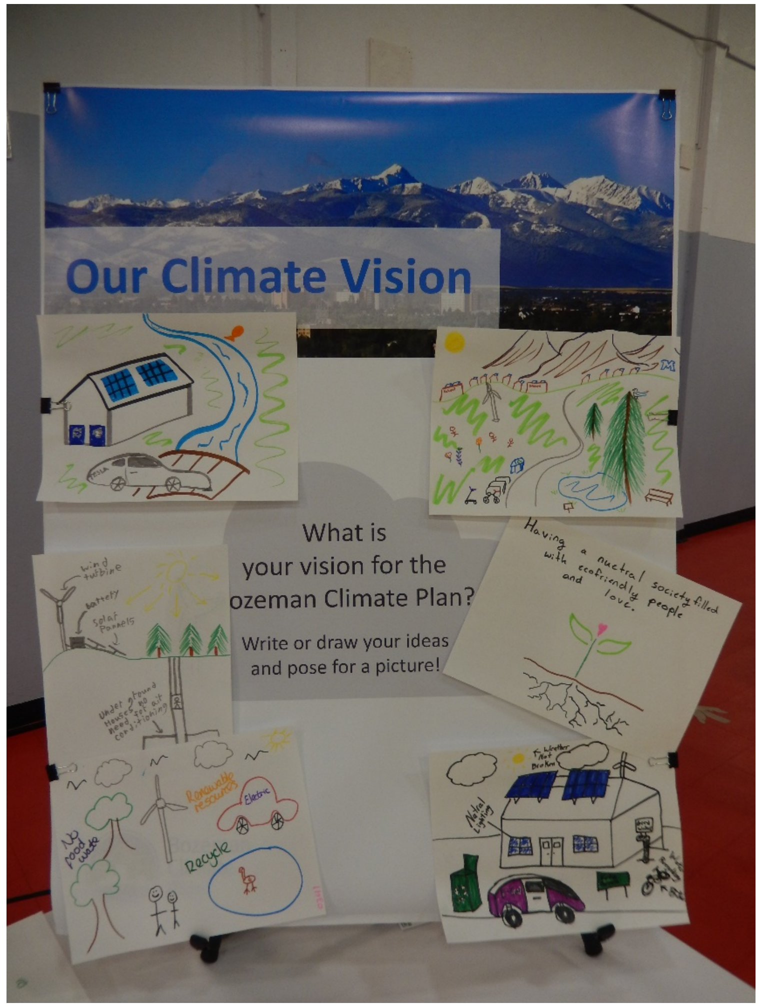
Action 4.J.8. Develop Bike and Car Share Programs

Bike, scooter, and car share programs help to increase access to alternative modes of transportation without significant investment on behalf of users. Vehicle share programs can all be used to improve first- and last-mile connections to and from transit. Car share programs in particular can help reduce the need for a second vehicle for households. As part of this action, the City will explore the feasibility of a community vehicle share program. Implementation of this program (or programs) may be City-led, a public-private partnership, or a private system.

For bike and scooter share programs, electric options can help improve mobility for some participants. However, if electric options are considered, the City will look to peer communities to identify best practices and policies for enhancing safety and reducing pedestrian-vehicular traffic conflicts. Similarly, a car share program could also showcase electric vehicles. Many studies have shown that drivers exposed to electric vehicles are significantly more likely to purchase one. Thus, including electric vehicles as part of a car share program could complement efforts identified in Solution K.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 3	• **City of Bozeman** • **Downtown Bozeman Partnership** • **Montana State University** • Bozeman Area Bicycle Advisory Board • Gallatin Valley Bicycle Club • High demand districts (e.g., Cannery District, Northeast Neighborhood)	• 4.3 Strategic Infrastructure Choices • 4.5 Housing and Transportation Choices • 6.2 Protect Local Air Quality • 6.3 Climate Action	• Increased transit ridership • Reduction in per capita vehicle miles traveled

Solution K. Decrease Direct Vehicle Emissions



Despite best efforts, some portion of community trips will always require a vehicle. Even though transit can be more efficient on a per-person basis, traditional diesel buses emit large amounts of pollutants and greenhouse gases. Thus, this solution focuses on promoting more efficient vehicles, vehicles that use alternative fuels, and electric vehicles.

Many barriers exist to the adoption of electric vehicles (EVs), including range anxiety, cost of procurement, and limited local availability. Some limited federal and state incentives are available to assist in the procurement of alternative fuel vehicles and electric vehicle supply infrastructure. Leading by example can be an important mechanism for spurring the private adoption of electric vehicles. Advertising electric fleet vehicles and making public charging stations visible and readily available can help build trust in community members. Public-private partnerships are another effective avenue for encouraging the expansion of electric vehicle supply infrastructure. Other ways to remove barriers can include requiring new construction to be pre-wired for electric vehicle charging or even requiring the provision of "preferred parking" for electric vehicles.

Equity

Ideally, all trips made via car, van, or public transit would do so using efficient, alternative fuel, and/or electric vehicles. Unfortunately, at this time, the price of these vehicles can be cost-prohibitive. For electric vehicles specifically, there is little to no market for used vehicles. One option to overcome this barrier is to support an electric vehicle car share, which would enable community members to access electric vehicles without the initial procurement costs. Electrifying transit fleets is another good means of providing access to cleaner means of transportation.

Human Health & Well-Being

Decreasing direct vehicle emissions can improve air quality, which can reduce the risk of associated illnesses, such as respiratory and heart disease.

City Assets

The City of Bozeman can lead by example by electrifying its light and medium duty fleet and by installing public charging stations at City facilities. The City can continue to encourage the conversion of all transit fleet vehicles to electric or hybrid options.

This solution primarily addresses Bozeman's greenhouse gas mitigation goals, while also providing supporting resiliency benefits.

Emissions Reduction Benefits

This is a supporting mitigation solution to achieve the 2025 and 2030 mitigation goals and a primary solution to achieve the 2050 goal.

Emission sources impacted include:

- Gasoline/Diesel
- Electricity

Resiliency Benefits

- Conserve natural resources
- Protect human health

Related Solutions

- Solution D. Increase Utility Clean Energy Mix
- Solution F. Increase Community-Based Distributed Renewable Energy Generation
- Solution G. Facilitate Compact Development Patterns
- Solution J. Increase Walking, Bicycling, Carpooling and Use of Transit
- Solution L. Improve Air Travel Efficiency

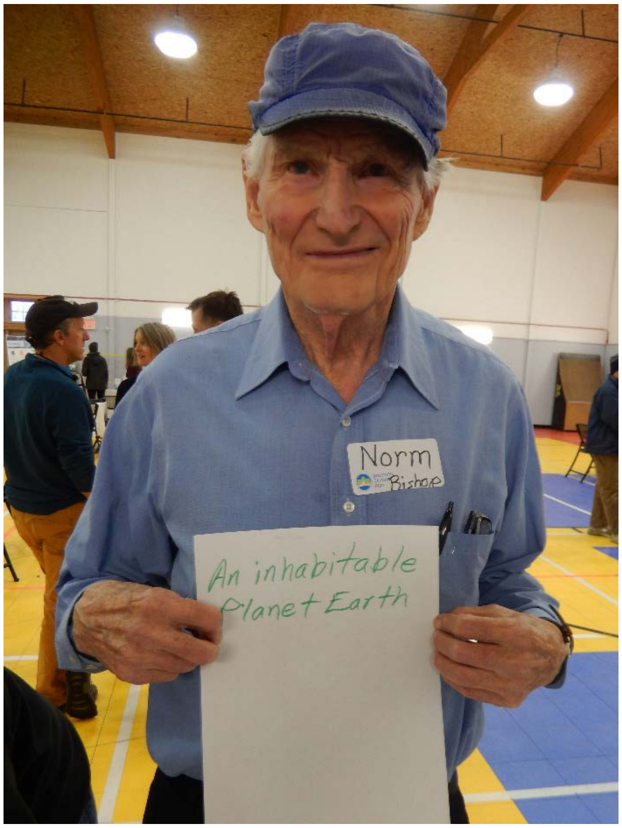
Action 4.K.1. Support Community EV Roadmap Development

An EV roadmap is a first step toward increasing the number of EVs in a community. An EV roadmap typically identifies a community vision, goal(s), and strategies to increase the total number of EVs on the road. This action seeks to build on the work completed to date by the Montana Solar Powered Community Transportation Initiative to draft a comprehensive EV roadmap for the City of Bozeman.

This roadmap will establish Bozeman's EV baseline, including approximate number of registered EVs in Bozeman and existing EV infrastructure. The roadmap will identify key steps the community can take to improve from that baseline. Strategies within the roadmap would include an evaluation of actions, such as requiring a certain percentage of parking spaces be EV-ready for new construction and coordinated utility distribution system planning. Key community stakeholders will include utility representatives, City staff, MSU representatives, and others listed below.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • NorthWestern Energy • Downtown Bozeman Partnership • Montana State University • Chamber of Commerce • Montana Dept. of Environmental Quality • Yellowstone Teton Clean Cities	• 4.3 Strategic Infrastructure Choices • 4.5 Housing and Transportation Choices • 6.2 Protect Local Air Quality • 6.3 Climate Action	• Increase % of EVs and PHEVs registered in Gallatin County • Reduce emissions per vehicle mile traveled

Action 4.K.2. Collaborate to Install Publicly Accessible EV Infrastructure



Though the majority of EV charging occurs at home, access to public EV charging infrastructure is still often cited as one of the biggest barriers to EV adoption. Providing publicly available charging infrastructure at City facilities, workplaces, and key destinations can significantly reduce barriers to adoption by signaling that EVs are a priority and that EV owners will always have access to charging infrastructure.

While the City may choose to provide some publicly available infrastructure, public-private partnerships and private investment will be crucial to sufficiently build out the necessary charging infrastructure. Charging stations should be right-sized to match the typical time spent in a location. For instance, Level 2 charging stations are appropriate for businesses or destinations where community members may typically spend 2 to 4 hours. Level 3, or DC fast chargers, are more appropriate for major travel corridors, such as I-90, to facilitate quick charging while travelers stop at rest stops, gas stations, etc. Montana State University will be a key partner as they embark on an EV charging station pilot in 2020.

As part of this action and in conjunction with Action 4.J.6., the City may explore charging stations for electric bikes, electric scooters, and EVs as part of a vehicle-sharing program.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • NorthWestern Energy • Downtown Bozeman Partnership • Montana State University • Chamber of Commerce • Montana Dept. of Environmental Quality • Yellowstone Teton Clean Cities	• 2.1 Business Growth • 4.3 Strategic Infrastructure Choices • 4.5 Housing and Transportation Choices • 6.2 Protect Local Air Quality • 6.3 Climate Action	• Number of public charging stations • Increase % of EVs and PHEVs registered in Gallatin County • Reduce emissions per vehicle mile traveled

Action 4.K.3. City Fleet and Transit EV Investment

One of the best ways Bozeman can lead by example is to invest in EVs for City fleet and Streamline Transit vehicles. As part of the EV roadmap, or under a separate effort, the City of Bozeman can identify a path for electrifying and modernizing the City's fleet. The first step in developing an action plan for fleet electrification is connecting with NorthWestern Energy. NorthWestern Energy can help the City identify areas with enough capacity to handle additional electricity demand and areas that may require additional investment. Once conversations with the utility have been initiated, the next step of fleet electrification is to consider a telematics study, which can help identify vehicles most eligible for EV replacement and the best EV models for the job. Following this step, the City may update their vehicle purchasing policy in conjunction with their vehicle replacement plan to meet electrification targets. Additional factors of fleet electrification include developing charging infrastructure and charging plans, conducting employee outreach and education, and coordinating training and maintenance. The Climate Mayor's EV Purchasing Collaborative may serve as a useful resource for preliminary efforts.

Preliminary fleet electrification activities should focus on the City's light-duty fleet, though the City may also consider and put policies in place to prepare for the electrification of heavy-duty vehicles as applicable. Where full electrification is not possible, the City may pursue other alternative fuel sources such as natural gas or biofuels.

Finally, the City will also work with Streamline Transit to conduct an updated feasibility analysis for electrifying transit vehicles.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Streamline Transit • City of Belgrade • Gallatin County • Montana State University • Bozeman Health • Big Sky Resort • Bridger Bowl	• 2.1 Business Growth • 4.3 Strategic Infrastructure Choices • 4.5 Housing and Transportation Choices • 6.2 Protect Local Air Quality • 6.3 Climate Action	• Improve light-duty city fleet average MPG • Reduce emissions per vehicle mile traveled • % of fleet converted to EV

Action 4.K.4. Advocate for EV Utility Rates, Incentives, Infrastructure, and Efficiency Standards

Advocacy will be an important aspect of implementation for this plan and can have broad and meaningful impacts toward reducing direct vehicle emissions. This action focuses on connecting partners and building community support to lead a broad range of advocacy efforts. Topics for EV advocacy may be split broadly into two categories: vehicle emissions standards and support for alternative fuel vehicles.

This plan encourages local advocates to engage state and federal policy makers to continue improving vehicle emissions standards (for instance, through a State-wide zero emission vehicle standard). Zero emission vehicles require manufacturers to manufacture a certain percent of zero emission vehicles as part of their fleet. In addition to reducing the overall potential emissions of a manufacturer's fleet, this would also increase market availability of efficient and alternative fuel vehicles. Alternatively, or in conjunction, supporters might advocate for state and federal tax credits and financial incentives for fuel-efficient and alternative fuel vehicles. Tax credits and financial incentives may help provide greater access to efficient and alternative-fuel vehicles. Ultimately, an uptick in these vehicles is expected to lower production costs in the long run as manufacturers realize economies of scale and technology continues to advance.

Finally, City of Bozeman partners could work with NorthWestern Energy to advocate for the development of an EV infrastructure plan and innovative funding models. Currently, NorthWestern Energy is exploring the possibility of an off-peak charging rate. EVs hold a lot of potential for evening out demand curves if the majority of charging occurs during off-peak hours. Off-peak charging incentives could produce a win-win for the utility and customers.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Streamline Transit • Montana Dept. of Environmental Quality • Montana Legislature • NorthWestern Energy • Public Service Commission	• 2.1 Business Growth • 4.3 Strategic Infrastructure Choices • 4.5 Housing and Transportation Choices • 6.2 Protect Local Air Quality • 6.3 Climate Action	• Improved state efficiency and emissions standards • Establishment of utility rate to incentivize off-peak EV charging

Action 4.K.5. Limit Wasteful Vehicle Emissions

Wasteful vehicle emissions occur when vehicles are circulating or idling unnecessarily. This action focuses on implementing policies and educational campaigns to reduce unnecessary emissions. Steps the City could take include establishing and enforcing an anti-idling policy. This policy could apply to semi-trucks, delivery trucks, transit buses, tour buses, construction equipment, and/or light-duty vehicles. The City might partner with local businesses to help enforce anti-idling policies. Adding signage in common locations, such as near pre-schools, can be another mechanism for encouraging drivers to turn off their vehicles.

Another common contributor to wasteful emissions is the result of circulating in search of parking. Installing smart parking signs, which tell travelers where spaces are available in a lot or garage, can reduce time spent circling for a spot. Of course, encouraging community members and visitors to rely on active modes and transit to destinations is another great way to avoid emissions associated with circulation. The City may also explore options to prohibit vehicles from driving exclusively for advertising.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Montana State University • Streamline Transit • Montana Dept. of Environmental Quality • Yellowstone Teton Clean Cities	• 6.2 Protect Local Air Quality • 6.3 Climate Action	• Reduce Particulate Matter (PM)

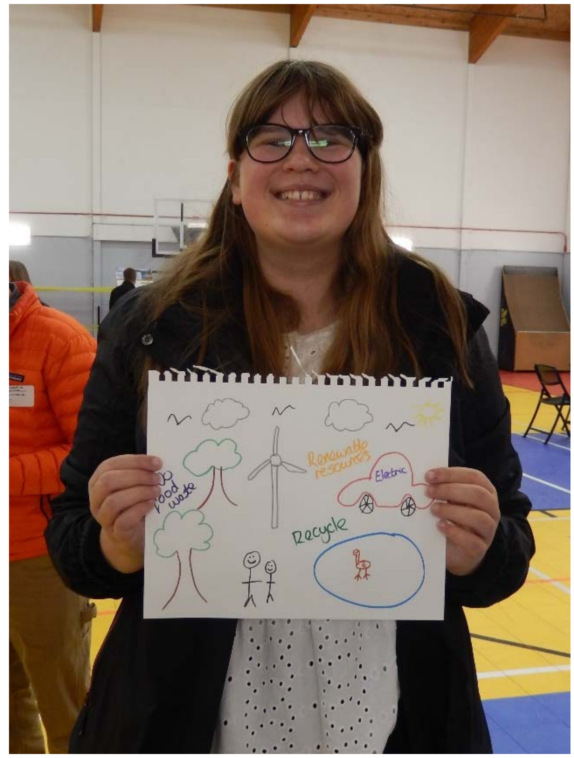
Action 4.K.6. Support EV Group Buy and Outreach

This action focuses on providing EV outreach and education to help address adoption barriers related to misinformation. As part of this action, the City can partner with Yellowstone-Teton Clean Cities to develop an educational campaign, organize a ride-and-drive, or organize group buys. Group buys are typically orchestrated in partnership with local dealerships and can significantly reduce the cost of purchasing an electric vehicle by offering a discount when multiple parties agree to purchase at once. Other educational partners may include the Montana Department of Environmental Quality and Western Transportation Institute. Topics of educational materials should include the financial benefits of EV ownership, information about various types of charging infrastructure, how to pick the right vehicle to suit individual needs, and environmental benefits, among others.

While replacing conventional internal combustion engine vehicles with EVs can reduce direct tailpipe emissions, EV adoption will not reduce congestion. Thus, efforts promoting EV use should still occur in conjunction with efforts promoting the reduction of trips overall and replacement of trips with alternative modes of transportation whenever possible.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 3	• **City of Bozeman** • Montana Dept. of Environmental Quality • Yellowstone-Teton Clean Cities • NorthWestern Energy • Car dealerships • Major employers	• 2.1 Business Growth • 4.5 Housing and Transportation Choices • 6.2 Protect Local Air Quality • 6.3 Climate Action	• Increase % of EVs and PHEVs registered in Gallatin County • Reduce emissions per vehicle mile traveled

Solution L. Improve Air Travel Efficiency



This solution is focused on supporting the development of alternatives to traditional air travel. It includes technology improvements to enhance the fuel efficiency of air fleets, as well as offering alternatives to air travel which could include shared rides and vehicles, shuttles and buses, rail transportation, and even more advanced technologies such as hyperloops.

Air travel and tourism provide tremendous economic benefits to the Bozeman community, and the Bozeman Yellowstone International Airport is the busiest in the state. This solution does not seek to eliminate or reduce visitation to Bozeman, but instead focuses on making travel to and from Bozeman more efficient and less impactful from an emissions perspective. Advancing this solution requires advocacy and standards for the airline industry to adopt new technologies and operate more efficiency.

In addition to more efficient air travel, this solution also includes the development of air travel alternatives at the regional scale. Alternatives like shuttles and buses necessitate collaboration with other regional partners including tourist destinations and attractions, such as nearby national and state parks, trailheads, ski areas, and Montana State University. Furthermore, such alternatives should also seek to provide alternatives to traditional vehicle travel and leverage electric or alternative fuel technology to create even further emissions reduction benefits.

Though airline travel can have a big impact on greenhouse gas emissions, resources earmarked for mitigating air travel are only relevant to community members with means and access to airline travel. Thus, the resources spent reducing a very specific portion of emissions could have a much larger, positive impact on equity issues elsewhere.

Still, mitigating emissions from air travel could reduce noise and air pollution associated with plane travel. Since it is likely that members of the community living proximate to the airport do not have the resources to relocate, reducing noise and air pollution could have a positive equity impact on those community members.

The City of Bozeman does not have authority over the airport operations (the airport is owned by the Gallatin Airport Authority) or operate a local bus system. However, the City can continue to invest in the Streamline system (operated by the Human Resources Development Council) and support connections to other regional transit services to provide other alternatives to regional air travel.

This solution primarily addresses Bozeman's greenhouse gas mitigation goals, while also providing some supporting resiliency benefits.

Emissions Reduction Benefits

This is a supporting mitigation solution to achieve the 2025, 2030, and 2050 mitigation goals.

Emission sources impacted include:

- Diesel

Resilience Benefits

- Conserve natural resources
- Protect human health

Related Solutions

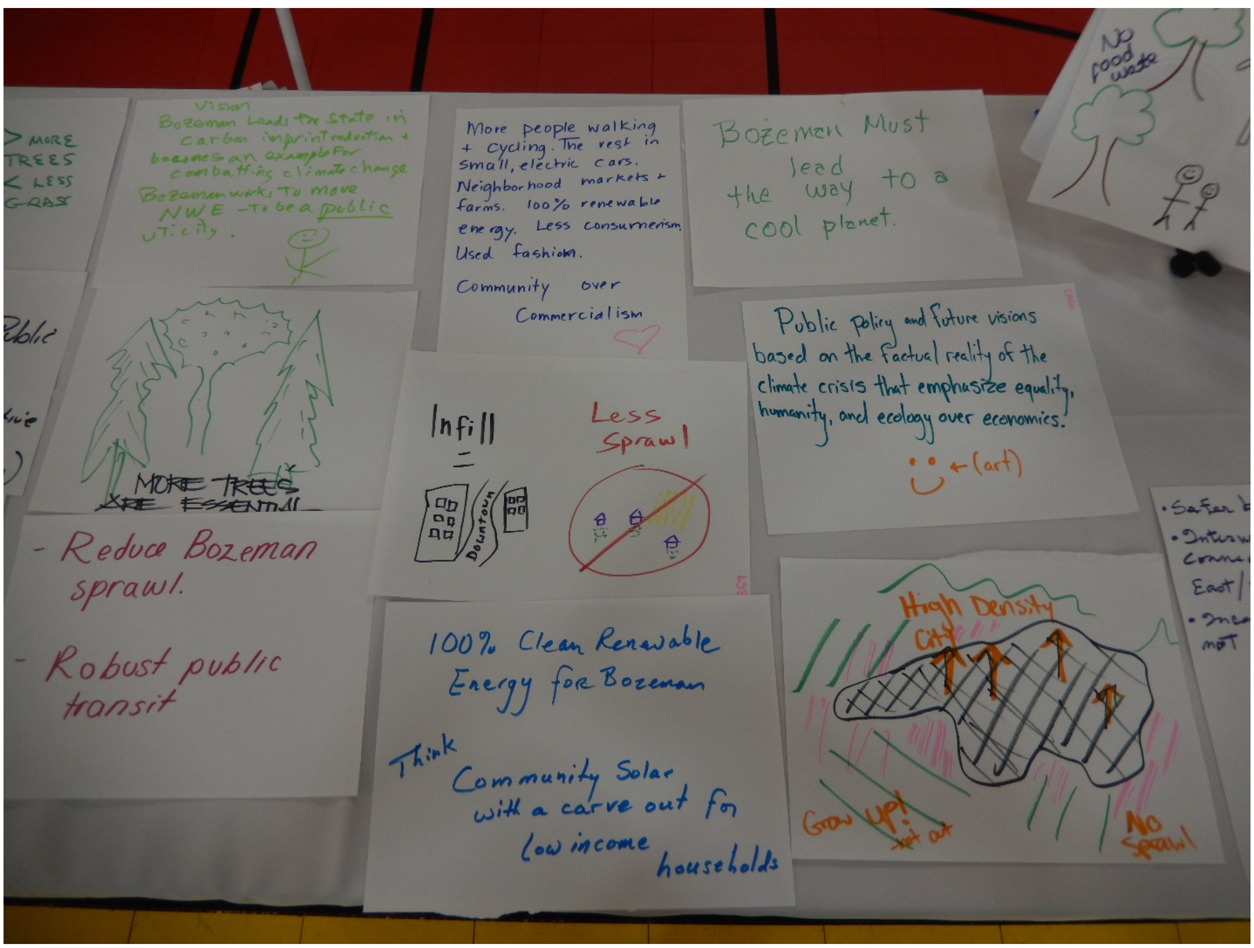
- Solution J. Increase Walking, Bicycling, Carpooling and Use of Transit

Action 4.L.1. Build Awareness of Air Travel Impacts and Alternatives

This action focuses on promoting alternative modes of travel in place of air travel. In partnership with City departments and community organizations, implementation of this action will include development of educational materials to increase awareness of the impacts of flying, share alternative modes of transportation, and encourage community members to remain local. For instance, short airline trips can be replaced with car, bus, or train travel. Air travel for business can be avoided if meetings are replaced with teleconferencing and other remote communication and engagement tools.

In particular, the City of Bozeman can lead by example by limiting meetings for which City employees must fly and even by limiting instances that require contractors to fly into Bozeman. Finally, the City of Bozeman can partner with the Bozeman Yellowstone International Airport to encouraging passengers to offset carbon emissions or through signage at the airport or through the ticket purchasing process.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Montana State University	• 6.2 Protect Local Air Quality • 6.3 Climate Action	• Reduced annual employee air travel miles



Action 4.L.2. Advocate for Increased Air Travel Efficiency

The carbon intensity of flying is significant, with varying levels of fuel efficiency and carbon intensity among commercial carriers. Currently, no local, state or federal fuel efficiency requirements exist for air travel in the United States.

This action is a call for the Bozeman community to advocate for stringent efficiency standards for air travel. City partners and community representatives can engage in state and federal-level advocacy to promote more fuel efficient and alternative-fuel air travel.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 3	• **City of Bozeman** • Bozeman Yellowstone International Airport • Montana State University • Chamber of Commerce • Montana Dept. of Environmental Quality	• 2.2 Infrastructure Investments • 6.2 Protect Local Air Quality • 6.3 Climate Action	• Reduced emissions per passenger trip

Increasing Resiliency to Climate Hazards

The following opportunities and considerations summarize how the solutions in this chapter can help improve resiliency to future climate hazards.

Opportunities	Considerations
Extreme Heat	
• Replacing or shading parking areas can mitigate urban heat island impacts. • Mitigating vehicle emissions could improve air pollution typically worsened on hot days	• Active forms of transportation (such as biking and walking) can be dangerous in extreme heat. • Active transportation routes should be designed with trees, shade structures, and with stations to refill water.
Flooding	
• EVs could release fewer pollutants than internal combustion engine vehicles during flood events. • A frequent and reliable transit system can provide community members with a safe mode of travel during flooding.	• Infrastructure built to support alternative modes of transportation must be designed to withstand flood events. • Active forms of transportation may be difficult or dangerous in the event of flooding.
Drought & Mountain Snowpack	
• Transit service to nearby ski areas may need to expand beyond winter months as ski resorts add activities in other seasons in anticipation of reduced snowpack.	• Severe droughts can lead to an increase in particulate matter, which can exacerbate respiratory issues for people walking and biking.
Wildfire	
• A robust transit system could help evacuate residents in the event of a wildfire.	• Active transportation may be limited during periods of poor air quality. • Severe wildfires can disrupt transit service and routes.
Winter Storms	
• A transit system can provide a convenient, safe, and affordable mode of transportation when snow/ice make walking, biking, or driving difficult or dangerous.	• Active transportation (e.g., biking and walking) could be disrupted by deep snow. • Active transportation infrastructure must be plowed regularly to maintain a diverse network of options. • Electric vehicles and buses may lose range in extremely cold weather.

Focus Area 5: Comprehensive & Sustainable WASTE REDUCTION

Emissions from landfilled waste accounted for 5% of Bozeman's emissions in 2016, and emissions from this sector are on the rise due to increases in construction waste and municipal solid waste. Despite these challenges, on a per-capita basis, total landfilled waste is declining, meaning that efforts to reduce, reuse, and recycle waste are showing signs of progress.

The solution in this chapter emphasizes a comprehensive and system-wide approach to address community waste. It takes into account the current realities of the waste industry and helps Bozeman reimagine a new type of economy which is circular in nature and that provides more sustainable long-term solutions for reducing and managing waste.

RELATED PROGRAMS & RESOURCES

- City Compost Collection
- Single-Stream Recycling
- Cleanup Bozeman Program

Emissions from the decomposition of waste make up a relatively small portion of the City's GHG emissions footprint, so the overall impact of these solutions on the City's emissions goals is limited. These solutions do however have strong impact on the regional and global emissions for the production and transportation of goods, which is not reflected in the City's inventory. In order to optimize the emission mitigation and resiliency potential, the Bozeman community will need to:

- **Solution M. Move Toward a Circular Economy and Zero Waste Community**
  Manage solid waste through a hierarchy of source reduction, reuse, recycling, composting and anaerobic digestion, and waste to energy conversion, with landfilling waste as a last resort.

Detailed waste-related targets for each goal year are shown in Table 10.

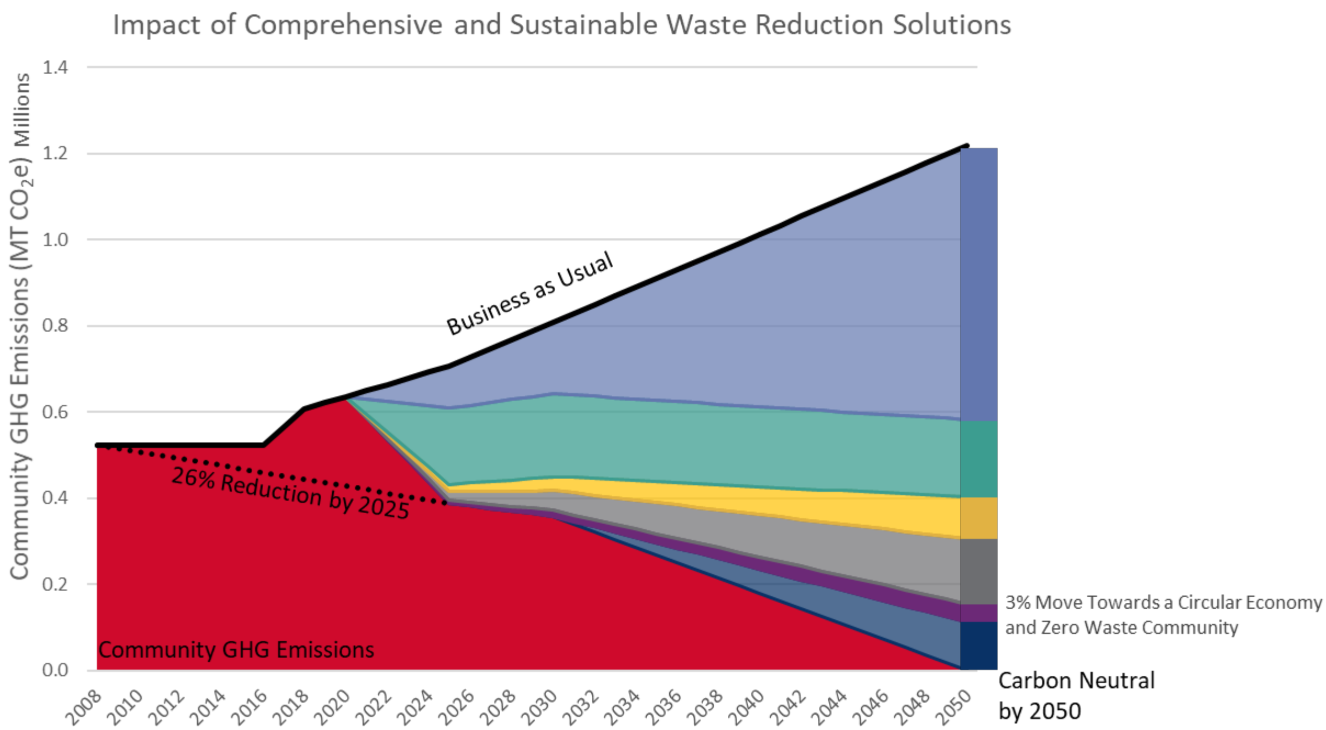
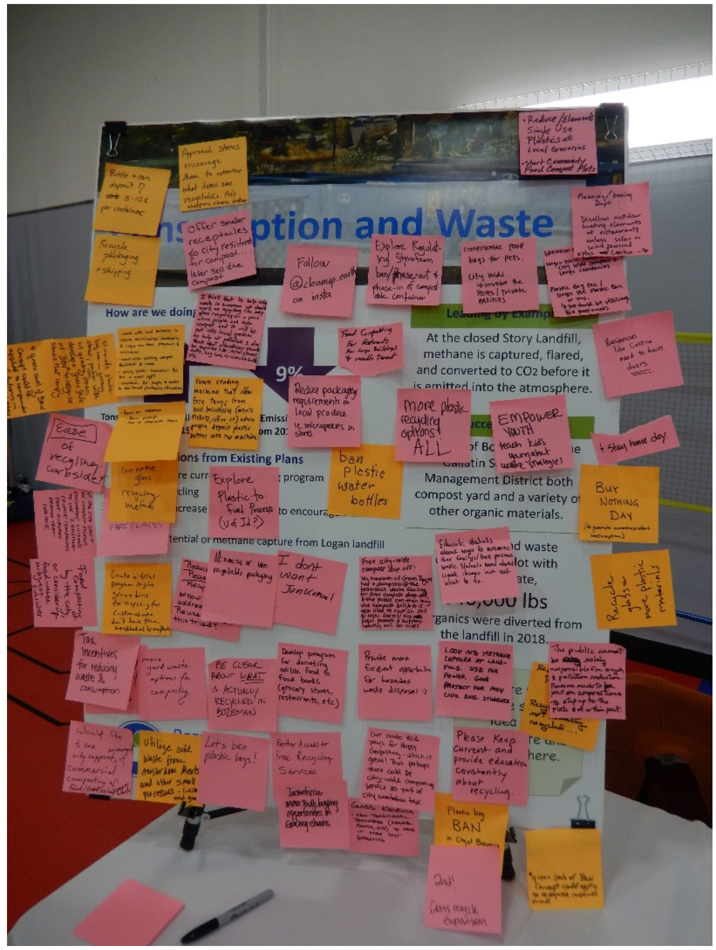


Figure 20. Projected waste GHG emissions reduction by solution

Table 10. Projected waste reduction mitigation targets

Solution	*Metric (annual savings)*	*Target* 2025 *Paris Accord*	*Target* 2030 *100% Net Clean Electricity*	*Target* 2050 *Carbon Neutral*
Solution M. Move Toward a Circular Economy and Zero Waste Community	GHG Emissions Savings (MT CO_2e)	7,100	14,100	42,300

Solution M. Move Toward a Circular Economy and Zero Waste Community



This solution focuses on the management of solid waste through a hierarchy of source reduction, reuse, recycling, composting and anaerobic digestion, and waste to energy conversion, with landfilling waste as a last resort. Waste reduction is strongly influenced by upstream production practices and the availability of services such as composting and recycling. As such, this solution requires an economy-wide effort—one that aims to eliminate waste and keep resources in use for as long as possible. This concept of a circular economy works in tandem with the goal of zero waste, which aims to prevent waste from being sent to landfills.

Source reduction is simply waste prevention, or the elimination of waste before it is created. This approach involves refining supply chain and purchasing practices to stop waste before it happens. A waste audit is a great way to begin to understand the amount and types of waste generated.

Reuse is the practice of using an item or product again, either for its originally intended use, or for a new use. Short of some resale shops and small-scale efforts, few comprehensive source reduction or reuse programs or community-scale activities currently exist in Bozeman.

In high-growth communities, total waste tonnage often tracks with increases in population and construction activities. Source reduction and reuse practices can help decouple trends in community growth from total landfilled waste trends.

The City of Bozeman offers single-stream curbside recycling services for customers. Recycling is more expensive than landfilling waste, and recent shifts in the global recycling market have resulted in changes at the local level in terms of what can be collected and effectively recycled. For comprehensive recycling services to be viable long-term, communities like Bozeman will need to work together to address recycling market imbalances and challenges. Reduction in waste is far more environmentally friendly, efficient, and cost-effective than relying heavily on recycling markets and services.

Composting and anaerobic digestion are processes that decompose organic materials such as yard waste and food scraps to produce a nutrient-rich soil amendment that helps restore soil health. The City of Bozeman offers seasonal yard waste compost collection to solid waste customers. One recent success story is a food waste composting pilot with Montana State University, through which more than 540,000 pounds of organic materials were diverted from the landfill in 2018.

Decomposing organic waste releases methane, a potent greenhouse gas. Methane capture and recovery projects can help reduce the methane emissions from landfills. Methane at Logan Landfill is currently vented and released into the atmosphere. Methane from the closed Story Landfill is captured and flared.

Equity

There are many equity perspectives to examine when considering circular economy and zero waste solutions. For many people, reducing consumption can free up financial resources and support economic mobility. However, system-wide creation of more durable goods and products can result in higher initial costs, which can limit access. With respect to recycling and composting, residents of multi-family properties, visitors, and commercial properties often face significantly more barriers to accessing these services than traditional curbside residential waste customers. To make recycling and composting recycling services viable, changes in waste rate structures or other disincentives are often necessary, which can be onerous for lower income or cost-burdened community members. Similarly, lowering the cost of trash services below the cost of providing services might pull municipal resources away from other key programs.

Human Health & Well-Being

Adequate sanitation services are critical to support community health and well-being. Beyond the immediate needs of proper waste collection and disposal, there are many other health and wellness benefits of a circular economy and zero waste practices, including but not limited to pollution reduction, portion control, fresh and local foods, and job creation.

City Assets

The City of Bozeman can advance circular economy and zero waste practices within municipal buildings and operations by providing composting and recycling infrastructure at all City facilities, examining supply chain and purchasing policies to reduce packaging and buy in bulk, limiting new purchases or purchasing products made with recycled content, and offering surplus and outdated equipment for sale through a state-wide auction.

This solution primarily addresses Bozeman's greenhouse gas mitigation goals, while also providing supporting resiliency benefits.

Emissions Reduction Benefits

This is a supporting mitigation solution to achieve the 2025, 2030, and 2050 mitigation goals.

Emission sources impacted include:

- Waste

Resiliency Benefits

- Conserve natural resources
- Increase social cohesion
- Protect human health

Related Solutions

- Solution G. Facilitate Compact Development Patterns
- Solution I. Enhance Social Infrastructure and Community Preparedness
- Solution N. Cultivate a Robust Local Food System

Action 5.M.1. Actively Promote Source Reduction, Recycling, and Repair

Providing source reduction, recycling, and repair resources and education to Bozeman citizens and businesses will be critical to achieving success in moving toward a Zero Waste Community and circular economy. Consumer purchasing decisions have a considerable impact on the community's overall waste stream.

The City of Bozeman has partnered with the Gallatin County Solid Waste Management District to host numerous "Fix-it Clinics" to extend the life of goods and encourage repair rather than disposal. The City will continue to partner with organizations, the business community, neighborhood groups, maker spaces, and homeowners' associations to promote and institute waste reduction and diversion best practices.

The City of Bozeman can increase awareness of zero waste practices by providing information and education to reduce residential and commercial waste and improve waste diversion practices. Additional ideas include promoting zero waste events, developing and piloting a quarterly or annual "Buy Nothing" day, encouraging neighborhood sponsored "community swaps," developing a program to reduce junk mail, and supporting the development of virtual or physical community "sharing sheds." The City will continue to partner with Montana State University to expand and promote programs that provide education and promote alternatives to disposable products and single use plastics. By promoting the use of resale stores, cooperative and bulk discount programs, and reuse/shared economy technology and services, the City of Bozeman can help consumers reduce their personal waste stream, and as a result, help build a circular economy and Zero Waste Community.

As of 2020, nearly 47% of City of Bozeman Solid Waste customers (blue bins) also had green recycling bins. The City will implement a recycling trial program to encourage the expansion of Bozeman's curbside recycling program. On a neighborhood-by-neighborhood basis, the City will provide all City of Bozeman curbside waste customers, who are not current curbside recycling customers, with a free trial of the program for 90 days. Offered on a rolling basis throughout the City, residences without green bins will have the opportunity to sample the convenience of single stream recycling and the City can expand the amount of recyclable material diverted from the landfill. The City can test the program in a few neighborhoods to determine the effectiveness and cost of a city-wide sampling program.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **Gallatin Solid Waste Management District** • **City of Bozeman** • Montana State University • Inter Neighborhood Council • Bozeman Climate Partners Working Group • Home Owner Associations • Bozeman Makerspace	• 1.1 Outreach • 6.3 Climate Action	• # of Zero Waste events or programs

Action 5.M.2. Expand Composting Services and Collection

When disposed of in a landfill, food waste and green waste create methane, a potent greenhouse gas. Expanding composting services and collection will enable the City to reduce the amount of compostable materials going into the landfill, thereby reducing methane production.

The City is currently expanding residential composting services from the existing yard waste collection to a community-wide curbside composting collection service that will include kitchen scraps and yard waste. There are several existing private composting businesses providing residential services; it is important that the City partner with private composters to expand capacity and collaborate with small businesses to craft programs. The City will provide information and education to reduce pesticide use, increasing the types of applications possible for finished compost. The City will explore requiring large food waste generators (e.g., restaurants and grocery stores) to compost and adopt updated development standards for commercial and multi-family recycling and composting infrastructure.

Biochar is a powerful soil amendment: it enhances plant growth and amplifies yields, which can increase food production, especially in areas with depleted soils or limited water resources. The City will partner with Montana State University to evaluate the waste stream inputs, operations, and application of biochar.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman** • Montana State University (e.g., Food Services, Facilities, Sustainability Office) • Bozeman Public School District • Private waste and composting companies • Restaurants • Grocers	• 2.1 Business Growth • 6.3 Climate Action • 6.5 Parks, Trails, & Open Space	• Tons of compost diverted from the landfill

Action 5.M.3. Improve Waste Policies, Services, and Operations

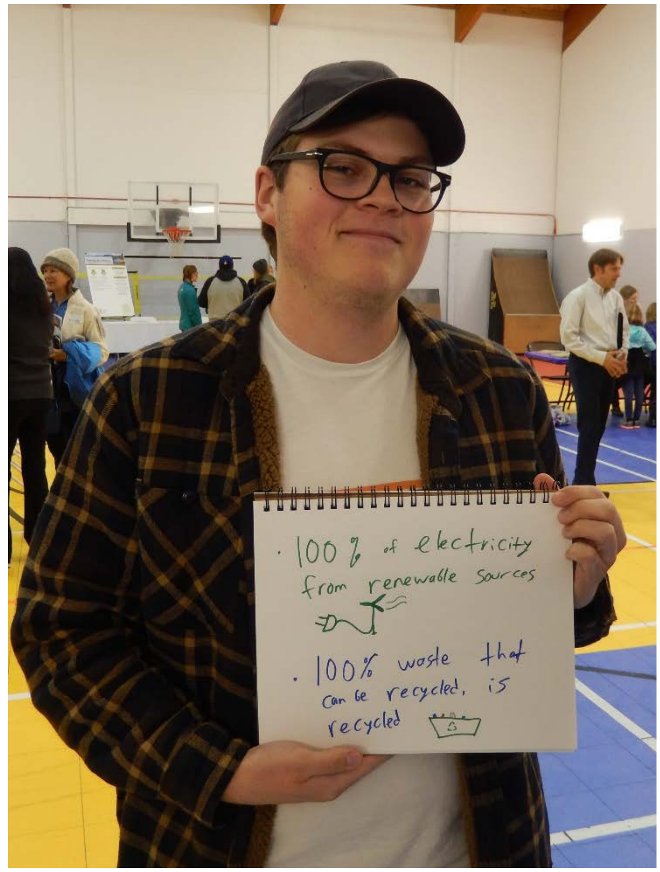
The City of Bozeman's waste policies, services, and operations can help reduce the community's waste stream. The City of Bozeman provides curbside recycling as an opt-in service and diverted nearly 1,800 tons from the landfill in fiscal year 2018.

The City will review and consider adopting updated development standards for commercial and multi-family recycling and composting infrastructure. The City of Bozeman will explore options to review current trash and recycling fees and shift the recycling program and other waste diversion services to an "opt out" system (i.e., customers are automatically enrolled in these services unless they decline to participate) to increase participation. The City will coordinate with Gallatin Solid Waste Management District to explore increasing the frequency of Household Hazardous Waste (HHW) collection events to further reduce the volume of HHW going into the landfill.

Transportation emissions associated with transporting waste from City limits to the Logan Landfill account for a significant portion of waste emissions. The City will support Gallatin Solid Waste Management District to identify viable options for constructing a waste transfer station to properly sort and divert waste, or consider more efficient options for transport to the Logan Landfill. To reduce emissions at the landfill, the City will partner with Gallatin Solid Waste Management District to continue to study new opportunities for landfill gas to energy conversion infrastructure.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **Gallatin Solid Waste Management District** • **City of Bozeman**	• 6.3 Climate Action	• Tons of recycling, compost, and HHW diverted from the landfill • Reduced transportation emissions from hauling waste and compost

Action 5.M.4. Support Construction Waste Diversion



While construction waste in itself does not generate a significant amount of greenhouse gases, the waste stream from construction sites need not take up valuable space in the landfill. Much of the material in construction waste is easily reusable or recyclable.

The City of Bozeman offers a comprehensive selection of roll-off containers for construction sites, including "wood only" recycling bins. The City will continue to educate and inform contractors of the economic benefits of separating out construction waste.

The City will develop guidelines and provide information and education to increase awareness in the development sector regarding construction waste reduction and diversion best practices (such as a "deconstruction handbook" that provides guidance on the benefits of deconstruction over demolition, available reuse and recycling resources, and local contractor information) and explore economic incentives for supporting demolition (deconstruction) businesses. The City will develop a recognition program for voluntary "green" construction and demolition practices and explore development incentives to achieve more sustainable construction and demolition waste management.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measure*
Level 3	• **City of Bozeman** • **Montana State University** • Southwest Montana Building Industry Association	• 6.3 Climate Action	• Tons of construction waste diverted from the landfill

Action 5.M.5. Encourage the Development of Material Markets

A significant challenge with developing a circular economy is the lack of market demand and available infrastructure to properly and cost-effectively turn recycled materials into usable products and get them back into circulation. Some materials, such as plastic, cannot truly be recycled and can only be downcycled (i.e., re-manufactured into a lesser quality product) and can only be downcycled one or two times before losing integrity. Other materials can be recycled over and over, but there is a lack of market demand for products that contain recycled content.

To encourage the development of material markets, the City will partner with Montana State University or other research institutes to conduct a study to better understand full lifecycle economics of current household and construction waste practices, and identify ways to create new economic opportunities to move Bozeman toward a more circular economy. The City will also identify regional partners to engage in reuse, reduction, recycling, and other waste/purchasing/supply chain efforts. Related to Action 6.N.4., the City and its' partners will encourage local food production and food recovery programs, and identify partnerships between food processors and potential end users to utilize processing waste.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 3	• **City of Bozeman** • Montana State University • Prospera Business Network • Gallatin Solid Waste Management District • City-County Local Food Council	• 2.1 Business Growth • 6.3 Climate Action	• Complete study of lifecycle economics of waste/recycling practices

Action 5.M.6. Develop Plans for Green Purchasing and Zero Waste Events for City Operations

The City of Bozeman has the potential to generate unnecessary waste by using single use or non-recyclable/non-compostable products, both within its own operations and in hosting community events. Environmentally Preferable Purchasing (EPP) or Green Purchasing and guidelines for hosting Zero Waste events can help mitigate this impact. Environmentally Preferable products are products and services that have a lower negative impact on human health or the environment compared to similar conventional products.

The City will analyze its purchasing policy and develop a plan to integrate source reduction and green purchasing into its requirements. Green purchasing policies could include requiring the City to source products that:

- are reusable, recyclable, or recycled content products,
- conserve energy, water, or natural resources,
- are lower carbon content,
- are made from sustainable and renewable resources,
- prevent pollution or contain fewer hazardous chemicals or toxic substances, and
- minimize waste or have minimal packaging.

Local food procurement and best practices for printing and disposable paper products will also be addressed in the purchasing policy.

The City will also develop a plan for hosting Zero Waste events for City operations. Strategies for Zero Waste events could include prohibiting single use, disposable items, providing on-site composting and recycling, providing clear signage and trained staff to ensure proper disposal of waste, sourcing event materials and food locally, encouraging attendees to bring their own containers and utensils, or investing in reusable catering materials, such as dishes, cups, and utensils.

The City will continue to explore avenues to limit or ban single use plastics for municipal operations and activities. The City has already eliminated the use of plastic water bottles for everything but emergency situations.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 3	• **City of Bozeman** • Montana State University • Southwest Montana Building Industry Association	• 6.3 Climate Action • 7.3 Best Practices, Creativity, & Foresight	• Updated green purchasing guidelines • # of Zero Waste city events

Increasing Resiliency to Climate Hazards

The following opportunities and considerations summarize how the solution in this chapter can help improve resiliency to future climate hazards.

Opportunities	Considerations
Extreme Heat	
• Composting occurs more quickly in warmer temperatures.	• A circular economy can reduce the need to transport goods in periods of extreme heat.
Flooding	
• Adding compost and biochar to soils can increase water holding capacity, which can prevent runoff during flood events.	• Flooding can cause high volumes of yard debris and organic materials that need disposal.
Drought & Mountain Snowpack	
• Waste reduction can include water waste reduction (e.g., reliance on grey water systems, drought tolerant landscaping, etc.) • Soil amended with compost can hold more water and decrease water demand for plantings.	• Water is essential to irrigate compost and clean/process recyclables. • Local production activities can be water-intensive.
Wildfire	
• Compost and biochar additions to soil can increase water holding capacity, making landscape more resistant to wildfires.	• A more localized economy could allow wildfire to disrupt wider sections of the Bozeman economy
Winter Storms	
• Less waste to collect could reduce the impact if waste collection services were disrupted.	• Composting can be challenging in cold or stormy weather. • Regular pick up of waste and recycling may still be necessary, and challenging, in event of severe winter storm.

Focus Area 6:

Regenerative Greenspace, Food Systems & NATURAL ENVIRONMENT

This chapter includes solutions that encompass the natural environment and systems, including land and water resources and food systems. While they present some opportunities for emissions mitigation through their connections to energy consumption, these solutions also provide emissions benefits in the form of carbon sequestration. Previous Bozeman greenhouse gas inventories have not accounted for carbon sequestration, but future inventories could aim to quantify their emission-related benefits.

RELATED PLANS & STUDIES

- Unified Development Code
- Bozeman Design Guidelines for Historic Preservation & Neighborhood Conservation Overlay District
- Triangle Community Plan

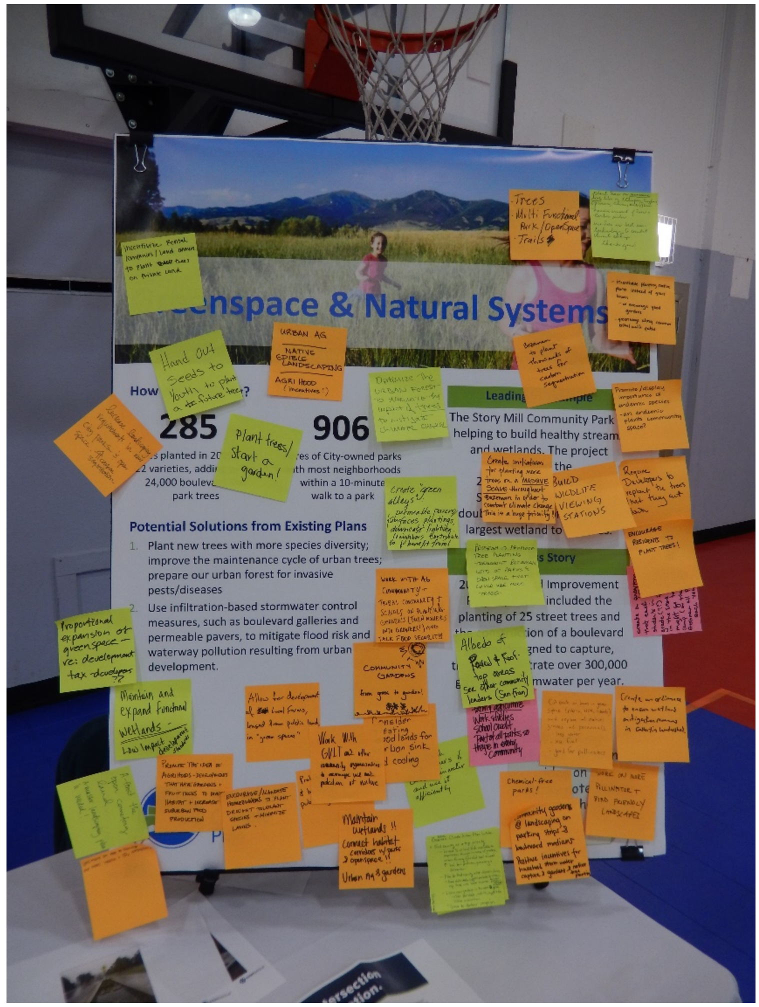
Bozeman currently boasts 906 acres of City-owned parks, with most neighborhoods within a 10-minute walk to a park. Additionally, the City has more than 24,000 trees in public boulevards and parks. The 14-acre wetland in Story Mill Community Park is the community's largest wetland, and it is an award-winning project that is helping to build healthy streams and wetlands.

Like many of the other chapters, the solutions in this chapter emphasize conservation and efficiency in resource use. These solutions also emphasize regeneration, or the renewal, restoration, and regrowth of systems to make them healthier and more resilient to disturbance and future climate impacts.

The solutions in this chapter help advance Bozeman's resiliency goals. The emissions mitigation benefits are limited and are accounted for with other related solutions. To maximize resiliency and emissions mitigation potential, the Bozeman community will need to:

- **Solution N. Cultivate a Robust Local Food System**
  Encourage and facilitate the development of all facets of a local food system, including but not limited to food production, access, processing, distribution, sales, resiliency, waste, education, and natural resources.
- **Solution O. Manage and Conserve Water Resources**
  Utilize a comprehensive approach to manage water resources at the watershed scale.
- **Solution P. Manage Land and Resources to Sequester Carbon**
  Encourage and implement practices that support the urban forest, strengthen carbon sinks, and improve carbon sequestration potential.

Solution N. Cultivate a Robust Local Food System



This solution recognizes that the natural system conversation is not complete without considering humans and our need to eat. This solution encourages the development of all facets of a local food system, including but not limited to food production, access, processing, distribution, sales, resiliency, waste, education, and natural resources. This complex web of processes required to get food from farm to table intersects with many other facets of this Climate Plan and brings to light the opportunity to address the food system holistically.

Food systems contribute heavily to greenhouse gas emissions. Retooling food systems presents many opportunities for emission reductions through improved agricultural practices, lower-carbon food products, production and transportation efficiency, and waste reduction. These approaches extend well beyond the typical urban farms and gardens, and necessitate transformations in economic, social, and environmental systems.

Climate change and the recent COVID-19 pandemic illuminate issues related to food system function and food security. In Montana, farmers are beginning to see climate change impacting productivity and nutritional quality, and their anxiety is increasing with this rise in uncertainty. Like communities across the country, Bozeman has an “on-demand” food supply that can be disrupted by events like a snowstorm that prevents truck deliveries. Similarly, the recent pandemic has transformed restaurant and food retail operations, interrupted agriculture production and harvesting, and increased food supply chain issues. A comprehensive food security assessment and plan could be a helpful next step to reduce food insecurity and improve food system resiliency.

A robust local food system is inherently a more equitable one, as it reduces food insecurity and improves food access and supply. It also creates a safety net in case of emergencies and supply chain disruptions.

From a health and well-being perspective, food stability and improved access to healthy food positively impacts health outcomes.

In terms of City assets, the City of Bozeman can facilitate food production on public properties, as well as support all aspects of a local food system through partnerships and its own purchasing practices.

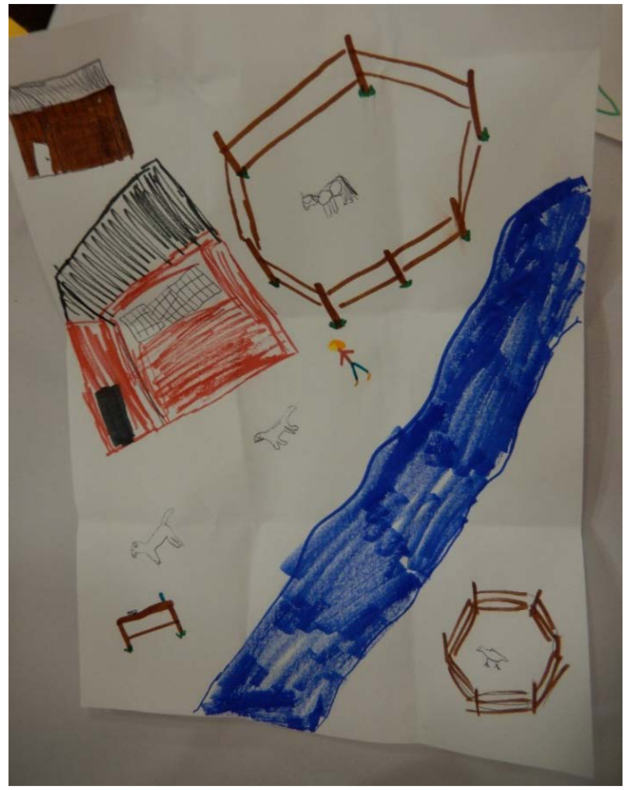
This solution primarily addresses Bozeman's climate resiliency goals, while also providing limited (unquantified) greenhouse gas emissions reduction and carbon sequestration benefits.

Resiliency Benefits
• Conserve natural resources • Increase social cohesion • Mitigate property and economic losses • Protect human health

Related Solutions

- Solution G. Facilitate Compact Development Patterns
- Solution I. Enhance Social Infrastructure and Community Preparedness
- Solution M. Move Toward a Circular Economy and Zero Waste Community
- Solution O. Manage and Conserve Water Resources
- Solution P. Manage Land and Resources to Sequester Carbon

Action 6.N.1. Support the Formation of a Local Food Council



To guide the process of cultivating a robust local food system in coordination and collaboration with our larger "foodshed", the City of Bozeman will explore the formation of a City-County Local Food Council. The City will work with community partners to establish a resolution and form a Local Food Council to advise both city and county government on local agricultural and food-related policy issues, lead education, advocacy, and programs to grow and enhance a local food system.

The Local Food Council will provide education to the community about food systems, the benefits of local foods, the carbon intensity of various food sources, low water use food source options, individual food production efforts, community gardens, and home gardening, edible landscapes, and season extension practices. It will develop and launch a comprehensive educational campaign to share composting benefits and best practices and help residents improve home composting practices. The Council will promote soil-building and healthy soil practices on private property (e.g., soil amendments, limiting pesticides and herbicides, etc.), and the importance of water use efficiency.

The Local Food Council will partner with local institutions to support increased engagement with our local food economy and greater understanding of the multiple benefits of procuring local and regional foods. It will support and encourage farmers markets, farm to table opportunities, cottage food businesses, and agriculture/gardening in public green spaces.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **Gallatin County** • **City of Bozeman** • **City of Belgrade** • Open & Local • Montana State University & Extension • Greater Gallatin United Way • Gallatin Valley Land Trust • Gallatin Valley Food Bank • Local farms • Gallatin Valley Farm to school	• 1.2 Community Engagement • 6.3 Climate Action • 6.6 Habitat	• Creation of a City-County food council

Action 6.N.2. Help Develop a Food System Assessment and Security Plan

To cultivate a local food system, the City must have a clear understanding of current practices and conditions, systemic challenges, available resources, community partners and stakeholders, and a plan for implementation.

The City will partner with community stakeholders to develop a comprehensive food system plan to be led by the Local Food Council. The scope of the food assessment will include food emergency response policies to encourage quick and resilient action, a map of regional food suppliers, avenues for supporting local food production, and avenues to connect community gardens to food banks. Middle income earners currently have poor access to local food as they are challenged by the higher prices and are not likely to access emergency food security programs. The plan will include both low- and middle-income individuals and families in the assessment.

The City will partner with Human Resources Development Council and Greater Gallatin United Way to invest in emergency food system planning and distribution, and with local grocery stores and other food sector businesses to identify opportunities to bolster resilience, such as investing in backup generators and mapping infrastructure that can be prepared to assist during emergencies.

Through the Local Food Council, the City will work to strengthen relationships with farmers, ranchers, processors, and other food sector businesses outside of City limits and to identify mechanisms for encouraging more diverse agricultural production and food supply chains, while identifying mechanisms to protect existing agricultural activity through land use planning and development regulations.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measure*
Level 2	• **Gallatin County** • **City of Bozeman** • **City of Belgrade** • HRDC • Open & Local • Montana State University • Greater Gallatin United Way • Gallatin Valley Land Trust • Gallatin Valley Farm to School • Local farms	• 3.2 Health and Safety Action • 6.3 Climate Action	• Completion of food security plan

Action 6.N.3 Encourage Local Agriculture and Preservation of Working Lands

A local food system is not possible without the preservation of agricultural land on which to grow food. As more and more agricultural land is converted to subdivisions in the Gallatin Valley, it is critical for the City of Bozeman to prioritize local agriculture and the preservation of working agriculture lands. The City of Bozeman, with input from the Local Food Council, will review existing land uses and the Community Plan, research effective metrics to evaluate urban agriculture, and consider targets to ensure that food production remains feasible within and near Bozeman. The City will encourage water use efficient appropriate and compatible new agricultural opportunities, such as incorporating garden pilot projects as part of City parks, supporting development design that integrates significant agricultural elements and opportunities (e.g., cluster development, community gardens, agricultural neighborhoods), and working with existing producers and land owners in the valley to encourage and support carbon sequestration practices on working lands.

The City will cultivate relationships with Montana State University, non-profits, and other local agencies to improve the management model for community gardens, emphasizing the need for responsible urban gardening, and organize community gardens in a manner conducive to re-investment in garden infrastructure.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **Gallatin County** • **City of Bozeman** • **City of Belgrade** • Open & Local • Montana State University • Greater Gallatin United Way • Gallatin Valley Land Trust • Local farms • Gallatin Valley Farm to School	• 2.1 Business Growth • 6.3 Climate Action • 6.6 Habitat	• Establishment of urban agriculture and local food production targets

Action 6.N.4 Support Local Food Production, Processing, and Distribution

Disruption of supply chains is a threat to Bozeman's ability to access healthy, affordable food. Bozeman is vulnerable to both availability of imported food and impacts to the transportation network and supply chain. By focusing on enhancing capacity for local food production, processing, and distribution, Bozeman can expand food security and increase resilience.

The City will work with the Local Food Council to strengthen the food supply chain by supporting farmers, ranchers, processors and other food sector businesses through City Economic Development programs and supporting the development of food processing and distribution centers within or proximate to City limits. While continuing to focus on water use efficiency, the City will review City codes to identify barriers to local food production and distribution (e.g., yard production, light industrial agriculture, hydroponics, etc.), expand the narrative in current code allowing for garden use, and explore code changes to encourage new or larger commercial opportunities. The City-County Food Council may identify opportunities to advocate for agricultural subsidies for producers contributing to community food security. This supports the local producers to sell locally, makes food affordable for low- and middle-income individuals, and supports the local economy.

The City will establish best practices for food enterprises and expand opportunities for home-based or cottage food entrepreneurs, such as jams and jellies, baked goods, dry herbs and seasonings, granolas and trail mixes.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measure*
Level 3	• **Gallatin County** • **City of Bozeman** • **City of Belgrade** • Open & Local • Montana State University • Greater Gallatin United Way • Gallatin Valley Land Trust • Local farms • Gallatin Valley Farm to School	• 2.1 Business Growth • 6.3 Climate Action • 6.6 Habitat	• Completion of food security plan

Solution O. Manage and Conserve Water Resources

This solution focuses on using a comprehensive approach to manage water resources at the watershed scale. This includes coordinating water resources across various sectors and applications, such as water supply, wastewater, reuse, stormwater, watersheds, wetlands, agriculture, and energy.

The production, conveyance, and treatment of water is highly energy intensive. Careful management and conservation of Bozeman's limited water resources helps reduce energy consumption and associated greenhouse gas emissions. Beyond this water-energy nexus, water conservation and management provide many other co-benefits, including supporting agricultural production, improving drought-preparedness and wildfire response, enhancing ecosystem function, extending the reach of current water supplies, generating utility cost savings, and reducing water bills for customers

Water availability is limited as the City of Bozeman relies on snowpack for its water supply. Roughly 80% of the City's water comes from snowmelt in the Gallatin Range, which feeds Sourdough Creek and Hyalite Reservoir. The other 20% comes from a developed spring at the headwaters of Lyman Creek. As the community grows and climate patterns shift, water supplies will likely become less reliable, with more of the community's precipitation coming in the form of rain instead of snow. Water conservation is critical to ensure that the City has an adequate supply of water in the future. The City of Bozeman offers conservation-oriented water rates, including a tiered -block structure for single-family customers. The City of Bozeman utilizes a radio-read metering system, and proposed amendments to the landscaping code may require heavy water users to implement irrigation submetering. Use of water catchment and grey water systems in Montana are subject to state and county regulations. Both present additional water conservation opportunities, but regulatory considerations currently limit viability.

Bozeman has a history of collaborating with Montana State University, University of Montana, other jurisdictions, and various partners on climate and water research, modeling, and forecasting. To further support water supply planning and forecasting, the City of Bozeman will be developing a water supply optimization tool and water conservation plan. The optimization tool will examine current water rights and historical yield, providing staff with the means to track and project current and future supply availability. The water conservation plan will evaluate the benefit/cost for existing and future program measures to identify a volume of water the city can expect to save through demand management. Together, these initiatives will provide the city with a more detailed and comprehensive look into the city's future water supply reliability and water demand patterns.

Efforts to maintain and improve water quality are also critical in Bozeman's urbanized area and the greater watershed upstream and downstream of the community. The City has invested in infrastructure and technology to treat water and wastewater, as well as managing stormwater and agricultural runoff. Preservation and expansion of environmental features like wetlands and riparian buffers help manage runoff naturally. Low impact development techniques and green infrastructure development can mimic and support natural systems and provide water quality benefits.

Equity

Measures to conserve water are more cost effective than securing additional water supplies, treatment, and/or building costly new infrastructure. There is a role for all Bozeman community members to implement water conservation practices. As with improvements for energy efficiency, some water conservation efforts are simple, low-cost opportunities, while others require more significant investments that may be out of reach for some community members. Likewise, modern urban water management techniques such as green infrastructure may prove to be costly investments that some cannot afford.

Human Health & Well-Being

Pricing structures for customer water use should reflect the true cost of the water being used. The Water Treatment Plant must be built to accommodate high water use on the hottest, driest day in the summer when lawn irrigation is at its peak. Pricing structures should minimize the cost of water for essential uses and increase cost of water for non-essential uses such as irrigation. Access to clean and healthy water is vital for all community members. Comprehensive management of water resources across the watershed will help ensure that water is not only available, but of high quality.

City Assets

The City of Bozeman can advance this solution by implementing energy-efficient practices in the production, conveyance, and treatment of water. The City can also continue to provide an integrated approach to water supply planning and management that includes water conservation as a primary strategy. Finally, the City of Bozeman can continue to lead by example in managing stormwater to protect water quality and can seek opportunities to incorporate Low Impact Development (LID) techniques with all City development projects.

This solution primarily addresses Bozeman's climate resiliency goals, while also providing limited (unquantified) greenhouse gas emissions reduction and carbon sequestration benefits.

Resiliency Benefits
• Conserve natural resources • Strengthen infrastructure to natural disaster • Mitigate property and economic losses • Protect human health

Related Solutions

- Solution A. Improve Efficiency of Existing Buildings
- Solution G. Facilitate Compact Development Patterns
- Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards
- Solution N. Cultivate a Robust Local Food System
- Solution P. Manage Land and Resources to Sequester Carbon

Action 6.O.1. Invest in Landscaping and Irrigation Upgrades at City Facilities

The City of Bozeman owns and manages 906 acres of City parks and numerous public facilities in the community. Investing in landscaping and irrigation upgrades at City parks and facilities will enable the City to lead by example with water conservation.

The City will update landscaping at City parks and facilities to be more drought tolerant, climate adaptive, and pollinator friendly, and update irrigation systems to include rain sensors, automatic controls, and high-efficiency equipment. The City will perform irrigation audits on City Facility and City Park irrigation systems to detect leaks and identify needed repairs, and program watering schedules and adjust programming seasonally for City properties to support water conservation goals. The City will also install drought tolerant landscaping in City medians to minimize or eliminate the need for irrigation systems.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measure*
Level 1	• **City of Bozeman**	• 4.3 Strategic Infrastructure Choices • 6.1 Clean Water Supplies • 6.3 Climate Action • 6.5 Parks, Trails, & Open Space	• Reduced water consumption for city facilities

Action 6.O.2. Build on the Success of Water Conservation Education and Incentives

The City of Bozeman's Water Conservation program is well-established and offers education, water-wise and drought management resources, free sprinkler assessments, rebates/incentives for commercial and residential indoor and outdoor water conservation, and a free online water use portal for tracking home water use. The Bozeman Climate Plan builds on this success and expands programming to further educate the community and conserve our limited water resources.

The City will provide information and resources to support more water efficient and resilient landscapes to private property owners, property management companies, and homeowners' associations, and develop a water-efficient landscape recognition program for private properties and/or common neighborhood areas. The City will initiate a turf lawn-to-resilient-landscape conversion program (drought tolerant, climate adaptive, pollinator friendly) to encourage lawn retrofits.

The City will continue to share information about Bozeman's Drought Meter and Drought Management Plan, and partner with large water users to develop drought contingency plans. This effort will be supported by the energy and water benchmarking program for commercial buildings (Action 1.A.4).

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • Montana State University • Gallatin Watershed Council • MT Institute on Ecosystems • Gallatin Local Water Quality District	• 1.1 Outreach • 6.1 Clean Water Supplies • 6.3 Climate Action	• Reduced gallons of water per capita per day • Reduced residential outdoor water use

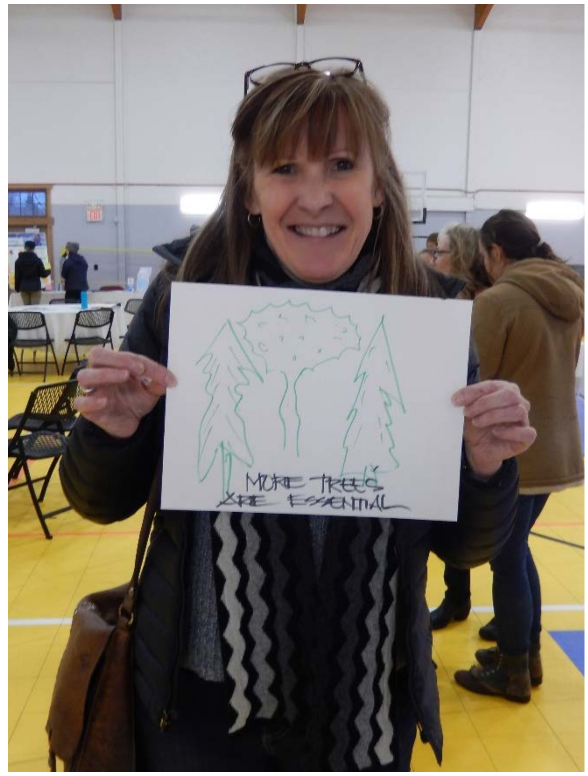
Action 6.O.3. Evaluate Additional Water Conservation Code and Water Rate Structure Adjustments

In addition to programming and physical infrastructure, the City of Bozeman has influence over consumer behavior through regulatory and financial mechanisms. To maximize opportunities for water management and conservation, the City will evaluate additional water conservation code and water rate structure adjustments.

The City will review City codes to identify opportunities to encourage or require more resilient water systems (e.g., greywater use, efficient irrigation and landscaping practices, etc.), bolster requirements for resilient landscapes, and identify contradictions between water conservation and vegetation requirements. The City will evaluate current water and waste code ordinances to align with water conservation goals and evaluate the feasibility of seasonal time-of-use irrigation restrictions or pricing. The City will evaluate the feasibility of requiring new developments to offset projected water demand through water conservation projects.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman**	• 4.1 Informed Conversation on Growth • 6.1 Clean Water Supplies • 6.3 Climate Action	• Reduced gallons of water per capita per day • Reduce outdoor water use

Solution P. Manage Land and Resources to Sequester Carbon



This solution encourages practices that support the urban forest, strengthen carbon sinks, and improve carbon sequestration potential. It includes management of land and resources on both public and private properties. This solution focuses on carbon sequestration and is closely related to other conservation and efficiency solutions.

Various land management practices and restoration techniques can support carbon sinks, the absorption of carbon dioxide from the atmosphere, and storage in natural systems like soil and plants. These opportunities include but are not limited to maintaining soil cover, minimizing soil disturbance and tilling, rotating crops, preserving natural and agricultural areas, protecting forests, managing grazing, restoring wetlands, reintroducing native species, removing invasive species, and implementing regenerative agriculture practices. Protecting existing trees, increasing the number of trees, and planning for tree replacement are all important practices to ensure a healthy and resilient urban forest.

In Bozeman's urban area, the tree canopy, public parks and open spaces, and private yards and properties all provide opportunities for carbon sinks and sequestration. At the community's edges, opportunities for carbon sequestration abound on agricultural properties and across nearby forests and open lands. Thoughtful stewardship of these lands and management of these resources provide many opportunities for Bozeman to naturally pull carbon emissions out of the atmosphere.

Equity

From an equity perspective, this solution needs careful consideration and implementation. Location and access by diverse segments of the community should be important drivers in the purchase of open space and/or development of other park or natural areas. Access needs to be balanced with the quality of the resources in order to maximize carbon sequestration benefits. Similarly, requirements for open space dedication within development projects could drive up development costs, making housing less affordable. With land and resource management on private property, renters may have less control over practices than owner-occupied properties, and some community members may be unable to afford investments into tree and yard maintenance.

Human Health & Well-Being

Enhancements to the urban tree canopy present significant opportunities to improve air quality and human health and increase resiliency through diversification of the tree canopy. Access to green spaces and the natural environment is important to physical and mental health.

City Assets

The City of Bozeman can implement best-in-class land management practices across its many parks and open space assets so that they can create a robust network of carbon sinks. The City can also continue to provide healthy, safe, and resilient community forest public spaces.

This solution primarily addresses Bozeman's climate resiliency goals, while also providing limited (unquantified) carbon sequestration benefits.

Resiliency Benefits
• Conserve natural resources • Harden infrastructure to natural disaster • Mitigate property and economic losses • Protect human health

Related Solutions

- Solution G. Facilitate Compact Development Patterns
- Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards
- Solution N. Cultivate a Robust Local Food System
- Solution O. Manage and Conserve Water Resources

Action 6.P.1. Protect Local Wetlands for Flood Resilience and Water Quality

Bozeman is predicted to experience increased urban flooding as a result of climate change. To mitigate this, the City can take actions to protect local wetlands for flood resilience and water quality.

The City will prioritize the protection, management, and restoration of wetlands. The City will enhance or construct wetlands to offset all losses and require that all wetland mitigation efforts are conducted within the impacted watershed. The City will study the wetland banking program and recommend reforms to encourage preservation of contiguous, high-value habitat and ecosystem services within the City of Bozeman.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measure*
Level 1	• **City of Bozeman** • **Gallatin County**	• 4.1 Informed Conversation on Growth • 6.1 Clean Water Supplies • 6.3 Climate Action • 6.6 Habitat	• Acres of wetlands preserved or mitigated within watershed

Action 6.P.2. Maintain and Expand the Urban Forest

Bozeman's urban forest is a valuable resource to the community, both aesthetically and from a public and environmental health standpoint. The Bozeman Urban Forest Management Plan (UFMP) aims to sustainably, holistically, and efficiently manage Bozeman's urban forest, and to illustrate the full expanse of benefits urban trees can provide. In public parks, open space, and rights of way, the City manages approximately 23,950 trees in the municipally owned urban forest (not including wild trees growing along stream banks and on private property or undeveloped City-owned property). The City is in the process of conducting a comprehensive inventory of Bozeman's urban forest. Bozeman's urban forest faces increasing threats from climate change and invasive pests. Addressing the health of the urban forest is among the highest priorities for the City.

The City of Bozeman will continue to grow the urban forest program to maintain and expand the urban forest on public and private property. The City will identify opportunities in City codes to promote the expansion, preservation, and maintenance of the urban forest, and prioritize planting low maintenance, drought tolerant tree species, and increasing species and age diversity. The City will explore the feasibility of launching and using an app to monitor tree health and carbon sequestration.

Education for the community on proper maintenance of the urban forest is critical. The City will increase public outreach related to proper tree care and maintenance, removal, and replacement on private property.

Priority	***Lead** & Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 1	• **City of Bozeman**	• 4.1 Informed Conversation on Growth • 6.3 Climate Action • 6.5 Parks, Trails, & Open Space • 6.6 Habitat	• Increased species diversity • Increased proactive tree maintenance

Action 6.P.3. Enhance Greenspace and Carbon Sequestration for New Development

The City of Bozeman has the capacity to help guide new development, especially with regard to shared open space and vegetation requirements. Greenspaces throughout the community can either sequester carbon, if designed thoughtfully and with intention, or they can inadvertently contribute to increased carbon emissions, through increased energy- and water-intensive activities such as mowing, irrigation, and fertilizing.

The City already requires dedicated park lands and open space in new residential developments. The City will review existing parkland requirements and determine the role of parks in the preservation of greenspace and carbon sequestration. The City will emphasize native landscapes in greenspace development and incentivize developers to enhance or construct wetlands in new developments, where appropriate. Trees in greenspace can provide cooling spaces, further mitigating the impacts of climate change.

The City will promote the installation and maintenance of greenways along bike and walking paths and incentivize the connection of habitat corridors as development and redevelopment occurs. The City will review requirements for parking lot, median, and boulevard landscaping to require climate-friendly landscaping.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measure*
Level 1	• **City of Bozeman**	• 6.3 Climate Action • 6.5 Parks, Trails, & Open Space • 6.6 Habitat	• Establish method to quantify urban carbon sequestration potential

Action 6.P.4. Provide Outreach on Water Pollution Prevention and Carbon Sequestration Strategies

Beyond the urban tree canopy, Bozeman's natural land and water resources have high potential for carbon sequestration and are at high risk for pollution. Management strategies that depend upon pesticides, herbicides, fungicides, and fertilizers can pollute water supplies and harm wildlife.

The City will partner with the Local Food Council, MSU, scientists, and other partners to explore the impacts of and alternatives to pesticides, herbicides, fungicides, and neonicotinoids on City properties and community wide. The City will emphasize best management practices and promote a phosphorous-free City. Whenever possible, the City will prioritize the installation of low maintenance landscapes to help reduce weed maintenance demands. The City will provide education and outreach to private property owners to minimize the use of or find alternatives to chemical applications where possible.

The City will provide information and education to help residents make improvements on private property related to carbon sequestration and land management and provide resources on investments into local/regional carbon offset programs. The City will quantify carbon sequestration potential and irrigation requirements of different types of vegetation, especially in evaluation of food production.

Producing biochar from agricultural wastes could sequester up to 12% of global carbon emissions (Woolf, D., Amonette, J., Stret-Perrott, F. et al., 2010). Biochar has myriad co-benefits: it improves soil health and crop yields and produces biogas as a by-product that could be refined into biodiesel for use in City vehicles. Biochar is an existing, low-tech technology that the City of Bozeman could utilize in addressing the community's agricultural and urban forest waste. The City will work with Gallatin County, the Local Food Council, and other partners, such as local farmers, to support production of biochar within Gallatin County. Additionally, the City could experiment with making its own biochar from municipal yard waste and sharing the process with local farmers. The City could also provide a market for purchasing biochar to incentivize local farmers to produce it.

Priority	***Lead*** *& Implementing Partners*	*Strategic Plan Alignment*	*Performance Measures*
Level 2	• **City of Bozeman** • **Montana State University Extension** • Local Farmers • MSU Engineering • MSU College of Agriculture • Non-profits (e.g., Gallatin Valley Land Trust, One Montana) • Western Sustainability Exchange	• 6.3 Climate Action • 6.5 Parks, Trails, & Open Space • 6.6 Habitat	• Reduction in Nitrogen and Phosphorous concentration in Water Reclamation Facility (WRF) effluent • Develop guidance on carbon sequestration and irrigation requirements for vegetation and food production

Increasing Resiliency to Climate Hazards

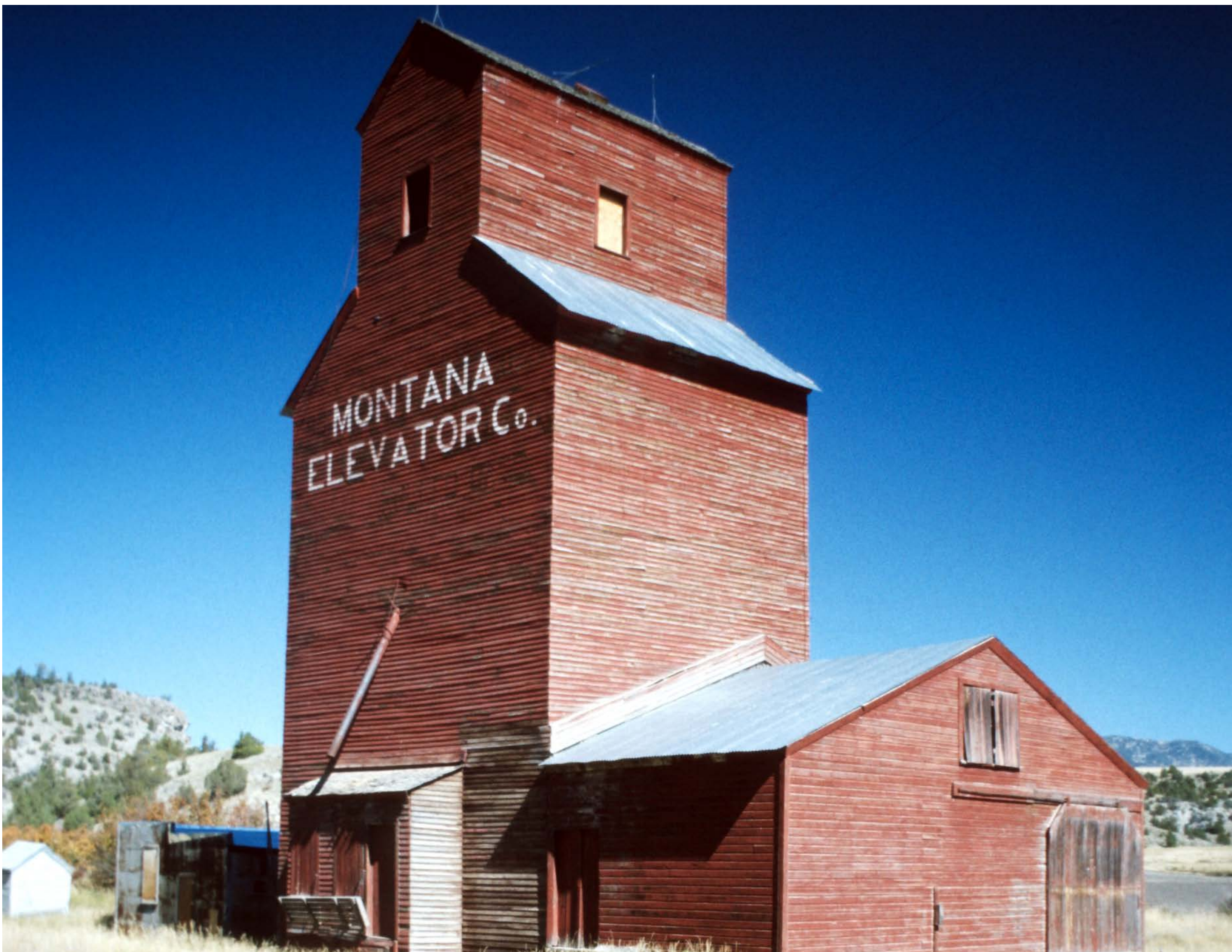
The following opportunities and considerations summarize how the solutions in this chapter can help improve resiliency to future climate hazards.

Opportunities	Considerations
Extreme Heat	
• Robust greenspaces and urban forests provide cooling benefits and decrease urban heat island effect. • A robust tree canopy reduces water demand of shaded vegetation and turf grass. • Healthy soils can help protect plants during extreme heat events. • Conserving water resources will help to ensure adequate supplies.	• Extreme heat could adversely affect agriculture productivity and food production. • Surface waters may experience losses due to evaporation.
Flooding	
• Protecting and restoring wetlands and riparian areas will improve flood and erosion control. • Healthy soils can store additional water and are less prone to erosion.	• Flooding could adversely affect agriculture productivity and food production.
Drought & Mountain Snowpack	
• Protecting/conserving water resources and increasing soil health will improve water holding capacity of natural systems. • Native and drought tolerant landscapes require less water than turf.	• Drought could adversely affect agriculture productivity and food production. • Increased tree canopy and greenspace could increase burnable fuel during drought.
Wildfire	
• Developing a diverse supply of water resources can increase resilience if wildfires threaten quality of surface water. • Local food production could provide food security if wildfires impact food supply chain. • Conserving water resources can help ensure supplies necessary to fight fires.	• Water may be limited for other critical uses, like local agriculture, if required for firefighting. • Water quality can be challenging to protect in the event of a wildfire.
Winter Storms	
• Local food systems can provide food during times of limited distribution. • Greenspaces can help to absorb runoff when snow melts. • Winter storms could bolster snowpack and resulting water resources.	• Severe winter storms could adversely affect agriculture productivity and food production.

Bozeman Climate Plan

CHAPTER 4:

IMPLEMENTATION GUIDE



This section provides a detail to guide implementation of this Climate Plan. The implementation guide includes three major sections: (1) implementation considerations, (2) activities to keep the plan on track, and (3) implementation playbook for strategies and actions.

IMPLEMENTATION CONSIDERATIONS

Forecasting and Risk

Implementing this Climate Action Plan is subject to conditions along the way that could affect progress toward goals. Some of these are large systemic conditions that the City and community can influence, but don't have direct control over. It's important to recognize these changing conditions in monitoring the plan's progress, adjusting course where needed along the way.

Population Growth

The business-as-usual forecast is based on constant per capita emissions, so variation in population growth has a significant impact on the projected community emission and therefore the amount of reductions needed to achieve the goals. Bozeman is a high-growth community and there is significant uncertainty in the future population projections as discussed in Chapter 2: Climate Trends & Goal Contributions.

All goals analyzed in this plan are based on the high-growth scenario to allow contingency planning for all growth scenarios.

Economic Conditions

- As shown in the next section, the solutions in this plan have a good financial performance as a whole, but many require upfront investment. Recessionary conditions can dampen the pace of investment and therefore progress toward the goal.
- Economic downturn may also slow the pace of new development reducing the overall community GHG emissions.
- Projected EV adoption rates may be slowed from current projections if there is a prolonged economic downturn.

Electric Utility Supply

- The plan relies heavily on NorthWestern Energy's ability to deliver significantly more clean energy to Bozeman. Spot market, short-term sales, market prices for renewable assets and state regulatory conditions all affect the utility-based clean energy solutions in this plan.
- The feasibility of distributed electricity generation projections may depend on local or national policy affecting net metering, allowed system size, and available incentives.
- Utility policy to support EV infrastructure may accelerate EV adoption.

Federal Policy and Investment

- Implementing a carbon tax or an emissions trading system will provide financial incentives for businesses to mitigate their emissions as well as provides governments financing for mitigation or adaptation programs. The implementation of a carbon tax could accelerate the implementation on many of the identified solutions.
- Production tax credits for large scale wind products are set to expire at the end of 2020 and it appears unlikely to be renewed. This may reduce investment in utility-scale wind programs.
- In 2018, the corporate average fuel economy (CAFÉ) standards were frozen at the 2021 level rather than continuing to improve fuel economy through 2025. The future of this program will likely influence auto maker incentives to continue to innovate and produce lower emissions vehicles.

To meet the community's 2050 goals, new emissions reduction or carbon sequestration solutions that are not currently commercially available must be developed. The development of these solutions will rely on consistent demand and support of research and innovation.

The above risks are illustrative and not exhaustive. Managing risks will require implementing all of the solutions in the plan which were built together as a system to achieve the goals. As a matter of best practice, new strategies should be considered during implementation to periodically freshen the solutions with the latest opportunities in cost-effective mitigation and resilience.

Economics of Climate Change

To reach the climate mitigation goals in this plan, significant investment will be required from the both the City of Bozeman and the community's businesses and residents. Before determining the cost effectiveness of various mitigation solutions and supporting strategies, we must first understand the cost of doing nothing. Figure 21 shows the cost of energy use in the community for the 2016 baseline, as well as the projected cost for each of the goal years assuming business as usual. This is a combined cost of fuel use from City operations, businesses, and residents as well as the social cost of carbon from the resulting emissions. The social cost of carbon is a metric that shows the cumulative cost of the negative impacts of Bozeman's GHG emissions including increased frequency and severity of natural disasters as discussed above (US Environmental Protection Agency, 2020). As the City and residents consider various investments to reduce the community's GHG emissions, the investment should be compared against the cost of doing nothing in order to net costs and benefits, keeping in mind that the social cost of carbon is higher in later goal years as the effects of climate change are more pronounced.

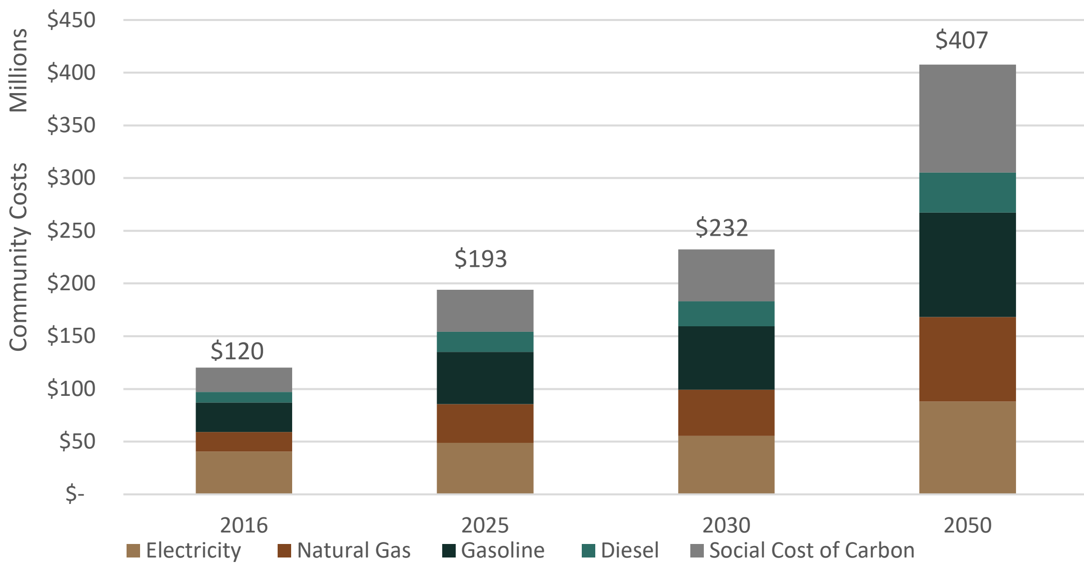


Figure 21. Projected cost of energy use and social cost of carbon under the business-as-usual high-growth scenario

The following table summarizes which solutions in this plan reduce each of the fuel components of the community GHG inventory shown in Figure 6. For each solution, total costs or cost savings are estimated, along with return on investment (ROI), through the year 2050. As shown in Table 11, the total estimated cost savings from these climate mitigation solutions is $217.5 million by 2050 in addition to avoiding the projected social impacts of carbon emissions estimated at $102 million in 2050.

In 2016 the cost of energy use in Bozeman was about $2,200 per resident. Under the business as usual scenario in Figure 21, this cost would rise to about $2,900 per resident in 2050 based on consistent energy consumption per person and the US Energy Information Administration's projections for energy costs in Montana (US Energy Information Administration, n.d.). By contrast, the average per person energy costs drops to $850 per person in 2050 based on the goal scenario outlined in this plan, a 70% reduction. While Bozeman homes and businesses will need to invest significantly to reduce energy use and shift to renewable energy sources, the end result will be more affordable energy costs.

Table 11. Projected fuel savings and return on investment per mitigation solution

Mitigation Solution	Emissions Source Impacted	2050 Cost Savings and Return on Investment[9]
A. Improve Efficiency of Existing Buildings	Electricity and Natural Gas	Cost Savings: $71.4 Million ROI: 4% for deep energy retrofits; 5.9% for medium retrofits (Nadel, 2020)
B. Achieve Net-Zero Energy New Construction	Electricity and Natural Gas	Cost Savings: $47.4 million ROI: 10-13% (Emerson & Sullivan, 2020)
C. Electrify Buildings	Natural Gas	Cost: $9.3 million ROI: None – varies based on existing systems
D. Increase Utility Clean Energy Mix	Electricity	Cost Savings: unknown ROI: not applicable
E. Increase Community Participation in Utility Green Power Programs	Electricity	Cost: Variable based on green power agreement ROI: not applicable
F. Increase Community-Based Distributed Renewable Energy Generation	Electricity	Cost Savings: $4.4 million ROI: 2.5%-5.6% for solar depending on net metering and electric TOU rates (Nadel, 2020)
G. Facilitate Compact Development Patterns	Gasoline and Diesel	Cost Savings: $37.9 million ROI: public investment; High ROI to solution participants reducing trips
J. Increase Walking, Bicycling, Carpooling and Use of Transit	Gasoline and Diesel	Cost Savings: $34.6 million ROI: public investment; High ROI to solution participants reducing trips
K. Decrease Direct Vehicle Emissions	Gasoline and Diesel	Cost Savings: $29.8 million ROI: public investment; High ROI to solution participants reducing trips
TOTAL		**TOTAL Cost Savings $215.7 million**

[9] Projected energy prices based on US Energy Information Administration projections for Montana.

Building efficiency, NZE construction, and distributed renewable energy generation all have a positive return on investment to the institutions, businesses, and residents making capital improvements to their facilities. These solutions also tend to improve property values. With decreasing costs of utility-scale renewable energy, utility-based renewable energy solutions may have a direct economic benefit to the community in the future, particularly when considering the societal cost of carbon. Utility-based solutions are also essential to achieve deep carbon reduction goals. When evaluating potential rate impacts or possible premiums for electricity through tariffs, it's therefore important to evaluate cost effectiveness in conjunction with energy efficiency.

Three solutions in this plan tackle emissions associated with gasoline and diesel consumption. Two of the solutions reduce the number of trips and/or trip length in conventional vehicles. These strategies rely heavily on public investment in transit, compact land use development, and walking and biking infrastructure. On the other hand, cost savings from these public investments are realized by residents and visitors who choose to participate in these solutions. The third solution targets replacing conventional vehicles with zero emission electric vehicles. The vehicle or fleet owner bears the incremental investment cost and realizes the savings directly. However, similar to most of the other mitigation solutions, the actual return on investment can varying depending on public policies, utility rebates, and tax incentives. To address this, the solutions within this plan include dedicated advocacy and policy strategies that can improve financial performance and therefore participation in these solutions community wide.

Achieving the climate resiliency goals in this plan will also require significant investment from all facets of the Bozeman community, including the City of Bozeman, residents, businesses, and others. Unlike the mitigation solutions, many of which yield a positive ROI that can be estimated using fuel savings over time, the costs of resiliency solutions are more difficult to quantify. This is because resiliency solutions involve transforming complex physical, social, and economic systems and decision-making. In terms of ROI of resiliency, is estimated that planning for resilience during normal times will yield $6 in economic benefits for every $1 spent (Multihazard Mitigation Council, 2017).

KEEPING THE PLAN ON TRACK

Plan Leadership

The **City of Bozeman Sustainability Program Manager** will continue to serve as the project manager for Climate Plan implementation. The Sustainability Program Manager will serve as a central coordinator of various implementation initiatives, establishing partnerships and delegating responsibility to others as necessary, and seeking approval from the City Commission as appropriate. The Sustainability Manager will also be responsible for continuing to convene stakeholder and leadership meetings, overseeing the Climate Plan monitoring and reporting activities, and initiating future Climate Plan amendments and updates.

A **Charter** will be developed within six months of plan adoption to further define specific actions and sub-actions needed to implement priority Level 1 items. The Charter will serve as a work plan with defined steps needed to accomplish action items with assigned responsible parties, timelines, budgets, and priorities. The Charter is intended to be a living document that changes over time to reflect subsequent changes and near term needs moving forward. Formation of a permanent **Climate Advisory Board** is recommended to guide annual implementation planning, support monitoring and reporting, and provide implementation leadership and assistance.

The **Bozeman City Commission** is responsible for adopting the Climate Plan and continuing to align City policies, decisions, and funding to support its implementation.

Monitoring and Reporting

Ongoing monitoring of progress and reporting of achievements is essential in keeping the Climate Plan current and on track to achieve the climate goals. Technology platforms such as Smartsheet and ArcGIS Hub will be used to create a real-time **Dashboard** to measure progress over time. Significant potential exists for developing informative public outreach solutions using modern tools to effectively engage the community. Monitoring and reporting activities will include the following:

- **Performance Monitoring:** updating and reporting on the community's GHG emissions every two years (i.e., 2020, 2022, 2024, etc.).
- **Implementation Monitoring:** providing an annual memorandum or report summarizing the status of each solution and associated implementation strategies (including achievements, challenges, and general progress). Where feasible, a triple bottom line analysis of activities, including environmental, social, and economic benefits will be included.

Amendments and Updates

Finally, because this Climate Plan is intended to provide a framework for emissions reduction for the next three decades through 2050, it is likely that amendments and updates to the Plan will be necessary. Every two years, the GHG emissions inventory will be updated, and this information will be provided to the City Commission and the general public. Shortfalls in meeting the City's GHG emissions reduction goals may necessitate changes to interim emissions reduction targets or more aggressive climate solutions, requiring an update to the charter or implementation playbook, which will be undertaken at the City Commission's discretion, with the general public having the opportunity to provide input.

Similarly, the implementation strategies identified in the implementation playbook will need to be updated as actions are completed, and new ideas and priorities emerge. On an annual basis, the implementation playbook will be reviewed and updated to reflect emerging or changing priorities, modifications in timelines and sequencing, updates to partners or resources, and removal of completed activities.

IMPLEMENTATION PLAYBOOK

The following table provides a summary of actions, organized by focus area and solution. Actions listed in *italic* text indicate that another entity beyond the City of Bozeman is responsible for leading implementation, with the City of Bozeman as a supporting implementation partner. See the detailed action narratives in Chapter 3 for more details about implementation roles for each action.

Actions listed in **bold** text indicate that they may require amendments to the City of Bozeman Unified Development Code (Chapter 38 of the City Code). The City Commission has established a twice-yearly cycle for UDC updates. It is recommended that implementation of these code-related amendments follow the established update cycle, to the extent practical and feasible.

Table 12. Summary of actions by focus area and priority level

Focus Area	*Action*	*Priority Level*
Focus Area 1. Healthy, Adaptive & Efficient Buildings	Solution A. Improve Efficiency of Existing Buildings	
	1.A.1. Increase Energy Efficiency at City Facilities	1
	1.A.2. Use Data and Price Signals to Advance Energy Efficiency	1
	1.A.3. Expand Energy Efficiency Information and Resources for Private Property	1
	1.A.4. Establish an Energy and Water Benchmarking Standard for Commercial Buildings	1
	1.A.5. Require Home Energy Labeling at Time of Listing	2
	1.A.6. Promote Energy Efficiency Financing and Investment	2
	1.A.7. Create a Rental Registry Program to Advance Renter Safety and Energy Efficiency	3
	Solution B. Achieve Net Zero Energy New Construction	
	1.B.1. Support High Performance Building Resources and Training for the Development Community	1
	1.B.2. Advocate for Adoption of State-Wide Net Zero Energy Code	1
	1.B.3. Encourage High Performance Construction for All Publicly Funded Buildings	2
	1.B.4. Analyze and Support Opportunities for District Energy	2
	1.B.5. **Offer a Voluntary Pathway & Incentives for Above-Code Construction**	3
	Solution C. Electrify Buildings	
	1.C.1. Advance Electrification Upgrades and Conversion Projects for City Facilities	2
	1.C.2. **Include an Electrification Component for Above-Code Construction**	3
	1.C.3. Support Outreach and Incentives for Electric Appliances and Equipment	3

Focus Area	Action	Priority Level
Focus Area 2. Responsible & Reliable Clean Energy Supply	Solution D. Increase Utility Clean Energy Mix	
	2.D.1. Complete a 100% Net Clean Electricity Community Feasibility Study	2
	2.D.2. Collaborative and Innovate Utility-Scale Solutions with Utility Provider	1
	2.D.3. Support Policies to Expand Renewable Energy and Just Transition Initiatives	1
	2.D.4. Encourage Philosophical Shift for Utility Provider	1
	Solution E. Develop and Promote Utility Green Power Programs	
	2.E.1. Advance Green Tariff Program Development and Participation	1
	Solution F. Increase Community-Based Distributed Renewable Energy Generation	
	2.F.1. Plan and Install Renewable Energy Projects for City Facilities	1
	2.F.2. Streamline Solar Permitting and Adopt Solar-Ready Code Provisions	1
	2.F.3. Advance Distributed Solar Policies	2
	2.F.4. Promote Education and Incentives for Distributed Renewable Energy and Storage	3
Focus Area 3. Vibrant & Resilient Neighborhoods	Solution G. Facilitate Compact Development Patterns	
	3.G.1. Continue Regional Coordination on Compact Growth and Sustainable Development	1
	3.G.2. Review Development Code to Enhance Compact and Sustainable Development	1
	3.G.3. Develop Sustainable Neighborhoods Outreach	2
	Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards	
	3.H.1. Plan for Resilience Hubs at Critical Facilities	1
	3.H.2. Advance Resilience in Development Code and Development Review	1
	3.H.3. Support Business and Residential Preparedness Outreach	2
	3.H.4. Incorporate Resilience into Infrastructure Plans	2
	Solution I. Enhance Social Infrastructure and Community Preparedness	
	3.I.1. Support Community and Neighborhood Resilience Programming	1

Focus Area	*Action*	*Priority Level*
Focus Area 4. Diverse & Accessible Transportation Options	Solution J. Increase Walking, Bicycling, Carpooling and Use of Transit	
	4.J.1. Prioritize Regional Multi-modal Planning and Connectivity	1
	4.J.2. Pursue Innovative Funding for Pedestrian and Bicycle Connections and Network	1
	4.J.3. Improve Maintenance of Multi-Modal Infrastructure	1
	4.J.4. Pursue Sustainable Transit Funding and Expansion	1
	4.J.5. Support Employee Trip Reduction Programs and Transportation Demand Management	1
	4.J.6. Support Regional Transit Service Coordination and Outreach	1
	4.J.7. Leverage Parking Policies to Encourage Other Modes of Transportation	2
	4.J.8. Develop Bike and Car Share Programs	3
	Solution K. Decrease Direct Vehicle Emissions	
	4.K.1. Support Community EV Roadmap Development	1
	4.K.2. Collaborate to Install Publicly Accessible EV Infrastructure	1
	4.K.3. City Fleet and Transit EV Investment	2
	4.K.4. Advocate for EV Utility Rates, Incentives, Infrastructure, and Efficiency Standards	2
	4.K.5. Limit Wasteful Vehicle Emissions	2
	4.K.6. Support EV Group Buy and Outreach	3
	Solution L. Improve Air Travel Efficiency	
	4.L.1. Build Awareness of Air Travel Impacts and Alternatives	2
	4.L.2. Advocate for Increased Air Travel Efficiency	3
Focus Area 5. Comprehensive & Sustainable Waste Reduction	Solution M. Move Toward a Circular Economy and Zero Waste Community	
	5.M.1. Actively Promote Source Reduction, Recycling, and Repair	1
	5.M.2. Expand Composting Services and Collection	1
	5.M.3. Improve Waste Policies, Services, and Operations	2
	5.M.4. Support Construction Waste Diversion	2
	5.M.5. Encourage the Development of Material Markets	3
	5.M.6. Develop Plans for Green Purchasing and Zero Waste Events for City Operations	3

Focus Area	*Action*	*Priority Level*
Focus Area 6. Regenerative Greenspace, Food Systems & Natural Environments	Solution N. Cultivate a Robust Local Food System	
	6.N.1. Support the Formation of a Local Food Council	1
	6.N.2. Help Develop a Food System Assessment and Security Plan	2
	6.N.3. Encourage Local Agriculture and Preservation of Working Lands	2
	6.N.4. Support Local Food Production, Processing, and Distribution	3
	Solution O. Manage and Conserve Water Resources	
	6.O.1. Invest in Landscaping and Irrigation Upgrades at City Facilities	1
	6.O.2. Build on the Success of Water Conservation Education and Incentives	2
	6.O.3. Evaluate Additional Water Conservation Code and Water Rate Structure Adjustments	2
	Solution P. Manage Land and Resources to Sequester Carbon	
	6.P.1. Protect Local Wetlands for Flood Resilience and Water Quality	1
	6.P.2. Maintain and Expand the Urban Forest	1
	6.P.3. Enhance Greenspace and Carbon Sequestration for New Development	1
	6.P.4. Provide Outreach on Water Pollution Prevention and Carbon Sequestration Strategies	2

COMMUNITY GUIDE TO IMPLEMENTATION

As the City and its partners create climate action policies, programs, and services, Bozeman residents and businesses can do their part by learning about their contribution to Bozeman's greenhouse gas emissions and becoming supportive adopters of new climate actions as they become available. Example action items that invite all Bozeman residents, businesses, and community organizations to get involved are outlined on the following pages.

Inviting Bozeman Residents to Join the Climate Effort!

Bozeman residents are encouraged to print a copy of this page and the next as a quick reference guide for how you can do your part in supporting the Climate Plan. See the more detailed action descriptions in the Climate Plan to learn more about how each action contributes to Bozeman's climate goas. Check off actions as you complete them!

Action 1.A.2.

Reduce energy usage during peak energy demand (4pm to 8pm)

Healthy, Adaptive & EFFICIENT BUILDINGS

Action 1.A.3.

Learn about and begin to make changes to your energy behaviors

Healthy, Adaptive & EFFICIENT BUILDINGS

Action 1.A.3.

Contact NorthWestern Energy, or other qualified auditor, to schedule a home energy audit and make efficiency improvements to your home

Healthy, Adaptive & EFFICIENT BUILDINGS

Action 1.B.2.

Advocate for more stringent state-wide energy and water efficiency regulations

Healthy, Adaptive & EFFICIENT BUILDINGS

Action 2.F.4.

Explore opportunities to install on-site renewable energy and storage on your property

Responsible & Reliable CLEAN ENERGY SUPPLY

Action 3.G.3.

Plan a neighborhood activity to help build social connections and improve community cohesion

Vibrant & Resilient NEIGHBORHOODS

Action 3.H.3.

Get to know your neighbors and swap contact information for times of need or emergency

Vibrant & Resilient NEIGHBORHOODS

Action 3.H.3.

Review City maps to understand if you are in a location that is vulnerable to flooding, fires, or other hazards and develop an emergency plan if a hazard event occurs

Vibrant & Resilient NEIGHBORHOODS

Actions 4.J.2. & 4.J.6.

Walk, bike, carpool, or take transit to destinations instead of driving alone

Diverse & Accessible TRANSPORTATION OPTIONS

Action 4.K.1.

Limit idling and combine trips when using a vehicle for transportation

Diverse & Accessible
TRANSPORTATION OPTIONS

Action 4.K.6.

Consider investing in an electric vehicle for your next vehicle purchase

Diverse & Accessible
TRANSPORTATION OPTIONS

Action 4.L.1.

Find alternatives to air travel, avoid binge flying, and/or purchase offsets for your next airline trip

Diverse & Accessible
TRANSPORTATION OPTIONS

Action 5.M.1.

Review your waste and consumption practices and look for opportunities to reduce, reuse, or share products

Comprehensive & Sustainable
WASTE REDUCTION

Action 6.N.2.

Volunteer at or donate to a local food bank

Regenerative Greenspace, Food Systems &
NATURAL ENVIRONMENT

Action 6.N.3.

Learn to garden and grow your own food

Regenerative Greenspace, Food Systems &
NATURAL ENVIRONMENT

Action 6.P.2.

Plant and maintain a tree

Regenerative Greenspace, Food Systems &
NATURAL ENVIRONMENT

Action 6.O.2.

Update irrigation equipment and landscaping to use less water

Regenerative Greenspace, Food Systems &
NATURAL ENVIRONMENT

Action 6.P.4.

Reduce pesticide and herbicide use

Regenerative Greenspace, Food Systems &
NATURAL ENVIRONMENT

Inviting Bozeman Businesses and Community Organizations to Join the Climate Effort!

- ☐ Reduce energy usage during peak energy demand (4pm to 8pm) (Action 1.A.2.)
- ☐ Become a City of Bozeman Energy Project Partner (Action 1.A.3.)
- ☐ Contact NorthWestern Energy to schedule an energy appraisal and implement appraisal recommendations (Action 1.A.3.)
- ☐ Monitor and benchmark your building's energy performance (Action 1.A.4.)
- ☐ Explore opportunities to install on-site renewable energy and storage on your property (Action 2.F.4.)
- ☐ Review City maps to understand if you are in a location that is vulnerable to flooding, fires, or other hazards and develop an emergency and continuity of operations plan if a hazard event occurs (Action 3.H.3.)
- ☐ Engage your employees or constituents in emergency preparedness planning, drills, and protocols (Action 3.H.3.)
- ☐ Provide options and incentives for employee telecommuting and alternatives to single-occupancy vehicle travel (e.g., bike to work days, preferred parking spots, carpool matching, bicycle racks, wellness programs, etc.) (Actions 4.J.2. and 4.J.5.)
- ☐ Install electric vehicle charging infrastructure for fleet, employee, and potentially public use (Action 4.K.2.)
- ☐ Convert fleet vehicles and equipment to electric or alternative fuel models (Action 4.K.3.)
- ☐ Establish and enforce employee idling policies when using personal or fleet vehicles for business use (Action 4.K.5.).
- ☐ Limit non-essential airline travel and/or purchase carbon offsets for airline trips (Action 4.L.1.)
- ☐ Review your supply chain and consumption practices and look for opportunities to use less packaging, reuse or recycle materials, and compost organic waste (Action 5.M.1.)
- ☐ Provide markets for recycled products by supporting suppliers and businesses that use recycled materials (Action 5.M.1.)
- ☐ Reuse or donate used equipment and goods (Action 5.M.1.)
- ☐ Donate unused food and right-size large catering orders (Action 5.M.1.)
- ☐ Plant and maintain trees (Action 6.P.2.)
- ☐ Purchase products that support growth of the local food system (Action 6.N.3.).
- ☐ Update irrigation equipment and landscaping to use less water (Action 6.O.2.)
- ☐ Reduce pesticide and herbicide use (Action 6.P.4.)

Bozeman Climate Plan

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Bozeman Climate Plan

APPENDICES

APPENDIX A: MITIGATION ANALYSIS SUMMARY

The anticipated impact of each solution on the business-as-usual fuel use and emissions for each goal year are shown in Table 13. Descriptions of assumptions made developing these scenarios are described in the sections below. Note that potential emissions reductions were not calculated for solutions that primarily present resiliency or carbon sequestration benefits.

Table 13: Contribution of each mitigation solution to Bozeman's mitigation goals based on pathway B

Solutions		2025 *Paris Accord*	2030 *100% Net Clean Electricity*	2050 *Carbon Neutral*
Healthy, Adaptive and Efficient Buildings				
A. Improve Efficiency of Existing Buildings	Electricity Savings (kWh)	75,000,000	149,000,000	447,000,000
	Natural Gas Savings (therms)	4,200,000	8,400,000	25,000,000
B. Achieve Net-Zero New Construction	Electricity Savings (kWh)	51,000,000	64,000,000	303,000,000
	Natural Gas Savings (therms)	2,700,000	3,400,000	16,200,000
C. Electrify Buildings	Natural Gas Savings (therms)	0	0	9,200,000
	Increased Electricity Use (kWh)	0	0	269,000,000
Responsible and Reliable Clean Energy Supply				
D. Increase Utility Clean Energy Mix	Increased Clean Electricity Production (kWh)	80,000,000	100,000,000	306,000,000
E. Develop and Promote Utility Green Power Programs	Increased Clean Electricity Production (kWh)	124,000,000	98,000,000	0
F. Increase Community-Based Distributed Renewable Energy	Increased Clean Electricity Production (kWh)	16,000,000	40,000,000	80,000,000
Vibrant and Resilient Neighborhoods				
G. Facilitate Compact Development Patterns	Gasoline Savings (gallons)	1,900,000	3,800,000	11,200,000

Solutions		2025 *Paris Accord*	Target 2030 *100% Net Clean Electricity*	2050 *Carbon Neutral*
Diverse and Accessible Transportation Options				
J. Increase Walking, Bicycling, Carpooling and Use of Transit	Gasoline Savings (gallons)	1,700,000	3,400,000	10,200,000
K. Decrease Direct Vehicle Emissions	Gasoline Savings (gallons)	800,000	2,700,000	10,600,000
	Increased Electricity Use (kWh)	6,000,000	19,000,000	74,000,000
L. Improve Air Travel Efficiency	GHG Emissions Reduction (MT CO2e)	1,000	2,000	5,900
Comprehensive and Sustainable Waste Reduction				
M. Move Toward a Circular Economy and Zero Waste Community	GHG Emissions Reduction (MT CO2e)	7,100	14,100	42,300
Future Technologies				
Future Solutions	GHG Emissions Reduction (MT CO2e)	0	0	113,000
Summary by Fuel				
Electricity Savings (kWh)		126,000,000	213,000,000	750,000,000
Additional Clean Energy Production (kWh)		220,000,000	238,000,000	386,000,000
Natural Gas Savings (therms)		6,900,000	11,800,000	50,400,000
Gasoline Savings (gallons)		4,400,000	9,900,000	32,000,000
Other Emissions Reductions		8,100	16,100	161,200
Total GHG Emissions Reduction ($MT\ CO_2e$)		**319,000**	**451,000**	**1,218,000**

QUANTITATIVE ANALYSIS METHODOLOGY

The sections below provide basic information on the methodology used to calculate the impact of the mitigation solutions outlined in this plan.

Baseline & Forecast

The 2016 GHG inventory detailed in the 2017 Bozeman Community Greenhouse Gas Emissions Report was used to establish a baseline GHG emissions inventory for the community. This is also the data, along with the baseline 2008 inventory, on which the council made the decisions about the mitigation goals for the community. To understand what the emissions in 2050 were likely to be if no action was taken, we took the per capita emissions from the 2016 inventory, and applied to the likely population range for 2050. To estimate community population in 2050, we reviewed the population projections from a variety of community planning efforts outlined in Figure 22 as well as the more recent population projections in the Demographic and Real Estate Marketing Assessment.

Estimate or Projection Source	2010	2014	2015	2020	2024	2025	2030	2034	2035	2040
U.S. Census Bureau/CEIC Estimate	37,280	41,660	--	--		--	--	--	--	--
Bozeman Community Plan	42,700	--	54,500	69,500	--	88,700	--	--	--	--
Bozeman Wastewater Collection Facilities Plan Update	--	41,056	--	--	55,176	--	--	63,964	--	--
Bozeman Integrated Water Resource Plan										
Moderate Projection (2% annual growth)	--	--	41,160	45,444	--	50,174	55,396	--	61,161	67,527
High Projection(3% annual growth)	--	--	42,383	49,133	--	56,959	66,031	--	76,548	88,740
Bozeman Water Facility Plan	42,700	--	54,500	69,500	--	88,700	--	--	--	--
Wastewater Facilities Plan/Bozeman Community Plan (2008)	44,500	--	56,800	72,500	--	92,500	--	--	--	--
Bozeman Fire Protection Master Plan										
Census Based Projection	--	34,029	--	--	37,747	--	--	--	--	--
Development Based Projection	--	42,400	--	--	49,400	--	--	--	--	--
Bozeman 20/20 Community Plan, 2001	39,600		43,120	46,600	--	--	--	--	--	--
RPA Projection @2.49%/yr (1970-2010 rate)			42,697	48,285		54,603	61,748		69,828	78,966

Figure 22. Various population projections for the City of Bozeman

These population estimates were graphed and a high and low growth scenario were established to encompass the majority of the data points as shown in Figure 23.

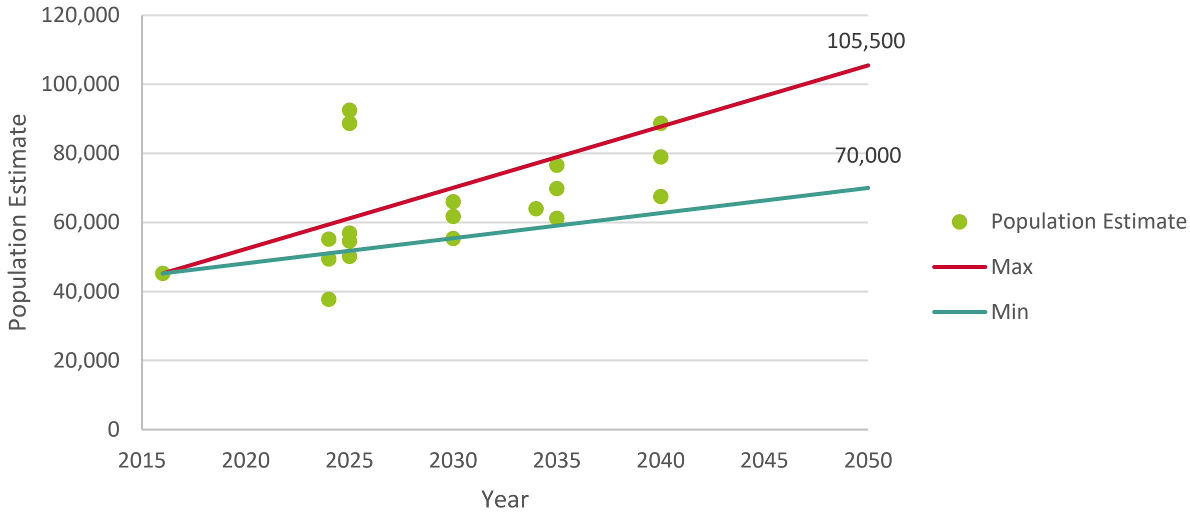

Figure 23. Estimated population growth scenarios

These growth scenarios estimate that the 2050 population will be between 70,000 and 105,500 people in 2050. These population estimates were multiplied by the 2016 per capita GHG emissions to calculate the business as usual emission between 1.2 million and 0.8 million MT CO_2e. To plan for worst case scenario, all calculations used to determine the required contribution of each solution to meet the City's goals were based on the high growth scenario.

Note: NorthWestern Energy provided revised emissions factors for the 2008 and 2016 electricity use in 2020. The impact of the updated emissions factor changes were reviewed and did not result in a material change in the total reported emissions for the 2008 and 2016 inventories. Based on guidance from the Global Protocol for Community-scale Greenhouse Gas Inventories, the baseline inventory was not recalculated. The projected 2050 business as usual GHG emissions estimates were updated to reflect the revised emissions factors.

Identifying and Prioritizing Solutions

To meet the City's ambitious carbon neutral by 2050 goal, solutions must be identified to address all sources of GHG emissions within the City including residential and commercial buildings, on-road transportation, aviation emissions, and waste. The solutions identified through stakeholder engagement are listed below and the solutions with significant or measurable mitigation potential for the sources noted are in bold:

- **Solution A. Improve Efficiency of Existing Buildings**
- **Solution B. Achieve Net Zero Energy New Construction**
- **Solution C. Electrify Buildings**
- **Solution D. Increase Utility Clean Energy Mix**
- **Solution E. Develop and Promote Utility Green Power Programs**
- **Solution F. Increase Community-Based Distributed Renewable Energy Generation**
- **Solution G. Facilitate Compact Development Patterns**
- Solution H. Reduce Vulnerability of Neighborhoods and Infrastructure to Natural Hazards
- Solution I. Enhance Social Infrastructure and Community Preparedness
- **Solution J. Increase Walking, Bicycling, Carpooling, and Use of Transit**
- **Solution K. Decrease Direct Vehicle Emissions**
- **Solution L. Improve Air Travel Efficiency**
- **Solution M. Move Toward a Circular Economy and Zero Waste Community**
- Solution N. Cultivate a Robust Local Food System
- Solution O. Manage and Conserve Water Resources
- Solution P. Manage Land and Resources to Sequester Carbon

Potential Impact by Solution

To understand the potential greenhouse gas emissions impact of each mitigation solution, a best in class estimate of potential impact was calculated from leading communities around the country. This was compared to the current efforts in Bozeman to provide a solution baseline. Recognizing that best in class examples are rapidly evolving as cities and states innovate and adopt new policies and technologies, using existing examples of best practices helps demonstrate the viability of the proposed solutions. The methodology is described below by solution including assumptions made and resources referenced. The potential emissions reduction based on the best in class scenarios described below were presented to the Climate Team during Workshop 2 to get their feedback on the potential emissions savings from each solution applied to Bozeman. Through a keypad polling exercise, the stakeholders were asked to consider climate, political will, and existing momentum to determine the emissions potential for each solution in Bozeman expressed as a percentage of the best in class scenario. This is expressed in a percentage of the best in class emissions reduction by scenario below and shown in Figure 24. Note that interactions between solutions has not been taken into account at this phase of analysis. This was considered in the scenarios and goal modeling.

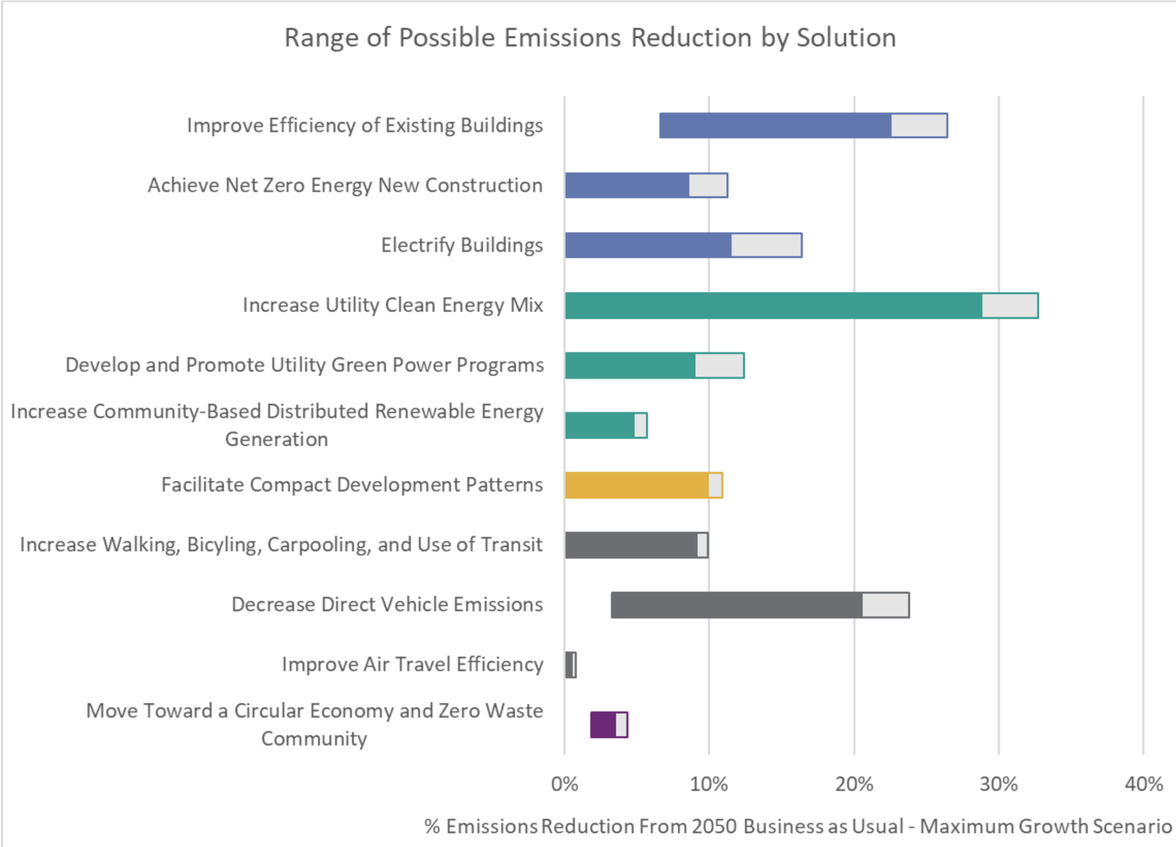

Figure 24. Estimated emissions reduction potential by solution

Note that Solutions H, I, N, O, and P were not analyzed in terms of greenhouse gas emissions reduction potential as they are primary resiliency and/or carbon sequestration solutions that cannot be modeled in this analysis.

Solution A. Improve Efficiency of Existing Buildings

Current Efforts

As part of NorthWestern Energy's 2019 Resource Procurement Plan, they evaluated the potential electricity savings from the current rebate program between 2015 and 2034 to be 7% of total use. This estimate was extrapolated to 2050 (NorthWestern Energy, 2019). Similar data were not readily available for natural gas savings, so the same level of savings was assumed for natural gas efficiency.

Best in Class

Arizona's second largest utility, Salt River Project, has the best in class energy efficiency programs in the mountain west with 2.1% energy savings in 2013 and reached similar levels in 2014, 2015, and 2018. Arizona was identified as the top performing state outside of California by the American Council for an Energy-Efficient Economy (Southwest Energy Efficiency Project, 2019). This annual savings estimate was projected to 2050. The total estimated energy reduction was compared to the expected electricity savings under the advanced energy efficiency scenario in an ACEEE study which considers the impact of deep penetration of existing energy efficiency technologies in 2050 (American Council for an Energy-Efficient Economy, 2012). Based on this study, the total energy efficiency savings was capped at 51% of baseline energy use.

Bozeman's Best

Based on feedback from stakeholders in workshop 2, the potential reduction of electricity and natural gas use from energy efficiency was assumed to be 85% of the best in class scenario.

Solution B. Achieve Net-Zero New Construction

Current Efforts

The number of new homes built to net-zero ready construction standards in 2019 was considered to be negligible.

Best in Class

The C40 Cities initiative to encourage cities to commit to work toward net zero construction by 2030 was considered to be best in class performance in this emerging field. To evaluate the impact of this commitment, the increase in electricity and natural gas use due to new buildings anticipated before and after 2030 based on the max growth scenario was estimated. Using the DOE definition of net zero ready home, the energy use of net zero buildings was estimated to be 50% of baseline energy use (US Department of Energy, n.d.). The adjusted energy use of new buildings was calculated assuming that 50% of buildings built between now and 2030 will be net zero energy ready (i.e., 50% energy use reduction)and after 2030, 100% of all buildings constructed will be net zero ready (i.e., 50% energy use reduction).

Bozeman's Best

Based on feedback from workshop participants, Bozeman's Best was analyzed as 76% of the best in class scenario. This means that 76% of all new buildings built after 2030 would be built to net zero standards (i.e., 50% energy use reduction) or that in effect all new construction achieves an overall average of 38% energy use reduction (76% of 50%). In addition, 38% (76% of 50%) of the buildings built between now and 2030 will be built to net zero standards (50% reduction) or that in effect all new construction achieves an overall average of 19% energy use reduction.

Solution C. Electrify Buildings

Current Efforts

It was assumed there are not currently any significant efforts to electrify building energy loads in Bozeman.

Best in Class

A report by the National Renewable Energy Laboratory (NREL) shows that 80% of commercial and residential natural gas loads could be electrified (National Renewable Energy Labratory, 2017). Based on the GHG inventory energy use data, there are no significant industrial loads in Bozeman to consider. The total potential savings assumes that the City reaches its 100% renewable electricity goal.

Bozeman's Best

The potential for electrification of residential and commercial loads in Bozeman was assumed to be 70% of the best in class scenario based on stakeholder feedback.

Solution D. Increase Utility Renewable Energy Resource Mix

Current Efforts

At workshop 2, NorthWestern Energy's carbon reduction plan presented in Our Vision for Montana was used as the baseline scenario. This report projects emissions factor reductions through 2045. It was assumed no significant changes would be implemented between 2045 and 2050.

Best in Class
Several utilities including Platte River Power Authority in Colorado, Idaho Power, and Xcel Energy have made public pledges to eliminate carbon from their generation mix by 2050 or sooner. The total potential GHG savings assumes business as usual growth in electric energy use from residential and commercial buildings. Additional electric load from building or transportation was not considered. The interdependencies between strategies was taken into account during scenario modeling (Figure 13).

Bozeman's Best
Based on stakeholder feedback, it was assumed by 2050 the carbon free generation sources for NorthWestern Energy will be 88% of its total energy generation.

Solution E. Develop and Promote Utility Green Power Programs
Current Efforts

NorthWestern Energy financial reports from E+ Green program were used to estimate the amount of electricity purchased through this program (NorthWestern Energy, 2018). The total was scaled to the city based on the total electricity generated by NorthWestern Energy compared to the electricity used by Bozeman.

Best in class:
Current green tariff programs from across the country were evaluated to understand the potential size of the program. The largest programs could serve the entire City, so the City is likely not limited by the amount of electricity they could procure by the mechanism. It was assumed that all of electricity from MSU (based on FY19 campus electricity use) and municipal operations (estimated as % of total community electricity) would be included. For the rest of the community energy use, it was estimated that 20% of homes and businesses would participate based on an NREL study on best in class green power program participation (National Renewable Enegy Labratory, 2017).

Bozeman's Best
It was assumed Bozeman could attain 72% of the best in class scenario based on stakeholder input.

Solution F. Increase Community-Based Distributed Renewable Energy Generation
Current Efforts
The program participation rates in NorthWestern Energy's E+ Renewable rebate program from the program evaluation report were used to estimate the number of community and rooftop solar systems in NorthWestern Energy's service territory (SBW Consulting, Inc., 2013). This estimate was scaled to Bozeman based on electricity use of the city compared to NorthWestern Energy generation.

Best in Class
Based on a 2017 NREL study of the potential behind the meter renewable energy generation potential was modeled on three different scenarios. In the central scenario modeled there a total of 850 MW of installed behind the meter in 2050 with incremental estimates of 250 MW in 2030 and 100 MW in 2025 (National Renewable Energy Laboratory, 2017). These estimates were scaled to Bozeman based on the percentage of NorthWestern's electricity generation delivered to Bozeman to calculate the best in class scenario. In addition to rooftop solar, 50 MW of community solar was used to estimate the community solar garden potential based on 100% Net Clean Energy goals set by the City of Fort Collins. For context, the median community solar installation in 2020 based on an NREL database was 2.1 MW (National

Renewable Energy Laboratory, 2020). This means that best in class performance is installing one community garden of similar size nearly every year between now and 2050.

Bozeman's Best

Based on stakeholder input it was assumed that distributed generation could reach 83% of the best in class scenario for community solar generation. For rooftop solar generation, the estimated production was not scaled because the central scenario from the 2017 NREL study was chosen rather than the high adoption scenario.

Solution G. Facilitate Compact Development Patterns

Current Efforts

While Bozeman has made considerable efforts to promote compact development patterns, there has been significant growth outside of City boundaries making it difficult to attribute significant reductions in vehicles miles traveled to compact development initiatives. Note that energy efficiency benefits from compact developments are captured in solution B.

Best in Class

Based on a study examining the effects of density on driving patterns there were a handful of top performing cities. The mean density increase from these cities was used to estimate the change in population density that could reasonably be expected in Bozeman over the 30-year planning horizon (DAVID BROWNSTONE, UNC Irvine, 2008). Based on the factors for density and proximity to downtown (or other commercial nodes), a separate study that evaluates the effect of various factors on Vehicle Miles Traveled (VMT) was used to estimate the reduction in miles traveled (Florida, 2017)

Bozeman's Best

Based on stakeholder feedback, we assumed that Bozeman could achieve 91% of the VMT reduction in the best in class scenario.

Solution J. Increase Walking, Bicycling, Carpooling and Use of Transit

Current Efforts

Based on US Census data, the percentage of Bozeman residents commuting using transit and active modes of transportation decreased between 2012 and 2016.

Best in Class

A study of the best performing cities as a percentage of citizens using public transit was reviewed to find a high performing city with similar characteristics. Ithaca, NY was chosen since it is a similar sized college town (rails-to-trails conservancy). At the time of the study, 28.8% chose to commute using transit and active modes of transportation.

Bozeman's Best

Based on workshop feedback it was assumed that Bozeman would be able to achieve 92% of best in class transit and active commuting or 26% of residents choosing to walk, bike, carpool, or use transit.

Solution K. Decrease Direct Vehicle Emissions

Current efforts

The increase in fuel efficiency from the Bozeman GHG inventory between 2012 and 2016 was extrapolated to 2050 at half the rate to account for changes in fuel efficiency regulations. The rate of fuel efficiency increase seen in Bozeman was similar to the national change, so it is likely due to fuel standards for vehicles. The percentage of EVs registered in MT in 2018 was <1%, so it was assumed to be negligible in Bozeman (Auto Alliance, 2018).

Best in Class

It was assumed that by 2050, vehicle fuel efficiency gains would be from EVs rather than more efficient Internal Combustion Vehicles (ICE). To understand best in class EV adoption, manufacturer predictions on the percentage of sales that would be EV were reviewed (MJB&A, 2019). Then the national vehicle replacement rate was used to understand what percent of the Bozeman vehicle fleet would likely be replaced by 2050 (Oak Ridge National Laboratory, 2019). Based on national EV sales projections to 2050 and the vehicle turnover rate, 70% of vehicles will be EVs by 2050 nationwide. Based on this, we assume that best in class cities will be nearly all EV, so best in class performance was set at 95% of light duty fleet being replaced by EVs. It is assumed that the City has reached its 100% renewable electricity goal to realize maximum benefit from EVs by 2050.

While technology is changing rapidly, it was assumed that the EV conversion is limited to light-duty vehicles since there is limited technology currently available for heavy duty applications and no commercially viable options for semi-trucks. Any improvements in this area between now and 2050 will be captured in the future solutions bucket.

Bozeman's Best

It was assumed that the community could reach 86% of the best in class scenario or about 82% of light duty vehicles converted to EVs based on stakeholder input during workshop 2.

Solution L. Improve Air Travel Efficiency

Current Efforts

There was a significant increase in airline emissions between 2012 and 2016 identified in the Bozeman GHG Inventory.

Best in Class

Based on Project Drawdown, there is the potential to increase airline fuel efficiency by 11% through installation of existing technologies (Project Drawdown, n.d.).

Bozeman's Best

Based on stakeholder feedback it was assumed that Bozeman would be able to reach about 62% of the best in class scenario or a 7% increase in fuel efficiency.

Solution M. Move Toward a Circular Economy and Zero Waste Community

Current efforts

The landfilled waste per person went down by 9% between 2012 and 2016. This rate of reduction was projected through 2050. Since recycling rates are starting to plateau across the country this projected waste reduction estimate was de-rated by 50%.

Best in Class

Austin, TX has developed a plan to reach 90% waste reduction by 2040 (Gary Liss & Associates, 2008).

Bozeman's Best

Based on stakeholder feedback, it was estimated that Bozeman could reach 80% of the best in class scenario or about 72% reduction in landfilled waste.

Scenarios & Goal Modeling

Next, a top-down analysis was done based on the business-as-usual projections, to create emissions reduction scenarios that allow the City to meet its emission reduction and clean electricity goals. The business as usual scenario was built using fuel use and emissions estimates from the 2016 GHG inventory scaled by population using the maximum population growth scenario. The scenarios in each of the goal years are detailed below.

2025: Paris Accord

To develop a scenario allowing the City to meet their 2025 commitment, it was assumed that all efficiency and electrification strategies are on pace to meet Bozeman's best scenario in 2050 with the following exceptions:

1. No meaningful building electrification is expected until after 2030.
2. EV transition for the community's light-duty fleet is assumed to have a more exponential uptake curve rather than the linear progression assumed for other strategies based on national sales trends over the last 10 years.

Next, the amount of clean electricity generation that would be required for Bozeman to meet its carbon reduction goal was calculated. For this report, we modeled two potential scenarios for generating the necessary clean electricity.

No Additional Utility Renewable Energy Generation - Pathway A: This pathway shows how the City can generate the amount of required clean electricity without any additional clean energy added to NorthWestern Energy's electricity generation mix. As of 2019, NorthWestern Energy delivered approximately 280 million kWh of clean electricity to Bozeman or about 56% of the projected clean electricity required in 2025. The amount of renewable electricity generated through distributed generation within the city remains unchanged at 3%, and the remaining 40% of required clean electricity is purchased through utility-scale green tariff or other procurement mechanism. This level of participation is slightly higher than the modeled best in class scenario, 40% vs. 38%. This means that success in meeting the 2025 goal without additional clean electricity from the utility will require significant community support for the green tariff or for other efficiency measures to outperform what has been projected.

Additional Utility Renewable Energy Generation - Pathway B: Under this pathway, NorthWestern Energy is able to leverage new clean energy generation opportunities to meet their stated carbon intensity goal. Assuming that the carbon intensity of the fossil fuel electricity generation mix remains constant, the utility mix will include 72% clean electricity. Based on the distributed generation installed capacity curve described in the Bozeman's Best Scenario, distributed generation can provide about 3% of the required clean electricity. This means that the City must work with NorthWestern Energy to establish an agreement to provide about 25% of the City's electricity needs through a green tariff. It is assumed that this green tariff agreement allows subscribers to replace a portion of NorthWestern's fossil fuel generation with clean electricity sources.

2030: 100% Net Clean Electricity

To meet this goal, only scenarios impacting electricity use were considered. Bozeman's best scenario was first applied for energy efficiency and net zero new construction solutions using a maximum annual implementation rate. This was used to determine the maximum energy use reduction that might be expected from these efficiency measures by 2030. It was assumed that there would be no significant building electrification efforts until the City is able to reach their 100% clean electricity goal to maximize the benefit from these efforts and give the technologies some time to mature. Next, the expected increase to electricity use from the electrification of the light duty vehicles in the Bozeman community was considered to determine the total amount of clean energy generation that would be required to meet Bozeman's projected electricity use.

The required clean electricity to meet the City's goal of 100% net clean electricity will be provided through the generation from the three renewable energy solutions (D-F). Again, we modeled two potential scenarios for generating the necessary clean electricity.

No Additional Utility Renewable Energy Generation - Pathway A: This pathway shows how the City can generate the amount of required clean electricity without any additional clean energy added to NorthWestern's electricity generation mix. As of 2019, NorthWestern Energy delivered about 280 million kWh of clean electricity to Bozeman or about 54% of the projected clean electricity required in 2030. The amount of renewable electricity generated through distributed generation within the city remains unchanged at 8%, and the remaining 38% of required clean electricity is purchased through utility-scale green tariff or other procurement mechanism. This level of participation is in line with the percentage participants in the community described in the Best in Class scenario.

Note: Other solutions shown in Table 13 not described above show the level of impact required to be on pace to achieve the City's 2050 goal.

Additional Utility Renewable Energy Generation - Pathway B: Under this pathway, NorthWestern Energy leverages new clean energy generation opportunities to meet their stated 2030 carbon intensity reduction goal. Assuming that the carbon intensity of the remaining fossil fuel generation remains constant, the utility mix will include 73% clean electricity. Based on the distributed generation installed capacity curve described in the Bozeman's Best Scenario, distributed generation can provide about 8% of the required clean electricity. The remaining 19% clean electricity will be purchased through utility-scale green tariff programs.

2050: Carbon Neutrality

For the 2050 scenario, the maximum expected savings from each energy efficiency and electrification solution (A-C, G, J-L, and M) shown above were applied to the projected business as usual (BAU) energy use and emissions from each GHG emissions source. This allowed the anticipated total energy requirements of the City to be determined.

The required clean electricity will be provided through the generation from the three clean energy solutions (D-F). The City is expected to have reached 100% net clean electricity by 2030 and will need to continue to add clean energy generation to meet growing electricity demand from building and transportation electrification as well as City growth. For this report, we modeled two potential scenarios for generating the necessary clean electricity.

Additional Utility Renewable Energy Generation - Pathway B: This pathway shows how the City can generate the amount of required clean electricity without any additional clean energy added to NorthWestern Energy's electricity generation mix. As of 2019, NorthWestern Energy delivered about 280 million kWh of clean electricity to Bozeman or about 42% of the projected clean electricity required in 2050. The amount of renewable electricity generated through distributed generation within the city produces an additional 20% of the required clean electricity. This leaves the remaining 38% of required clean electricity to be purchased through utility-scale green tariff or other procurement mechanism. While this scenario doesn't rely on the utility to add additional clean electricity generation, it does require the City and the Utility to work together to contract clean electricity through the green tariff solution.

The remaining fuel use and GHG emissions required will need to be addressed through future opportunities or technologies that will likely arise between the writing of this report and 2050. The remaining energy use and emissions that will need to be addressed by future opportunities include:

1. Natural gas use from existing facilities that cannot be electrified
2. Heavy duty vehicle emissions
3. Remaining landfill emissions
4. Aviation transportation emissions

These future opportunities may include local carbon sequestration and/or biofuels.

Additional Utility Renewable Energy Generation - Pathway B: Under this pathway, NorthWestern Energy leverages new clean energy generation opportunities to reach 88% clean energy by 2050, which exceeds their stated carbon intensity reduction goal. Distributed generation in the City generates the remaining 12% of the required renewable energy. It should be noted that the defined Bozeman's Best distributed generation, about 60 MW of rooftop solar and 40 MW of community solar gardens, could generate more than the required clean energy. This leaves some redundancy in case efficiency solutions do not perform as well as currently modeled. Community solar development is dependent upon a change in state-law and/or a new tariff for utility-owned community solar (as specified in Action 2.F.3).

APPENDIX B: EXISTING PLAN SUMMARY

2020 BOZEMAN YELLOWSTONE INTERNATIONAL AIRPORT MASTER PLAN

The Master Plan for the Bozeman Yellowstone International Airport (BZN) was undertaken by the Gallatin Airport Authority to outline a 20-year long range, orderly direction for airport development to yield a safe, efficient, economical, and environmentally acceptable air transportation facility.

https://bozemanairport.com/reports-and-statistics

BOZEMAN COMMUNITY PLAN (2020 – DRAFT UNDERWAY)

The Community Plan will serve as a document to help guide the City on how to grow and develop within the context of rapidly changing land development and economic conditions. The Community Plan is oriented around seven themes: a city of neighborhoods, a city bolstered by downtown and complementary districts, a city influenced by our natural environment, parks, and open lands, a city that prioritizes accessibility and mobility choices, a city powered by its creative, innovative, and entrepreneurial economy, and a city engaged in regional coordination.

https://www.bozeman.net/city-projects/bozeman-community-plan-update

BOZEMAN DROUGHT MANAGEMENT PLAN (2017)

The Bozeman Drought Management Plan focuses primarily on short- and long-term actions to help mitigate potential drought. Uniquely, the plan recognizes the potential impacts of climate change on Bozeman's water system and the importance of planning for the additional uncertainty associated with this future change. Bozeman is expected to experience longer and more frequent drought periods in the future, partially associated with decreased snowpack.

https://www.bozeman.net/Home/ShowDocument?id=4791

BOZEMAN GUIDELINES FOR HISTORIC PRESERVATION & THE NEIGHBORHOOD CONSERVATION OVERLAY DISTRICT

The City of Bozeman adopted the Neighborhood Conservation Overlay District (NCOD) in 1991. The goal of the zoning overlay is to conserve neighborhood character, protect the integrity of historic structures and provide public notice to adjoining property owners of potential changes. The Bozeman Design Guidelines for Historic Preservation & Neighborhood Conservation Overlay District provide guidelines for appropriate work within the NCOD, Historic Districts and Individually Listed Properties. The guidelines address topics such as commercial and residential building height, materials, and utilities.

https://library.municode.com/mt/bozeman/codes/code_of_ordinances?nodeId=PTIICOOR_CH38UNDECO_ART3ZODILAUS_DIV38.340OVDIST_PT1NECOOVDIHIPR

BOZEMAN PARKS, RECREATION, OPEN SPACE, AND TRAILS PLAN (2007)

The Parks, Recreation, Open Space, and Trails Plan presents a roadmap for providing an equitable, well maintained park system to serve all of Bozeman's residents. While the plan does not include any mention of climate change, the implementation of this plan may be seen as a mitigation tool by encouraging the protection and creation of greenspace and natural systems and encouraging the use of multi-modal trail systems.

https://www.bozeman.net/home/showdocument?id=3284

CITY OF BOZEMAN AND MSU STORMWATER MANAGEMENT PLAN (2019)

The Stormwater Management Plan describes the plans and programs in place to improve water quality, comply with environmental regulations, and improve urban flood resiliency and climate change preparedness.

https://www.bozeman.net/home/showdocument?id=5681

CITY OF BOZEMAN STRATEGIC PLAN (2018)

With a shorter implementation horizon than the Community Plan, the interactive City of Bozeman Strategic plan puts many of the high-level ideas of broader community and area plans into an actionable format. The Bozeman Strategic Plan provides several actionable strategies oriented around community safety and risk abatement, well-planned growth, and a sustainable environment, among others. While many of these strategies relate to Climate Plan focus areas, including City Assets, Transportation, and Greenspace, the Strategic Plan also includes one goal explicitly linked to Climate Change: "Reduce community and municipal Greenhouse Gas (GHG) emissions, increase the supply of clean and renewable energy; foster related businesses."

https://strategic-plan-bozeman.opendata.arcgis.com/

CLIMATE VULNERABILITY ASSESSMENT AND RESILIENCE STRATEGY (2019)

This plan inventoried key City infrastructure to evaluate Bozeman's climate-induced municipal vulnerability and identified key actions to maintain and enhance resilience. Heat, flooding, drought, declining snowpack, wildfire, and winter storms were the top climate hazards against which vulnerability was assessed.

https://www.bozeman.net/home/showdocument?id=8958

COMMUNITY CLIMATE ACTION PLAN (2011)

The Community Climate Action Plan extends beyond municipal stewardship and action to identify climate goals and objectives for all of Bozeman. This plan establishes a goal of "reducing community-wide greenhouse gas emissions 10% below 2008 levels by 2025." The 2011 Climate Action Plan includes a list of strategies oriented toward mitigating climate impacts through more efficient transportation, energy, buildings, and waste systems. This Climate Plan effort is an update to the Community Climate Action Plan.

https://weblink.bozeman.net/WebLink/DocView.aspx?id234376&dbid=0&repo=BOZEMAN&cr=1

COMMUNITY GREENHOUSE GAS EMISSIONS REPORT (2017)

The Community Greenhouse Gas Emissions Report is primarily an inventory of Bozeman's emissions, focused on transportation, buildings, waste, and water/wastewater. However, the end of this report includes a brief recommendations section. Recommendations include pursuing electrification of the vehicle fleet, commercial and residential building efficiency, and the completion of a vulnerability assessment and Climate Plan. The vulnerability assessment was completed in 2019 and the Climate Plan is currently underway.

https://www.bozeman.net/home/showdocument?id=5418

DOWNTOWN BOZEMAN IMPROVEMENT PLAN (2019)

The Downtown Bozeman Improvement Plan identifies climate change as an important backdrop for community planning efforts. Several focus areas of the plan, such as Heart of a Thriving Bozeman, Walkable and Accessible, and Connected to Nature & Culture, include recommendations that tie directly into the framework of the Bozeman Climate Plan (e.g., Community Development, Transportation, and Greenspace and Natural Systems).

https://s3.us-west-2.amazonaws.com/dba-2021/Resource-PDFs/2019_DBIP_with_Appendix_FINAL_ADOPTED_4-15-19.pdf

ECONOMIC DEVELOPMENT STRATEGY (2017)

Though the Economic Development Strategy does not explicitly consider climate change, several of the plan's objectives and recommended actions fit into the Climate Plan's focus areas. Climate-related strategies include efficient density and community development and a focus on multi-modal transportation. The Economic Development Strategy also elevates the need to protect and enhance the outdoor and tourism industry, which is one of Bozeman's most climate-vulnerable economic drivers.

http://weblink.bozeman.net/WebLink8/0/doc/120846/Electronic.aspx

GALLATIN COUNTY HAZARD MITIGATION PLAN (2018)

Similar to the Climate Vulnerability Assessment and Resilience Strategy (2019), the Gallatin County Hazard Mitigation Plan focuses on preparedness for hazards, including wildfire, earthquakes, and flooding. While the plan is not focused specifically on climate-induced hazards, it does recognize the role of climate change in exacerbating hazards in the future, especially wildfires.

https://www.readygallatin.com/wp-content/uploads/2019/06/FINAL-DRAFT-Gallatin-County-Hazard-Mit-Plan_05-30-2019_plus-MSU-Annex-CWPP_low_res.pdf

INTEGRATED WATER RESOURCES PLAN (2013)

The Integrated Water Resource Plan serves to "guide water supply and water use policy for the next fifty years." Strategies focus primarily on demand reduction and supply diversification. While the Plan is primarily a technical document to ensure the provision of water for Bozeman into the future, it recognizes the potential impact of climate change on water systems in the future.

https://www.bozeman.net/home/showdocument?id=836

MIDTOWN ACTION PLAN

The Midtown Action Plan aims to attract targeted private investment to the Midtown Urban Renewal District by removing barriers to development through strategic infrastructure investments and incentives. The objective of the Midtown Urban Renewal District is to promote economic development, multimodal transportation, support innovative infrastructure, promote unified human scale urban design, and support compatible urban density mixed land uses through infill, increased building density, retail, housing, and multimodal amenities.

https://midtownbozeman.org/uploads/Documents/Action-Plan-V10.pdf

MUNICIPAL CLIMATE ACTION PLAN (2008)

The Municipal Climate Action Plan establishes a city-operations climate goal of "reducing municipal greenhouse gas emissions 15% below 2008 levels by 2020." Similar to the Community Climate Action Plan, the Municipal Climate Action Plan includes a list of strategies oriented toward mitigating climate impacts through efficient land use, energy, wastewater, and transportation planning.

https://www.bozeman.net/home/showdocument?id=3140

MUNICIPAL GREENHOUSE GAS EMISSIONS (2012)

The Municipal Greenhouse Gas Emissions is an inventory of transportation, building, water/ wastewater, and waste emission. The inventory concludes with an actionable framework with the following three priorities for municipal operations: energy efficiency and conservation, high energy performance standards for all new facilities and infrastructure, and a renewable energy plan.

https://weblink.bozeman.net/WebLink/DocView.aspx?id=234375&dbid=0&repo=BOZEMAN

NORTHWESTERN ENERGY ELECTRIC SUPPLY RESOURCE PROCUREMENT PLAN (2019)

In this plan, Bozeman's electricity provider, NorthWestern Energy (NWE), summarized its current power supply, future needs and its strategy for procuring additional resources to address supply needs. Concurrent with Bozeman's Climate Planning timeline, NWE will be conducting an independent competitive procurement process to secure up to 400 MW of peaking capacity. NWE analysis indicates that thermal resources are the lowest-cost resource but notes that renewable resources and energy storage costs are declining. The procurement process will consider all resources – natural gas, renewable energy and even "demand side" resources that reduce supply needs through energy efficiency.

https://www.northwesternenergy.com/docs/default-source/default-document-library/about-us/regulatory/2019-plan/complete-plan.pdf?sfvrsn=2fe04519_7

REDESIGN STREAMLINE 2020

Bozeman's fare-free transit system, Streamline, launched the Redesign Streamline planning process in November 2019. The project involves a comprehensive study of Streamline service and operations, culminating in a Transit Development Plan that details recommendations and implementation strategies for near-and long-term service changes to meet the growing community needs.

https://streamlinebus.com/about/redesign-streamline-2020/

TRANSPORTATION MASTER PLAN (2017)

The Transportation Master Plan recognizes the importance of reducing transportation-related carbon emissions as part of a triple bottom line approach to evaluating transportation decisions. While the Plan focuses holistically on Bozeman's mobility, which includes options for vehicle mobility, the Transportation Master Plan includes several strategies and recommendations focused on Transportation Demand Management (TDM) and encourages the use of alternative modes of transportation, such as biking, walking, and taking transit.

http://weblink.bozeman.net/WebLink8/0/doc/122828/Electronic.aspx

TRIANGLE COMMUNITY PLAN

With population growth and fast-paced land development, the Triangle area of Gallatin County, which is loosely described as the areas between Bozeman, Four corners, and Belgrade, is experience change. The Planning Coordination Committee (PCC) developed the Triangle Community Plan to coordinate land use development patterns, deliver community services, and infrastructure, and protect important environmental resources, all in a manner that supports community values and vision while responding to rapid growth pressures.

https://gallatincomt.virtualtownhall.net/planning-community-development/pages/triangle-community-plan

UNIFIED DEVELOPMENT CODE

The Unified Development Code (commonly called the "UDC") is a set of regulations aimed to protect the public health, safety and general welfare. These regulations recognize and balance the various rights and responsibilities relating to land ownership, use and development. The UDC is just one chapter of the Bozeman Municipal Code but it covers a wide range of topics such as setbacks, building height, allowed uses, landscaping, affordable housing and parking. Some regulations apply city-wide while others are district-specific.

https://library.municode.com/mt/bozeman/codes/code_of_ordinances?nodeId=PTIICOOR_CH38UNDEC O

https://www.bozeman.net/home/showdocument?id=8932

URBAN FORESTRY MANAGEMENT PLAN (2016)

One of the foundational goals of the Urban Forestry Management Plan is to “develop [an] urban forest resilient to climate change and invasive pests.” The plan also recognizes the increasing threats associated with a changing climate and iterates the importance of a healthy urban forest to combat climate change. Plan strategies include building Bozeman’s urban canopy infrastructure, bolstering maintenance protocol, and focusing on outreach and education.

https://www.bozeman.net/home/showdocument?id=3621

APPENDIX C: CLIMATE TEAM WORKSHOP SUMMARIES

APPENDIX D: COMMUNITY FORUM SUMMARIES

Appendices C & D are available online via City of Bozeman documents:

https://weblink.bozeman.net/WebLink/DocView.aspx?id=254708&dbid=0&repo=BOZEMAN

For technical assistance, contact the Bozeman Sustainability Division at (406) 582-2317